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Supplementary information

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Mosaic integration of spatial multi-omics with SpaMosaic

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Supplementary Information

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Supplementary Note 1: Benchmarking on real spatial multi-omics data with simulated mosaic configurations

Integration of intra-assay embryonic mouse brain sections across developmental stages (Supp. Fig. S11-S13)
Here we used an experimental dataset consisting of mouse embryonic brain sagittal section data profiled with Misar-seq[1] that jointly captures gene expression (RNA) and chromatin accessibility (ATAC). We selected the E13.5, E15.5, and E18.5 sections, and removed the ATAC modality of E13.5 and RNA modality of E18.5 to simulate mosaic data availability (**Supp. Fig. S11a**). Examination of the UMAPs of each modality showed the presence of strong batch effects between the different time points in both modalities (**Supp. Fig. S12a**). Following the same procedure in the previous example, we compared SpaMosaic with the other single-cell mosaic integration methods as well as spatial domain (niche) identification methods. We benchmarked SpaMosaic against CLUE[2], Cobolt[3], MIDAS[4], MultiVI[5], scMoMaT[6], and StabMap[7].

We first considered the single modality clustering methods. The baseline Leiden clustering successfully identified major brain regions like the mantle zone of dorsal pallium (DPallm; cluster 7 in RNA, 3 in ATAC), ventricular zones of dorsal pallium (DPallv; cluster 10 in RNA, 4 in ATAC), hindbrain (cluster 1 in both modalities), and subpallium (cluster 9 in RNA, 5 in ATAC) across sections, but showed noisy cluster boundaries. For CellCharter and BANKSY, which are tailored for spatial omics, their clusters showed improved spatial coherence compared to the Leiden clustering (**Supp. Fig. S13a and S12b**). However, the two methods showed markedly different performance in distinguishing known anatomical regions. BANKSY's performance was largely poor, merging distinct brain regions with other structures and showing inconsistent cluster assignments for most anatomical regions across sections. For example, BANKSY merged DPallv and DPallm with the hindbrain and midbrain regions in the E13.5 and E15.5 sections. In contrast, CellCharter achieved more biologically relevant segmentation of major brain structures. It successfully identified the DPallm and DPallv in both modalities of the E15.5 section, and accurately identified the subpallium (cluster 10), thalamus (cluster 6), and hypothalamus (cluster 7) in the ATAC modality. However, CellCharter's limitation in handling mosaic datasets led to misaligned clusters between RNA and ATAC within the same section, making it difficult to achieve unified clustering across the entire dataset.

Next, we examined the mosaic integration methods' output. Among the mosaic integration methods, SpaMosaic's spatial clusters had the least noise and best matched the Allen Brain Atlas reference[8] in terms of the brain regions captured (**Supp. Fig. S11b,c**). Across all developmental stages, SpaMosaic successfully identified the most comprehensive set of brain regions with well-defined boundaries, including the midbrain, hindbrain, diencephalon, subpallium, hypothalamus, thalamus, DPallm, and DPallv (detailed cluster assignments shown in **Supp. Fig. S11b**). In comparison, the other mosaic integration methods showed varying degrees of limitations. MultiVI identified the subpallium across all three sections and thalamus in E15.5 and E18.5, but frequently merged other distinct brain regions (e.g., DPallm with DPallv, hindbrain with midbrain). The performance of CLUE, Cobolt, and MIDAS were also poor, only capturing certain major brain regions like the DPallm, DPallv, thalamus, and subpallium. Their remaining clusters were highly noisy and could not be matched to the reference structures. Furthermore, these methods produced inconsistent cluster assignments across sections. For example, CLUE assigned the hindbrain region to clusters 1, 2, and 7 in the three sections, respectively. scMoMaT successfully captured the DPallm, DPallv, subpallium, and thalamus in the E15.5 and E18.5 sections but failed to delineate the DPallm and DPallv in E13.5. StabMap identified the DPallm and DPallv in the E15.5 section, while its clusters in the E13.5 and E18.5 sections could not be matched to known anatomical references.

Our visual comparison of SpaMosaic's clusters with the Allen Brain Atlas' reference structures showed high concordance but we followed up with further validation using known marker genes (**Supp. Fig. S11d**). For instance, the cartilage marker *Sox9*[9] was highly expressed in cluster 13 (RNA, sections 1 and 2), while the midbrain markers *En2*[1]

and *Otx2*[10] were enriched in cluster 2 (RNA, sections 1 and 2). Among the clusters, 9 and 12 in E13.5 were particularly notable. Comparison with the Allen Brain Atlas revealed that these clusters corresponded well to the DPallm and DPallv regions (**Supp. Fig. S11c**), which were neither captured by CellCharter and BANKSY, nor annotated in the original publication (**Supp. Fig. S13b**). Among the other mosaic integration methods, CLUE, Cobolt, and MIDAS could also demarcate these regions. To confirm their identities, we visualized the expression of markers *Emx1*, *Id4*, *Nr2f1*, and *Pax6* that regulate mouse neocortex development[11-13] and are highly expressed in the DPallv or pallium region. We find them highly expressed in cluster 12, both in RNA count and gene activity scores derived from the ATAC modality (**Supp. Fig. S11d,e**). Comparison with ISH staining samples from the Allen Brain Atlas[14] also showed matching locations, confirming cluster 12's accuracy in capturing the DPallv (**Supp. Fig. S11e**).

Finally, we quantitatively evaluated the mosaic integration methods' performance in three aspects: continuity of spatial domains identified (PAS and CHAOS), batch mixing (iLISI), and modality alignment (Fraction of Samples Closer than True Match, FOSCTTM[15] and Matching Score, MS[16]). Higher scores indicate better performance for iLISI and MS, while the reverse is true for FOSCTTM, PAS, and CHAOS. To calculate the FOSCTTM and MS metrics, we created a new dataset using the E15.5 and E18.5 sections and split the RNA and ATAC modalities of the E18.5 section to simulate two new sections. The ground-truth pairing between the RNA and ATAC modalities of the E18.5 section was used to evaluate modality alignment. In terms of continuity of identified spatial domains (PAS and CHAOS), SpaMosaic scored the lowest for all sections in PAS, and for E15.5 and E18.5 in CHAOS (**Supp. Fig. S11f**). For batch mixing (iLISI), MIDAS was first, while Cobolt was second and scMoMaT and StabMap closely behind at third place. SpaMosaic was ranked fourth but close to scMoMaT and StabMap in performance (**Supp. Fig. S11g**). These results prompted us to investigate the batch mixing by examining the UMAPs colored by section and clusters (**Supp. Fig. S12c**). MIDAS indeed mixed the three sections very evenly, but the clusters also showed substantial overlapping. The remaining methods had better cluster separation but lower batch mixing. For modality alignment, SpaMosaic was the best at both FOSCTTM and MS metrics (**Supp. Fig. S11h**), highlighting its capabilities in cross-modality alignment.

Integration of postnatal mouse brain sections with section-size imbalance (Supp. Fig. S14-S16)

In this scenario, we employed a postnatal mouse brain dataset (spatial transcriptome and histone modification co-sequencing[17] + spatial Cut&Tag-seq[18]; spRH+spC&T), which contains data of RNA and a different epigenomic modality, the H3K4me3 histone modification. This compiled dataset consists of three sections acquired from different studies (**Supp. Fig. S14a**). The first section consists of the transcriptome and histone modification (H3K4me3) profiles of a P22 mouse brain section in a capture area with 100×100 pixels, each 20 μm in diameter[17]. For the second section, the original experiment acquired the spatial transcriptome and histone modification (H3K27ac) of a P22 mouse brain section with a capture array of 100×100 pixels, each 20 μm in diameter[17], of which we only retained the spatial transcriptome. For the third section, we selected the spatial histone modification (H3K4me3) profile of a P21 mouse brain section, acquired on a 50×50 array with 20 μm pixels[18]. These three datasets were acquired from mouse brain sections at different ages and with different capture area sizes, which posed additional challenges for data integration. From the UMAP plots, we can easily observe batch effect present, especially between section 1 and section 3 in their histone modification profiles (**Supp. Fig. S15a**). Again, we benchmarked SpaMosaic alongside the other state-of-the-art integration and spatial domain (niche) identification methods. Although Cobolt's publication only applied the method on RNA and ATAC modalities, there is no methodological restrictions on its generalizability towards other omics modalities. Therefore, we tested it alongside the other integration algorithms.

Using the Allen Brain Atlas[8] as reference, we found that SpaMosaic produced clustering results that aligned more closely with known anatomical structures compared to the other mosaic integration methods (**Supp. Fig. S14b and Supp. Fig. S16a**). In sections 1 and 2, we identified the cortex (clusters 2, 3, 4, 5, and 9), lateral ventricle (VL) (cluster 12), nucleus accumbens (ACB, cluster 1), genu of corpus callosum (CCG) and anterior commissure of olfactory limb (ACO) in cluster 7, caudoputamen (CP, cluster 13), and cluster 11 capturing the lateral septal complex (LS). In section 3, we could identify the VL (cluster 12), CCG (cluster 7), cortex (clusters 2, 3, 4, and 5), and CP (clusters 8 and 13). In contrast, the competing methods produced noisier clusters and were comparatively less successful at capturing the

brain structures. All methods, except scMoMaT and StabMap, captured the cortex layers in sections 1 and 2; however, none of them were able to explicitly separate the ACB and CP. Notably, Cobolt identified cortical layers as clusters 4, 5, and 11, demonstrating finer cortical stratification compared to SpaMosaic. For section 3, we could not unambiguously match the competing methods' output clusters to the reference. We also tested BANKSY and CellCharter, which achieved more coherent spatial clustering compared to the other mosaic integration methods (**Supp. Fig. S16b**). However, BANKSY identified only a single cortex layer and did not distinguish between the ACB and CP. CellCharter's output revealed spatial patterns similar to SpaMosaic in the RNA modality of sections 1 and 2, but it split the CP into two clusters and failed to differentiate the ACB and CP in section 1's H3K4me3 modality. On section 3, CellCharter produced more coherent spatial clustering than SpaMosaic. However, the cortex was assigned to different clusters in sections 1 and 3, preventing proper cross-section correspondence. The UMAP plots confirmed this misalignment (**Supp. Fig. S15b**).

In addition to referencing the Allen Brain Atlas, RNA expression of marker genes for the six brain regions in sections 1 and 2 was visualized (**Supp. Fig. S14c**). The expected markers were highly expressed in the corresponding SpaMosaic's clusters, affirming their identities. For instance, the striatum markers *Pde10a*[17], *Adcy5*[19], and *Foxp1*[20] were highly expressed in cluster 13 (CP), which is part of the striatum. We also observed the markers of CCG, *Sox10*[17], *Mbp*[17], and *Tspan2*[17] to be highly expressed in cluster 7 that contains the CCG structure. For section 3 that lacked RNA modality, we visualized the spatial plots and violin plots of gene activity scores derived from the histone marks of relevant marker genes (**Supp. Fig. S14d**). The gene activity scores showed high levels of noise, but we could observe region-specific expression patterns of key markers, such as *Satb2*[17] in the cortex and *Adcy5* in the CP/striatum. Violin plots also illustrated that gene activity patterns aligned with the identified spatial domains. To further verify the identified brain structures, we used Seurat[21] to integrate section 3's gene activity scores (GAS) with a published scRNA-seq mouse nervous system dataset[22] and transferred the annotated cell types to section 3. We then visualized the spatial distribution of four cell types, each with a unique spatial distribution (**Supp. Fig. S14e**). We found the mature oligodendrocytes (MOL1) enriched in the identified CCG region and the medium spiny neurons (MSN2) located only within the CP/striatum, which is consistent with existing studies[18,22]. We also observed that the TEGLU8 excitatory neurons were enriched in cluster 5 (cortex) that matched the upper cortical layer, and the TEGLU3 excitatory neurons were enriched in cluster 4 (cortex) that encompassed the deep cortical layer[22].

We next employed the metrics PAS and CHAOS to assess the spatial continuity of the identified domains, and iLISI to evaluate batch integration. Quantitative evaluation of modality alignment was not possible due to the lack of another section with paired RNA and H3K4me3 histone modification measurements. For spatial domain identification, SpaMosaic clearly outperformed the competing methods for all three sections (**Supp. Fig. S14f**). In terms of batch mixing, Cobolt and MultiVI ranked as the top two methods, followed by SpaMosaic, which slightly outperformed scMoMaT (**Supp. Fig. S14g**). We further examined the UMAP visualizations colored by the clustering labels and section labels (**Supp. Fig. S15c**). For the cluster labelled plots, the consistent grouping and clear separation of the SpaMosaic's clusters corroborated with its PAS and CHAOS scores, especially for sections 1 and 2. In contrast, Cobolt's clusters showed observable mixing while scMoMaT's output showed intra-cluster separations. In the visualization with section labels, the even mixing achieved by Cobolt and MultiVI matched their iLISI scores, while SpaMosaic, StabMap, and CLUE achieved better mixing than the remaining methods albeit with small local clusters of batch specific spots still visible.

Mosaic integration of human lymph node dataset (RNA+protein) (Supp. Fig. S17-S19)

Here we benchmarked SpaMosaic and competing methods on an in-house generated human lymph node dataset. Lymph nodes are small organs found throughout the human body as part of the lymphatic system. Within the lymph node, the cortex and medulla harbors concentrations of different immune cells like T and B cells that become activated during an infection. In particular, the B cells congregations in the cortex are called follicles. The organ is surrounded by a protective capsule of fibrous tissue, followed by the subcapsular sinus, a space through which lymphatic fluids flow through. The dataset has three tissue sections where the first and second sections are sequential. The 10x Genomics

Visium RNA and protein co-profiling technology was used to acquire the transcriptome and protein profiles for each section. To create a mosaic dataset, we removed the protein and RNA profiles of the second and third section, respectively (**Supp. Fig. S17a**). Prior to integration, we examined the UMAP plots for batch effects. In the RNA modality, we could observe the presence of mild batch effects, but much stronger batch effects were visible in the protein modality between sections 1 and 3 (**Supp. Fig. S17b**). Such levels of batch effects may be attributed to sections 1 and 2 being sequential while sections 1 and 3 were non-sequential. For performance assessment, we had the accompanying H&E images (**Supp. Fig. S17c**) annotated by an expert to provide the ground truth. Unlike the other datasets used for benchmarking in the main text, this lymph node dataset presents a unique challenge due to its intricate cell type-centric structures and a different combination of data modalities, namely RNA and protein.

We first visually inspected the results obtained from the different methods. For SpaMosaic, it captured many of the expected lymph node structures across the three sections (**Supp. Fig. S18a**). This includes the follicles (cluster 10), cortex (cluster 8), adipose tissue (cluster 6), and capsule (cluster 1), which we matched to the manual annotations. For the remaining clusters, they primarily matched the annotated medullar regions. Cobolt and StabMap produced similar outputs as SpaMosaic, capturing the follicles (clusters 4 and 9), cortex (clusters 7 and 5), and adipose tissue (clusters 4 and 8). CLUE also detected similar structures, except for the follicles that were incorrectly grouped with other structures (cluster 3). scMoMaT demarcated the cortex (cluster 7) across three sections but was less effective at capturing other structures. MIDAS captured the cortex of each individual section but failed to link the clusters across all three sections. In addition to the mosaic integration methods, we also tested the spatial clustering algorithms. Although BANKSY captured more coherent spatial domains compared to the competing single-cell mosaic integration methods, we could only annotate two of its clusters (clusters 4 and 5), which match the ground truth's adipose tissue (**Supp. Fig. S18b**). In contrast, CellCharter not only produced coherent spatial domains but also identified a broader range of structures. These included the cortex (cluster 6) and adipose tissue (cluster 9 in the protein modality and cluster 3 in the RNA modality) (**Supp. Fig. S18b**). However, it demarcated the follicles (cluster 1) in the RNA modality of section 1 but failed to capture them in the RNA modality of section 2 or in the protein modality.

In addition to comparing against the expert annotation, we further validated SpaMosaic's clusters by visualizing the RNA and protein expression of known tissue specific markers (**Supp. Fig. S18c,d)**. *MARCO* and *FLT4*, which are markers of lymphatic endothelial cells that are enriched in the human lymph medullary region[23] had high RNA expression in clusters 4 and 5. CD163 and CD27, markers of macrophages and plasma cells that are enriched in medullary region[24], had high protein expression in clusters, 3, 7, and 9. *PPARG*, *PLIN1*, and *ADIPOQ*, which are highly expressed in adipocytes[25-27], had high RNA expression in cluster 6. The markers of follicular cells[28-30], *CR2*, *MS4A1*, *CXCR4*, and *CXCL13*, had high RNA and protein expression in cluster 10. *PECAM1* and *ACTA2*, with reported expression in the capsule or adventitia[31], had high RNA and protein expression in cluster 1. We also found markers of T-zone reticular cells[31], *CXCL9*, and *CCL19*, to have high RNA expression in cluster 8. Finally, T cell markers CD3E, CD4, and CCR7 had high protein expression in clusters 2 and 8.

We next quantitatively evaluated SpaMosaic's performance in three aspects: spatial domain identification and continuity (ARI, PAS, and CHAOS), batch mixing (iLISI), and modality alignment (FOSCTTM and MS). In terms of spatial domain identification, CLUE achieved the highest ARI, while SpaMosaic was the second highest (**Supp. Fig. S19a**). SpaMosaic ranked fourth in overall PAS score, while its overall CHAOS score ranked second (**Supp. Fig. S19b**). Unlike its performance on the other presented examples, SpaMosaic did not achieve a clear advantage in the spatial continuity metrics for this lymph node dataset. This may be primarily due to its noisy identification of the large medullary region located at the center of the sections. For batch mixing, SpaMosaic had the top iLISI score, followed by Cobolt (**Supp. Fig. S19c**). We also visually inspected the UMAPs colored by section, cluster, and manual annotation (**Supp. Fig. S19d**). SpaMosaic mixed the three sections uniformly, matching its iLISI score, while maintaining separation between the manually annotated clusters. In contrast, MIDAS and scMoMaT failed to mix the three sections with visible separation between the sections. Finally, we assessed modality alignment with FOSCTTM and MS. To calculate the FOSCTTM and MS metrics, we created a new mosaic dataset using the first and second sections, where we split the RNA and protein profiles of the second section to obtain two new sections. For both metrics, SpaMosaic was the top performer (**Supp. Fig. S19e**).

Mosaic integration of human tonsil dataset (RNA+protein) (Supp. Fig. S20-S22)

In this example, we benchmarked SpaMosaic and the competing methods on an in-house generated human tonsil dataset. Like lymph nodes, tonsils are lymphoid organs which play a key role in human immunity. They are irregularly structured with B cell follicles embedded in T cell enriched zones that are further surrounded by epithelial and connective tissues. In these follicles, transient germinal centers form to facilitate B cell activation, proliferation, and somatic hypermutation during an immune response. The dataset consists of three sections generated using the 10x Genomics Visium RNA and protein co-profiling technology. The first and second sections were two sequential sections, and transcriptome and protein (ADT) profiles were acquired for all three sections. To create a mosaic dataset, we removed the protein and RNA profiles of the second and third section, respectively (**Supp. Fig. S20a**). We first examined the UMAP plots for batch effects. We did not find obvious batch effects in the RNA modality, likely due to sections 1 and 2 being neighboring sections. However, strong batch effects were present in the protein modality for sections 1 and 3, which were non-sequential (**Supp. Fig. S20b**). We also employed expert annotation of the accompanying H&E images to assist in annotating the subsequent analyses (**Supp. Fig. S20a,c**).

Among the mosaic integration methods, Cobolt, StabMap, and SpaMosaic successfully identified the germinal centers (cluster 4), which were surrounded by follicle mantle zones (cluster 3 for Cobolt, cluster 1 for SpaMosaic and cluster 2 for StabMap in sections 1 and 2; **Supp. Fig. S21a**). This demonstrated that, despite applying spatial smoothing, SpaMosaic was still capable of capturing relatively small structures in the tissue sections. In addition, SpaMosaic identified other key regions like the T-cell enriched zone (cluster 3, corresponding to the tonsillar parenchyma) and connective and epithelial tissues (cluster 2). Among the competing methods, Cobolt produced results most similar to those of SpaMosaic. For StabMap, its performance on section 3 was poor with the captured germinal centers merging with the surrounding regions. CLUE, MIDAS, and scMoMaT could capture only certain structures while missing out on the rest. For the spatial clustering methods MIDAS, BANKSY, and CellCharter, they also failed to detect the germinal centers, either by merging them with the surrounding tissues or by omitting them entirely (**Supp. Fig. S21b**).

To annotate SpaMosaic's clusters, we plotted the relevant cell type marker expression in the RNA and protein modalities (**Supp. Fig. S21c,d**). With the B cell markers *CD19*[29], we identified the B cell enriched regions, namely clusters 1 (follicles) and 4 (germinal centers). We also found high RNA expression of *IGHD* (naïve B cells)[29] and BCL2 protein expression (naive and memory B cells)[32] in cluster 1. For cluster 4, we found high RNA expression of *BCL2A1* and *LMO2* that mark germinal center B cells[28], expression of PDCD1 protein that regulates B cell survival in germinal centers[28], as well as both RNA and protein expression of *CR2* which is a marker for mature B cells and follicular dendritic cells[29]. To identify the T cell enriched zones, we plotted the canonical T cell markers of *CD3E*, *CD4*, and *CD8A*, and found their RNA and protein to be highly expressed in cluster 3. To annotate the epithelial tissues, we examined the expression of the canonical epithelial marker EPCAM, which was highly expressed in cluster 2. We also note in cluster 2 the RNA expression of *CD9*[29], which identifies B cells and plasma cell subsets within tonsillar tissues like the crypts.

We next quantitatively evaluated SpaMosaic's performance in three aspects: spatial domain identification and continuity (ARI, PAS, and CHAOS), batch mixing (iLISI), and modality alignment (FOSCTTM and MS). In terms of ARI, SpaMosaic was the highest and followed by CLUE, StabMap, and StabMap (**Supp. Fig. S22a**). For the spatial continuity metrics, CLUE, MIDAS, and scMoMaT scored the best in PAS, while CLUE was the best in terms of CHAOS (**Supp. Fig. S22b**). We believe their top performance in both metrics was due to their merging of tissue structures in their clusters. For batch mixing assessment with iLISI, SpaMosaic was the leading method (**Supp. Fig. S22c**). We also cross checked by examining the UMAP plots of the method outputs colored by the section, cluster, and manual annotation labels (**Supp. Fig. S22d**). SpaMosaic, Cobolt, and CLUE showed high levels of section mixing, which was also reflected in their high iLISI scores. To compute the FOSCTTM and MS metrics, we created a new mosaic dataset using the first and second sections, where we split the RNA and protein profiles of the second section to obtain two new sections. For both metrics, SpaMosaic was again the best performing method (**Supp. Fig. S22e**).

Supplementary Note 2: Benchmarking imputation performance

Imputation experiment settings

After data integration, imputation of missing data modalities is a task that some mosaic data integration methods

including SpaMosaic can accomplish. Here we assessed the performance of SpaMosaic, MIDAS, MultiVI, and BABEL at imputing ATAC and protein modalities based on the RNA profiles. Four datasets were used for assessment, the embryonic mouse brain (Misar) dataset (three sections at the E13.5, E15.5, and E18.5 stages, acquired with Misar-seq, RNA and ATAC modalities), postnatal mouse brain (three sections, RNA and ATAC modalities), human tonsil (three sections, RNA and protein modality) and human lymph node (three sections, RNA and protein modality) datasets. For each dataset, we performed three-fold cross validation. Specifically, we removed the ATAC or protein expression profiles from one section and used the other two sections as training datasets. We then evaluated the imputation performance on the section that had any modality removed. Each section was employed once as the test set.

Benchmarking metrics

To quantitatively assess the ATAC and protein imputation results, we used the Pearson correlation coefficients (PCC) and Correlation matrix distance (CMD)[33,34]. For ATAC data imputation, we additionally used the area under the receiver operating characteristic curve (AUROC).

PCC. For two vectors x, y, PCC is defined as:

$$PCC(x,y) = \frac{\sum_1^m (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_1^m (x_i - \bar{x})^2}\sqrt{\sum_1^m (y_i - \bar{y})^2}}$$

For each modality, we calculated both spot-spot PCC and feature-feature PCC. For spot-spot PCC, m is the number of feature dimensions, and x, y denotes the feature profiles of two spots where $\bar{x} = \sum_1^m x_i$ and $\bar{y} = \sum_1^m y_i$. For feature-feature PCC, m is the number of spots in the test set, and x, y are features within all spots. We then summarized the results by computing the average value for the spot-spot PCC and feature-figure PCC across all spots and features, respectively. The PCC has a range of [-1, 1], where 1 indicates perfect positive correlation between two variables while -1 indicates perfect negative correlation and 0 indicates no correlation between the two vectors.

CMD. CMD measures the difference between two correlation matrices X and Y, which is defined as[33,34]:

$$CMD(X,Y) = 1 - \frac{trace(XY)}{\|X\|_F \|Y\|_F}$$

where $\|\cdot\|_F$ denotes the Frobenius norm of a matrix, $trace(XY)$ is the sum of entries on the main diagonal of matrix XY. Similar to PCC, we calculated both spot-spot CMD and feature-feature CMD. For spot-spot CMD, X and Y denote the spot-spot PCC matrices calculated using the imputed profiles and ground truth, respectively. For feature-feature CMD, X and Y denote the feature-feature PCC matrices calculated using the imputed profiles and ground truth, respectively. CMD has a range of [0, 1], where 0 indicates perfect alignment of two correlation matrices.

AUROC. AUROC assesses performance in binary prediction tasks. Before computing the AUROC, we binarized the target peak count matrix using a threshold of 1. AUROC ranges from 0 to 1, where a value of 1 indicates perfect prediction.

kNN smoothing for ATAC. As the peak counts of ATAC data are noisy and sparse, its direct usage in performance assessment may result in underestimation of the imputation's efficacy. Therefore, we adopted the approach of Tal et al.[5] in *k*NN smoothing of the raw peak count matrix. Specifically, we took the top 50 principal components of the expression data and computed the *k*NN graph ($k = 50$). Thereafter, we computed the average of the neighbors' expression values for each spot. We then computed the average values of the spot-spot/peak-peak PCC and AUROC metrics.

Benchmarking methods for imputation

MIDAS. We ran MIDAS using the procedure described for mosaic integration, except for setting the command parameter '--act' to '*translate*'.

TotalVI[35]. We followed the procedure as described in TotalVI's tutorials (scVI version 0.19.0) (https://docs.scvi-tools.org/en/stable/tutorials/notebooks/multimodal/totalVI.html). The raw count matrices were used as input to TotalVI.

The configuration parameters were set as: $latent_distribution = 'normal'$, $n_layers_decoder = 2$, while the other training parameters were left at their defaults.

MultiVI[5]. We ran MultiVI by following the pipeline described in its tutorial (scVI version 0.19.0) (https://docs.scvi-tools.org/en/stable/tutorials/notebooks/multimodal/MultiVI_tutorial.html). Raw count matrices were used as input and the training epoch was set to 100 while the other parameters were left at their default values.

BABEL[36]. We ran BABEL by following the tutorial for the DANCE toolkit (version 1.0.1)[37] (https://github.com/OmicsML/dance/tree/main/examples/multi_modality/predict_modality/babel.py). Raw count matrices were used as input and all parameters were set to their defaults.

Benchmarking results

The quantitative benchmarking results on the four datasets are presented in **Supp. Fig. S23**. For the postnatal mouse brain dataset ATAC prediction task, MIDAS achieved the highest AUROC scores, followed by SpaMosaic (**Supp. Fig. S23a**). For spot-spot evaluation, SpaMosaic achieved the highest PCC and the lowest CMD (lower CMD scores indicate better performance) while for peak-peak evaluation, MIDAS achieved the highest PCC and the lowest CMD, followed by SpaMosaic and Babel. The embryonic mouse dataset prediction also showed similar results. These findings highlighted distinct strengths of different methods: SpaMosaic's imputations better preserved spot-spot relations while MIDAS's imputations better preserved the peak-peak relations. The reason is that SpaMosaic's imputation is based on spatial-aware embeddings that better preserve spot-spot spatial relations, while MIDAS's imputations are based on neural networks, which are not constrained by spatial information and can better adapt to feature patterns. Next, we recomputed the benchmarking metrics with the smoothed ground truth peak count matrices. With the smoothed ground truth, all methods' performance improved substantially (**Supp. Fig. S23b**). For example, on the postnatal mouse brain dataset, SpaMosaic's spot-spot PCC improved by 34.4% and AUROC improved by 52.3%, while MIDAS's spot-spot PCC improved by 34.6% and AUROC improved by 32.4%. In this scenario, both SpaMosaic and MIDAS scored the highest AUROC, and SpaMosaic was still the top performer at preserving spot-spot relations while MIDAS remained the best in preserving peak-peak relations. For the protein modality prediction task with the lymph node and tonsil, the methods' performance was broadly comparable (**Supp. Fig. S23c**). Only a few statistical differences were found, like BABEL's poorer performance at protein-protein PCC.

Supplementary Note 3: Detailed comparison of integration methods on postnatal mouse brain (spRA+spRH)

We first visually examined the clusters obtained from the mosaic integration methods (**Fig. 3b of the main text**). SpaMosaic's clusters clearly captured the different brain regions that we could match to the reference Allen Brain Atlas[8] (**Supp. Fig. S2b**). For example, we identified the diagonal band nucleus (NDB, cluster 6), CCG and ACO (cluster 2), CP (clusters 4 and 7), LS (cluster 9), ACB (cluster 11), and VL (cluster 3). Plotting previously described brain markers, we found them highly expressed in their respective annotated clusters, such as *Mbp* and *Tspan2* in the CCG, *Stard5*[38] in the ACB, and *Trpc4* and *Dgkg*[39] in the LS (**Fig. 3c**). For the cortex layers, we found *Fezf2*[17] highly expressed in the deep cortical layers (clusters 12 and 13) and *Cux2*[17] in the upper layers (cluster 8). In general, the competing methods captured noisier clusters, except for CLUE. However, CLUE's clusters captured fewer brain regions than SpaMosaic, being unable to separate the CP and ACB or even distinguish the cortex layers. For Cobolt, scMoMaT, StabMap, and UINMF, they produced noisier clusters, and none could separate the CP and ACB. While we could discern cortex layers in Cobolt's and UINMF's clusters, scMoMaT's and StabMap's clusters were even noisier with no cortex layers identifiable. We next examined the outputs of BANKSY and CellCharter. Both identified more continuous spatial domains compared to the mosaic integration methods, especially for the RNA modality (**Supp. Fig. S24**). However, BANKSY captured only a single cortex layer and failed to distinguish between regions such as ACB and CP, or LS and NDB. CellCharter (RNA) recovered spatial structures broadly consistent with the Allen Brain reference, including the cortex layers (clusters 1, 4,

13, and 14), CCG and ACO (cluster 9), CP (cluster 15), VL (cluster 6), and LS (cluster 2), but mixed NDB with the cortex layers (cluster 1). Additionally, the cortical regions were assigned to different clusters by CellCharter (RNA) in section 4 compared to the other three sections, which is consistent with the UMAP visualizations, despite the inputs having been batch corrected with scVI[40] (**Supp. Fig. S25a**). For the epigenomic modalities, CellCharter produced considerably noisier spatial domains, likely due to high intrinsic data noise and the absence of cross-section and cross-modality integration.

We next quantified spatial continuity of the identified domains with the PAS and CHAOS metrics and batch integration with iLISI. For PAS, SpaMosaic scored a joint first with CLUE on sections 1 and 3, first on section 2, and second on section 4 (**Fig. 3d**). Taken together, SpaMosaic and CLUE performed similarly but CLUE identified fewer brain regions as discussed in our earlier analysis. For CHAOS, SpaMosaic attained the lowest scores for all sections. For batch mixing assessed with iLISI, Cobolt was top with scMoMaT second and SpaMosaic a close third with UINMF (**Fig. 3e**). We also visually inspected the UMAPs (**Supp. Fig. S25b**), where we see that the batch mixing broadly reflects the metrics. For example, the batches were well mixed for Cobolt's output, while there was clear batch specific separation for CLUE.

Supplementary Note 4: Mosaic integration across platforms with vastly different resolutions and scales

We further benchmarked SpaMosaic using two mosaic integration scenarios of mouse brain acquired with different technologies, size, and resolutions (**Supp. Fig. S26a**). Scenario 1 uses an adult mouse brain dataset composed of three sections profiled by distinct technologies: two RNA only sections acquired with the 10x Genomics Visium HD[41] and 10x Genomics Visium[42], respectively, and a third section captured by Spatial Mux-seq with paired RNA and ATAC data[43] (**Supp. Fig. S27a**). For the Visium HD section with a pixel diameter of 2 μm, we performed 8×8 binning to obtain a 16 μm resolution for subsequent analysis. In Scenario 2, we employed a set of postnatal mouse brain data consisting of four sections profiled by three technologies: one section acquired with DBiT-ARP-seq (DBiT) for paired RNA and ATAC data[44], two Spatial RNA-ATAC-seq (spRA) sections with paired RNA and ATAC data[17], and one Spatial ATAC-seq (spA) section with ATAC only data[45] (**Supp. Fig. S27b**). The number of spots per section varied substantially, ranging from 2,707 to 98,917 in scenario 1 and from 2,497 to 9,787 in scenario 2 (**Supp. Fig. S26b**). UMAP visualizations of both settings revealed strong batch effects across technologies and sections (**Supp. Fig. S28a,b**).

In scenario 1, by comparing the clustering results with the Allen Brain Atlas reference[8] (**Supp. Fig. S29a**), we observed that SpaMosaic accurately recapitulated key brain structures across the sections with different resolutions (16μm, 50μm, 55μm), including the cortex, hippocampus, striatum, hypothalamus, VL, and thalamus (detailed cluster assignments shown in **Supp. Fig. S27c**). SpaMosaic also resolved multiple cortical layers across all three sections. To further validate these anatomical assignments, we analyzed the expression patterns of region-specific marker genes across all sections (**Supp. Fig. S30a**). Given the large differences in spot counts between sections, we analyzed the Visium and Spatial Mux-seq sections separately (**Supp. Fig. S30b,c**). Overall, the marker gene expression patterns consistently corroborated the anatomical region assignments. For the competing methods, their clustering results on the Visium HD section broadly matched the reference anatomical structures, albeit with fewer regions demarcated than SpaMosaic and with higher noise. However, these methods showed reduced performance on the lower-resolution Visium and Spatial Mux-seq sections. Specifically, for the Visium section, CLUE failed to identify anatomical structures, while MIDAS only captured the hippocampus (cluster 15). StabMap, scMoMaT, and UINMF showed improved brain structure recognition over CLUE and MIDAS but exhibited high noise levels and lacked cortical layer differentiation. Cobolt and MultiVI separated the cortical layers, though they resolved fewer layers than SpaMosaic. Both methods also identified a distinct dentate gyrus region (Cobolt: cluster 14; MultiVI: cluster 16), whereas SpaMosaic merged this region with cluster 2. However, Cobolt's and MultiVI's outputs exhibited higher noise levels overall, resulting in less sharply defined spatial domains. For the Spatial Mux-seq section, scMoMaT's clusters showed poor correspondence with the

reference structures due to substantial noise. CLUE and UINMF could only distinguish between the cortical and the non-cortical regions, and the cortical regions showed inconsistent cluster assignments across different sections. Cobolt, MIDAS, and MultiVI recovered anatomical structures comparable to those identified by SpaMosaic, albeit with high noise and reduced cortical layer differentiation. The spatial clustering method BANKSY also produced noisy spatial domains across all sections, particularly on the Spatial Mux-seq section (**Supp. Fig. S31**). It could not segregate the thalamus and hypothalamus and failed to resolve the cortical layers in the Visium HD section. CellCharter performed better by recapitulating coherent clusters on the Visium HD and Spatial Mux-seq (RNA) section that aligned with anatomical references. However, CellCharter failed to distinguish the hippocampus from surrounding regions in the Visium section and produced highly noisy clustering results on the Spatial Mux-seq ATAC modality.

In scenario 2, SpaMosaic's clusters corresponded well to the Allen Brain Atlas reference[8] (**Supp. Fig. S29b**), accurately capturing major brain regions including the cortex, CP, VL, and ACB across four sections (detailed cluster assignments shown in **Supp. Fig. S27d**). The marker gene expression and gene activity score patterns consistently supported the anatomical region assignments (**Supp. Fig. S32a,b**). Notably, SpaMosaic resolved more detailed cortical layers in sections 1 and 3 compared to other methods. Layer-specific marker gene expression was visualized to confirm the biological identity of these clusters (**Supp. Fig. S32c**). Specifically, the deeper layer markers[46] (*Hs3st4*, *Tox*, *Sdk2*) were predominantly expressed in layers 5 and 6 (clusters 1 and 9), while the upper cortical markers[47] (*Rorb*, *Rab3c*, *Cux2*) showed enriched expression in layers 2/3 and 4 (clusters 2 and 14), with spatial distributions aligning well with the cluster boundaries. Among the competing methods, CLUE and Cobolt captured the CCG, CP, and several cortical layers across the first three sections. Comparatively, the other methods' output showed limited correspondence with reference anatomical structures, with only a few identifiable regions such as the CCG being captured by MIDAS, MultiVI, and StabMap. On the more challenging ATAC-only section 4, all competing methods showed substantially noisy outputs, with CLUE failing to capture any meaningful spatial domains. For the spatial domain (niche) identification methods, BANKSY and CellCharter, identified more coherent domains than the other integration methods, but BANKSY was unable to resolve the cortical layers (**Supp. Fig. S33**). Furthermore, BANKSY and CellCharter produced inconsistent clustering of the cortical and CP regions across sections 1–3, suggesting suboptimal integration.

Finally, we applied PAS, CHAOS and iLISI for quantitative performance assessment. In scenario 1, SpaMosaic outperformed all other methods in terms of spatial continuity with the lowest PAS and CHAOS scores (**Supp. Fig. S27e**). With respect to batch integration, all methods achieved low iLISI scores (below 0.05) due to the substantial imbalance in spot numbers. Nevertheless, UMAP visualizations revealed that all methods except CLUE achieved partial mixing of sections, though the extreme numerical imbalance prevented uniform integration (**Supp. Fig. S28c**). In scenario 2, SpaMosaic also achieved the lowest PAS and CHAOS scores for spatial domain continuity (**Supp. Fig. S27g**). Regarding batch integration, UINMF obtained the highest iLISI score, closely followed by MIDAS and Cobolt (**Supp. Fig. S27h**). The UMAP visualizations revealed a trade-off between section integration and cluster preservation: methods like MIDAS, Cobolt, and MultiVI achieved better cross-section mixing but at the cost of cluster boundaries becoming blurred (**Supp. Fig. S28d**). In contrast, SpaMosaic showed less uniform section mixing but maintained clearer cluster separation.

Supplementary Note 5: Ablation study on graph construction strategies

Here we present a study of SpaMosaic variants to evaluate the contribution of SpaMosaic's model components towards its performance. The first variant is SpaMosaic (non-spatial), where the GNN component is replaced by a conventional multi-layer perceptron (MLP) while all other parts remain the same as the original SpaMosaic. The second variant is SpaMosaic (spatial-only), which retains the GNN but excludes the expression adjacency relationship when constructing the cell-cell graph, allowing us to investigate how the model performs when the GNN considers only spatial information. We assessed SpaMosaic and the two variants on both simulated data and three real datasets, the human lymph node, embryonic mouse brain (Misar), and embryonic mouse brain (Misar+Stereo) datasets.

For the simulation dataset, SpaMosaic and SpaMosaic (spatial-only) achieved similar ARI scores, both

outperforming SpaMosaic (non-spatial) by a clear margin (**Supp. Fig. S34a**). This highlighted the importance of incorporating spatial information for improving clustering performance on this dataset. In the spatial domain continuity evaluation, SpaMosaic (spatial-only) obtained PAS scores comparable to SpaMosaic on sections 1 and 2, while scoring lower in PAS on section 3 (**Supp. Fig. S34b**). Both methods exhibited nearly identical CHAOS scores and substantially outperformed SpaMosaic (non-spatial) on both PAS and CHAOS metrics. In terms of batch integration, SpaMosaic attained the highest iLISI scores, followed by SpaMosaic (spatial-only) and SpaMosaic (non-spatial) (**Supp. Fig. S34c**). For modality alignment, SpaMosaic achieved the highest label transfer F1-scores, with SpaMosaic (spatial-only) ranking second (**Supp. Fig. S34d**). Consistent with these quantitative metrics, spatial plots of the clusters further demonstrated that the patterns identified by SpaMosaic and SpaMosaic (spatial-only) were more closely aligned with the factor labels (**Supp. Fig. S34e**). The UMAP plots also revealed that SpaMosaic produced clearer factor separation and more effective batch mixing than SpaMosaic (spatial-only), supporting its higher iLISI and F1-scores (**Supp. Fig. S34f**).

For the human lymph node dataset, all three models achieved similar ARI scores, with SpaMosaic (spatial-only) performing marginally better (**Supp. Fig. S35a**). Consistent with the previous dataset, SpaMosaic (spatial-only) attained the lowest PAS and CHAOS scores, followed by SpaMosaic (**Supp. Fig. S35b**). The improved scores of SpaMosaic (spatial-only) over the other two methods stemmed from its more coherent clustering within the medullary regions, whereas SpaMosaic and SpaMosaic (non-spatial) captured more dispersed clusters due to the noisy expression data (**Supp. Fig. S35c**). In terms of batch integration, SpaMosaic achieved the highest iLISI score, closely followed by SpaMosaic (non-spatial) (**Supp. Fig. S35d**). This suggested that incorporating cross-section proximity improved integration. Considering the results of all three SpaMosaic variants, it appeared that incorporating expression adjacency relationships helped to mitigate these batch effects. UMAP plots further confirmed that SpaMosaic and SpaMosaic (non-spatial) achieved more uniform mixing across sections, whereas SpaMosaic (spatial-only) exhibited some section-specific separation (**Supp. Fig. S35e**).

With the embryonic mouse brain (Misar) dataset, SpaMosaic (spatial-only) was again the best overall in terms of PAS and CHAOS, except for section 2's CHAOS score (**Supp. Fig. S36a**). SpaMosaic was second in all except for section 2's CHAOS score where it edged out SpaMosaic (spatial-only). Both SpaMosaic (spatial-only) and SpaMosaic outperformed SpaMosaic (non-spatial) by a clear margin in terms of PAS, again demonstrating that incorporating spatial information improves spatial clustering continuity. For batch integration, SpaMosaic (non-spatial) attained the highest iLISI score and demonstrated the most effective batch mixing in UMAP visualization, followed by SpaMosaic (**Supp. Fig. S36b,c**). However, despite this higher iLISI score, spatial plots of the clustering revealed that SpaMosaic (non-spatial) failed to consistently align key regions across the three sections. For example, cluster 11 (hindbrain) in sections 2 and 3 was absent in section 1, and cluster 10 (mantle zone of dorsal pallium, DPallm) in sections 2 and 3 did not appear in section 1 (**Supp. Fig. S36d**). In contrast, SpaMosaic produced more coherent and more consistent cluster patterns across all three sections.

For the embryonic mouse brain (Misar+Stereo) dataset, SpaMosaic achieved the lowest PAS scores on 4 out of 6 sections, with SpaMosaic (spatial-only) ranking second; both substantially outperformed SpaMosaic (non-spatial) (**Supp. Fig. S37a**). Both SpaMosaic and SpaMosaic (spatial-only) also achieved comparable CHAOS values, again outperforming SpaMosaic (non-spatial). In terms of batch integration, SpaMosaic (non-spatial) attained the highest iLISI score (**Supp. Fig. S37b**). We speculate that this is because spatial proximity encoding preserves spatial relationships by grouping nearby spots together, which reduces batch mixing. From the spatial plots, we observed that SpaMosaic (non-spatial) identified clustering patterns that appeared noisier compared to those from SpaMosaic and SpaMosaic (spatial-only) (**Supp. Fig. S37c**). For example, cluster 10 (hindbrain) was more dispersed across sections and lacked a clear spatial pattern. This was also reflected in its comparatively poorer PAS and CHAOS scores. The UMAP visualizations showed that SpaMosaic (non-spatial) indeed produced more uniform embeddings in terms of section and hence a higher iLISI score (**Supp. Fig. S37d**).

Overall, our testing with simulated and real datasets confirmed that incorporating spatial information can substantially enhance the spatial continuity of identified spatial domains and improve domain consistency across sections, making them more closely aligned with the underlying anatomical structure. However, we note that this can be accompanied by a trade-off in poorer batch integration on the local scale as measured by the iLISI metric. Meanwhile, considering cross-section expression adjacency in the cell-cell graph improves representation consistency across

sections and thus batch integration, although it may slightly compromise the spatial clustering coherence within individual sections. This may be attributed to the fact that spatial-only information enforces local continuity, whereas integrating cross-section expression adjacency introduces a global structure at the cost of local smoothness.

Supplementary Note 6: Sensitivity analysis of *k* in spatial adjacency graph construction

The parameter *k* determines the neighborhood size in the spatial *k*-nearest neighbor (*k*NN) graph, directly controlling the extent of local information smoothing and propagation across spot embeddings. Intuitively, one might expect that the optimal *k* depends on the characteristics of specific technology, such as 10x Genomics Visium or Stereo-seq, where each spot is arranged in a hexagonal lattice with six direct neighbors. To investigate how the choice of *k* influences the model outcomes, we conducted evaluation on the simulated data, human lymph node, embryonic mouse brain (Misar), and embryonic mouse brain (Misar+Stereo) seq datasets with *k* set to 4, 6, 8, 10, and 20. We note that SpaMosaic performs random sampling of neighboring spots when the number of available neighbors is greater than the target *k*. While a fixed random seed could be set to ensure reproducibility, we deliberately varied the random seed to investigate whether different sampling results would substantially impact the outcomes. Therefore, for each *k* value, we repeated the experiment three times, each time using a randomly selected new seed.

We observed similar trends for the quantitative metrics across four datasets. The ARI scores remained relatively stable when *k* ranged from 4 to 10, but showed a noticeable decline at *k*=20 for the simulated and human lymph node datasets (**Supp. Fig. S38a,b**). In the case of the simulated dataset, the spatial clustering plots suggested an intuitive explanation: increasing *k* causes each spatial factor to expand its cluster, particularly in sections 2 and 3, leading to a deviation from the ground-truth labels (**Supp. Fig. S39a**). In terms of spatial clustering continuity, larger *k* values led to lower PAS scores for all test datasets (**Supp. Fig. S38a-d**). This is expected as a larger *k* introduces more neighboring spots into the smoothing process, making it more likely that most neighbors share the same cluster label. However, the CHAOS scores did not show substantial fluctuations (**Supp. Fig. S38a-d**). This is because PAS and CHAOS evaluate different aspects of the clustering results. Specifically, PAS assesses whether the majority of spots within a local spatial neighborhood belong to different clusters, thereby reflecting a broad spatial context around each spot. In contrast, CHAOS measures the nearest distance from each spot to another spot within the same cluster. As a result, CHAOS primarily reflects the presence of isolated points in the clustering outcome. From this perspective, varying *k* has a similar effect on reducing clustering outliers across four datasets. For batch integration, a larger *k* generally led to lower iLISI scores across the four test datasets (**Supp. Fig. S38a-d**). The UMAP visualization further supported the result that increasing *k* led to reduced section mixing, and this trend was also observed in other datasets (**Supp. Fig. S39b and Supp. Fig. S40-S42**). These phenomena were likely due to the incorporation of too many neighbors in the *k*NN graph, which resulted in over-smoothing of nearby spot representations and increased inconsistencies across batches. Importantly, across the four datasets, triplicate runs with different random seeds for each *k* value produced highly consistent results, with only minor variation across all quantitative metrics (**Supp. Fig. S38**). This consistency demonstrated that the random sampling of neighbors inherent in SpaMosaic's *k*NN graph construction did not introduce substantial variability to the experimental outcomes. Moreover, *k*=10 achieved a well-balanced performance in terms of clustering accuracy, spatial continuity, and batch integration across four datasets. Hence, we adopted *k*=10 for all examples presented in our manuscript.

Supplementary Note 7: Benchmarking different graph network architectures

Details of benchmarking model architecture

In SpaMosaic's model, a spot-spot graph is created for each modality to learn information on the relationships between spots in terms of expression similarity and spatial proximity. Prior to finalizing the model, we considered four different graph neural architectures. In the first architecture, we considered homogeneous graphs and employed the graph attention network (GAT)[48,49] to automatically learn the different relationships between spots. We adopted the same approach of Dong et al.[49], which is described as follows. The encoder layer in layer k is defined as:

$$z_i^k = \sigma(\sum_{j \in S_i} att_{ij}^k (W_k z_j^{k-1}))$$

where W_k is the trainable weight matrix, σ is the non-linear activation, S_i denotes the neighbor set of spot i (including itself), att_{ij}^k is the edge weight between spot i and j which is the output of k-th graph attention layer and z^0 denotes the model inputs. We highlight here that the final layer does not employ any attention mechanism. The k-th attention layer is defined as[49]:

$$att_{ij}^k = \frac{\exp(e_{ij}^k)}{\sum_{j \in S_i} \exp(e_{ij}^k)}$$

where $e_{ij}^k = \text{Sigmoid}\left(v_s^{k^T}(W_k z_i^{k-1}) + v_r^{k^T}(W_k z_j^{k-1})\right)$ with v_s^k and v_r^k being trainable weight vectors[49]. We also followed the same approach of Dong et al. in setting up the decoder which shares the same weight matrices W_k and attentions scores att_{ij}^k as the encoder.

Alternatively, the spot-spot graphs can be heterogeneous. In this second approach, we considered two types of edges in the graph and employed the general heterogeneous graph transformer (HGT)[50] to learn the heterogeneous edges. In brief, HGT uses node- and edge-type dependent parameters to characterize the heterogeneous attention over each edge, achieving dedicated representations for different types of nodes and edges[50]. We used the implementation of HGT in the Pytorch Geometric library[51] and followed the example to build the whole network. Moreover, based on the framework of light graph convolution network (LGCN)[52,53], we proposed a third model architecture. In traditional LGCNs, information is propagated and aggregated through:

$$\tilde{X} = LGCN(A, X) = Concat(X, \hat{A}X, \hat{A}^2X, \dots, \hat{A}^lX)$$

where $\hat{A} = \tilde{D}^{-\frac{1}{2}}(A + I)\tilde{D}^{-\frac{1}{2}}$, $\tilde{D}$ is the diagonal degree matrix of $A + I$, and l denotes the number of layers. The embeddings are finally transformed by a group of nonlinear transformations a multiple layer perceptron (MLP):

$$Z = f_{mlp}(\tilde{X})$$

Our modified HG-LGCN formulation then take the following form:

$$\tilde{X} = LGCN^1(A, X) = Concat(X, f(X), f\big(f(X)\big) \dots, f^l(X))$$

where $f(X) = Concat(\hat{A}^{intra}X, \hat{A}^{inter}X)$. The key idea of this modification is to construct two propagation paths between spots, and the output of current layer is recursively applied to subsequent layers. Thus, information propagation in this model architecture is structured like a complete binary tree. The fourth and last architecture is the weighted LGCN (WLGCN) that is described in the method section.

Benchmarking results

We first searched for the best combination of hyper-parameters for each architecture on a simulated dataset (simulation parameters: $b^{rna} = 4$, $s^{rna} = 1/15$, $p^{rna} = 0$, $b^{pro} = 2.0$, $s^{pro} = 1/20$). The hyper-parameters optimized were the number of layers, the dimension of hidden layers, the output dimensions and the number of nearest neighbors used in constructing the spot-spot adjacency graph. We selected the parameter combination that achieved the highest ARI scores for the whole dataset.

Using the first replicate of simulation setting 1 as an example, we first examined the UMAP plots of the outputs from the different architectures (**Supp. Fig. S43a**). The output of all methods showed good separation of clusters except HGT. In terms of batch mixing, WLGCN and HG-LGCN achieved the best mixing visually, while local concentrations of batch specific spots were clearly visible for HGT and GAT. We hypothesized that attention-based architectures preferentially amplify signals within highly homogeneous, intra-section neighborhoods, thereby restricting information propagation across sections and leading to poorer batch mixing. We also examined the spatial plots of the outputs (**Supp. Fig. S43b**). HGT achieved the worst performance, being unable to capture factor 1 (orange) in sections 1 and 2, and factor 3 in section 3, while the other three methods were able to correctly capture the factors.

We further quantitatively evaluated the four architectures on the data generated with all five simulation settings in three aspects: spatial domain identification and continuity (ARI, PAS, and CHAOS), batch mixing (iLISI) and modality alignment (F1-score of label transfer between section 2 and 3). In terms of ARI, WLGCN, HG-LGCN, and GAT showed similar performance and were better than HGT (**Supp. Fig. S43c**). For the PAS and CHAOS metrics, the GAT, WLGCN,

and HG-LGCN architectures were overall superior to HGT. Among these top performers, GAT achieved the best scores across most datasets, though its lead over WLGCN and HG-LGCN was relatively small (**Supp. Fig. S43d**). Assessing integration with the iLISI metric, HG-LGCN attained the highest iLISI scores with WLGCN as a close second (**Supp. Fig. S43e**). GAT and HGT lagged behind, and these were consistent with their UMAP plots. Finally, the label transfer F1-scores showed similarly good results for WLGCN, HG-LGCN, and GAT, while HGT again performed worse (**Supp. Fig. S43f**). In summary, both WLGCN and HG-LGCN achieved similarly superior performance. However, unlike HG-LGCN whose feature dimensionality grows exponentially with the number of layers (2^L), WLGCN increases only linearly ($L+1$), resulting in substantially lower time and memory costs for deeper networks. This efficiency makes WLGCN more practical for large graphs without sacrificing accuracy. Therefore, we selected WLGCN for our final model.

Supplementary Note 8: Detailed definitions of benchmarking metrics for mosaic integration

The benchmarking employed in this work encompassed four aspects: spatial domain identification accuracy, continuity of identified spatial domains, batch correction, and modality alignment. Spatial domain identification accuracy was assessed using the adjusted rand index (ARI), while identified spatial domain's continuity was evaluated using PAS and CHAOS[54]. Batch correction was assessed using the graph integration local inverse Simpson's index (iLISI)[55], and modality alignment was assessed using Fraction of Samples Closer than True Match (FOSCTTM)[15], matching score (MS)[16], and label transfer F1 score[4].

CHAOS CHAOS evaluates the spatial continuity of spatial clusters. To calculate CHAOS, we first construct a 1-nearest neighbor (1-NN) graph for each cluster based on spatial coordinates. Specifically, for each spot i in cluster k, we identify the nearest spot within the same cluster k based on Euclidean distance, obtaining the distance d_{ki}. We then compute the average of these minimum distances across all spots:

$$CHAOS = \frac{\sum_{k=1}^{K}\sum_{i=1}^{n_k} d_{ki}}{N}$$

where K represents the number of clusters and n_k represents the number of spots within cluster k. A lower CHAOS score indicates better continuity of the spatial domains.

PAS PAS is also used to evaluate the spatial continuity of output from spatial domain identification methods. In brief, the PAS score is calculated as the percentage of spots whose cluster label differs from that of at least six of their ten spatial neighbors.

Graph iLISI The original iLISI[56] was initially conceived to assess the degree of local batch mixing in single-cell data, and Graph iLISI is an extension for assessing graph-based outputs. Graph iLISI uses a graph-based distance metric to determine the nearest neighbor list and avoids skews in graph-based integration outputs[4]. Its scaled output ranges from 0 to 1, where 0 indicates strong separation of batches while 1 indicates perfect batch mixing.

FOSCTTM FOSCTTM quantifies the degree of alignment between embeddings of different modalities for the same location. For each modality pair (m_1, m_2), FOSCTTM is defined as:

$$FOSCTTM = \frac{1}{2N}\left(\sum_i \frac{N_i^{m_1}}{N} + \sum_i \frac{N_i^{m_2}}{N}\right)$$

where N is the number of spots that are simultaneously measured in modality m_1 and m_2, $N_i^{m_1}$ denotes the number of spots in modality m_1 that are closer to $z_i^{m_2}$ than $z_i^{m_1}$ to $z_i^{m_2}$, and $N_i^{m_2}$ is the number of spots in modality m_2 that are closer to $z_i^{m_1}$ than $z_i^{m_2}$ to $z_i^{m_1}$. The computed value ranges from 0 to 1, where 1 indicates perfect modality matching.

MS Like FOSCTTM, MS quantifies the degree of modality alignment and is calculated using spots with multimodal measurements. For two modalities, m_1 and m_2, a cross-modality matching matrix P is first constructed by computing the Jaccard index of cross-modality nearest neighbors in the modal-aligned embedding space[2]:

$$P_{m_1,m_2}(i,j) = \frac{|\left(N^{m_2}(z_i^{m_1}) \cap N^{m_2}(z_j^{m_2})\right) \cup \left(N^{m_1}(z_j^{m_2}) \cap N^{m_1}(z_i^{m_1})\right)|}{|\left(N^{m_2}(z_i^{m_1}) \cup N^{m_2}\left(z_j^{m_2}\right)\right) \cup \left(N^{m_1}\left(z_j^{m_2}\right) \cup N^{m_1}(z_i^{m_1})\right)|}$$

where $N^{m_2}(z_i^{m_1})$ denotes the set of spot $z_i^{m_1}$'s nearest neighbors in modality m_2, $N^{m_2}(z_j^{m_2})$ denotes the set of spot $z_j^{m_2}$'s nearest neighbors in modality m_2 (including itself), and vice versa for $N^{m_1}(z_j^{m_2})$ and $N^{m_1}(z_i^{m_1})$. Following the approach of CLUE, the number of nearest neighbors is set to 1 (i.e., $P(i,j) = 1$ if spot i and j are mutual nearest neighbors). Finally, MS is computed by:

$$MS = \frac{1}{N}\sum_i \sum_j \tilde{P}_{m_1,m_2}(i,j) \cdot \delta_{i,j}$$

where $\tilde{P}$ denotes the row-normalized P, $\delta_{i,j} = 1$ if spot i and j represent the same spot and 0 otherwise, and N is the number of spots. The computed MS ranges between 0 and 1 where 1 indicates perfect alignment.

Label transfer F1 score This metric assesses the alignment of labeled cell types or spatial domains between sections and is not restricted to spots with multimodal measurements. Therefore, it can test for the transfer accuracy of cell type or spatial domain labels from one section to another. For each pair of modalities, we first built a *k*NN graph between their embeddings and the transferred labels from one to another based on the *k*NN classifier (*k*=10). Thereafter, we calculated the F1 score by comparing the transferred labels with the ground truth. For each pair of modalities, we then averaged the F1 scores from both transfer directions.

Supplementary Note 9: Impact of clustering method choice on benchmarking

The use of different clustering algorithms downstream of data integration leads to distinct sets of clusters which in turn can affect the benchmarking results. To validate the robustness of our benchmarking framework and its results, we examined each method's performance with different downstream clustering algorithms (Kmeans, Mclust, and Leiden). We conducted these analyses on four datasets: simulated data, the human lymph node dataset, the embryonic mouse brain (Misar), and embryonic mouse brain (Misar+Stereo) datasets. To evaluate performance, we used the PAS and CHAOS metrics, and additionally reported ARI scores when ground truth labels were available.

For the simulated dataset, **Supp. Fig. S44a-c** panels show an example of spatial clustering results for SpaMosaic and competing methods applied with Kmeans, Mclust, and Leiden algorithms, respectively. SpaMosaic consistently identified the five spatial domains with high accuracy and superior spatial continuity across all clustering methods. This superior performance was reflected in its highest ARI and lowest PAS scores across all five simulated datasets, regardless of the clustering method used (**Supp. Fig. S44d-f**). Among the competing methods, Mclust did not work well with CLUE as only one spatial factor was captured in sections 1 and 3. On the other hand, CLUE paired with either Leiden and Kmeans was able to capture the factors and was second to SpaMosaic in terms of ARI. For the PAS metric, either CLUE or MIDAS were second to SpaMosaic. For CHAOS, SpaMosaic was consistently first across three clustering methods on datasets 4 and 5, and ranked in the top two with Kmeans and Mclust on datasets 1, 2, and 3.

For the human lymph node dataset, **Supp. Fig. S45a-c** panels illustrate the spatial clustering results for SpaMosaic and other methods using Kmeans, Mclust, and Leiden, respectively. CLUE, Cobolt, and scMoMaT all showed notable performance degradation with Mclust, as observable from the spatial plots and ARI scores, while the other methods were relatively robust to the choice of clustering method. When Kmeans was used, StabMap achieved the highest ARI and relatively low PAS and CHAOS values (**Supp. Fig. S45d**). With Mclust, SpaMosaic achieved the highest ARI score. When Leiden was used, MIDAS yielded the best performance in terms of both ARI and spatial continuity metrics (**Supp. Fig. S45d**).

For the embryonic mouse brain (Misar) dataset, **Supp. Fig. S46a-c** panels show that SpaMosaic demonstrated robust performance across all clustering strategies in capturing fine brain structures such as mantle zone of the dorsal pallium (DPallm) and ventricular zone of dorsal pallium (DPallv), and consistently producing coherent clustering results. The performance consistency is also observed in the metrics where SpaMosaic achieved the lowest PAS and CHAOS scores except for the CHAOS metric with Mclust where CLUE and MIDAS edge it out. Among the three clustering options, SpaMosaic with Mclust scored lowest for PAS while SpaMosaic with Leiden scored lowest for CHAOS (**Supp.**

Fig. S46d). For the other methods, scMoMaT performed best with Leiden, while CLUE, Cobolt, MIDAS, StabMap, and MultiVI performed better when Mclust was used. Kmeans generally showed the worst overall performance for most methods with this dataset.

Finally, the embryonic mouse brain (Misar+Stereo) results are shown in **Supp. Fig. S47a,b and Supp. Fig. S48a**. SpaMosaic generated consistently coherent spatial domains across all clustering methods. SpaMosaic also scored well in the metrics with consistently the lowest or second lowest PAS and CHAOS scores across the three clustering algorithms (**Supp. Fig. S48b**). For the other methods, the choice of clustering algorithm had mixed impact. Taking MIDAS with Mclust as an example, it grouped the three Misar-seq slices into three large clusters, leading to suboptimal results. There was also no clear optimal clustering algorithm as measured with the metrics. Leiden was optimal for CLUE, scMoMaT, and MultiVI, while Cobolt was less sensitive to the choice of clustering algorithm.

In summary, this analysis demonstrated that while the choice of clustering algorithm can affect the performance of most methods, SpaMosaic remained robust and accurate regardless of the clustering method tested. As Mclust performed slightly better for SpaMosaic across datasets compared to the other clustering methods, we retained its use with SpaMosaic. For the competing methods, the impact of clustering algorithm choice was more varied. There was no consistently superior method and exhaustive testing of optimal clustering algorithms for each method and each dataset is out of the scope of this work.

Supplementary Note 10: Effect of Harmony preprocessing on benchmarking results

Here we study the impact of preprocessing steps on mosaic integration methods, specifically batch integration of the input data prior to learning a joint embedding. For SpaMosaic, Harmony is one of the preprocessing steps employed. A popular single-cell data integration tool, Harmony takes in low dimensional embeddings like PCA as input and outputs a batch corrected embedding of the same dimensions. Among the competing methods, only CLUE accepts such low dimension embeddings as input, while the other methods require raw counts or normalized expression matrices as input. Therefore, we could only additionally consider the combination of CLUE with Harmony (CLUE+prep). We thus conducted a separate comparison of SpaMosaic, default CLUE and CLUE with Harmony (CLUE+prep), to assess the impact of Harmony as a preprocessing step. We applied the methods on three datasets, simulated data, the human lymph node, and embryonic mouse brain (Misar) datasets. We then quantitatively assessed the outputs with metrics that covered three aspects: spatial domain identification and continuity (using ARI, PAS, and CHAOS), batch correction (iLISI), and modality alignment (label transfer F1-score).

For the simulated data, SpaMosaic outperformed all other methods by a clear margin across all metrics and in all five simulation settings (**Supp. Fig. S49a-d**). Compared to CLUE, CLUE+prep achieved higher ARI scores on 4 out of 5 datasets, but its PAS and CHAOS scores were poorer on 3 out of 5, indicating reduced spatial continuity. Despite using batch-corrected low-dimensional inputs, CLUE+prep did not show noticeably improved iLISI scores over CLUE, indicating that simply applying batch correction to each modality is not sufficient for better data integration. In addition, CLUE+prep achieved lower label transfer F1-scores than CLUE across all datasets. Spatial clustering and UMAP plots also highlighted SpaMosaic's superior performance in both clustering quality and data integration, with CLUE+prep and CLUE showing little difference (**Supp. Fig. S49e,f**).

For the human lymph node dataset, CLUE+prep achieved the highest ARI score, while SpaMosaic and CLUE obtained comparable scores (**Supp. Fig. S50a**). However, only SpaMosaic and CLUE+prep successfully delineated the follicle structures that are small regions embedded within the cortex (**Supp. Fig. S50b**). In terms of spatial continuity, SpaMosaic attained the lowest CHAOS scores on sections 1 and 2, with CLUE and CLUE+prep obtaining similar scores (**Supp. Fig. S50c**). However, the PAS metric revealed a different trend: CLUE achieved the lowest PAS scores across all three sections, whereas SpaMosaic obtained the highest PAS scores on sections 1 and 3. This discrepancy may be attributed to the spatial clustering within the medullary regions, where SpaMosaic's clusters were more dispersed (**Supp. Fig. S50b**). In terms of batch integration, SpaMosaic achieved the highest iLISI score with CLUE+prep being second (**Supp. Fig. S50d**). UMAP plots also showed that SpaMosaic achieved good batch mixing of the three sections, while CLUE+prep achieved better mixing than CLUE (**Supp. Fig. S50e**).

For the embryonic mouse brain (Misar) dataset, SpaMosaic attained the lowest PAS scores across all sections and

the lowest CHAOS scores on sections 2 and 3, suggesting improved spatial continuity over the other two methods (**Supp. Fig. S51a**). Meanwhile, CLUE+prep yielded higher PAS and CHAOS scores compared to CLUE. In terms of batch integration, CLUE+prep attained the highest iLISI score, with SpaMosaic achieving a close second (**Supp. Fig. S51b**). Spatial plots of the clustering results further demonstrated that spatial patterns in SpaMosaic's clusters were more coherent and more consistent across three sections (**Supp. Fig. S51c**). For example, SpaMosaic consistently assigned the hindbrain and diencephalon regions to cluster 6 across all three sections, with this cluster exhibiting clear spatial domain boundaries. In contrast, CLUE assigned the hindbrain and diencephalon regions to different clusters (1, 2, and 7, respectively) that appeared dispersed, while CLUE+prep showed similar inconsistency (clusters 1, 1, and 2, respectively) with equally dispersed spatial patterns. Examination of the UMAP plots confirmed that the sections were better integrated in the outputs of SpaMosaic and CLUE+prep than that of CLUE (**Supp. Fig. S51d**).

In summary, applying batch correction to each modality before integration presents a trade-off: it can improve performance in terms of clustering accuracy (ARI) and batch integration (iLISI), but often at the expense of spatial continuity as measured with metrics (PAS and CHAOS) and without clear benefits for modality alignment. In our comparative analysis of CLUE, CLUE+prep and SpaMosaic, SpaMosaic consistently outperformed the others on simulated data. On experimentally acquired datasets, SpaMosaic achieved superior spatial domain coherence and cross-section clustering consistency, while maintaining batch mixing performance comparable to CLUE. Although CLUE+prep showed the highest iLISI scores in embryonic mouse brain (Misar) dataset, this came with noticeably poorer spatial coherence and cross-section clustering consistency.

Supplementary Note 11: Evaluation of time and memory consumption

Here we present a study that evaluates the computational performance of SpaMosaic and competing methods. All evaluation tests were performed on a server equipped with a 36-core Intel Core i9-10980XE CPU, 256 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU with 24GB of GPU memory. We recorded the CPU time usage, GPU time usage, wall time, peak Random Access Memory (RAM) usage, and peak GPU memory consumption of the methods while varying the number of CPU cores/threads. To assess performance in diverse mosaic integration scenarios, we used three datasets: embryonic mouse brain (Misar), embryonic mouse brain (Misar+Stereo), and mouse embryo datasets. The embryonic mouse brain dataset consists of one multimodal section and two single-modality sections with different modalities. The embryonic mouse brain dataset contains six slices, all with RNA profiles, but only three of them also include an ATAC modality. The mouse embryo presents a more complex setting where no section contains all modalities, and no modality is shared across all sections. For each scenario, we ran each method under different settings of CPU core counts and maximum number of threads allowed, namely, 1, 2, 4, 8, and 16. We note that the server was shared with other users and therefore the performance metrics may have been subject to some variability.

We first considered CPU usage by the different methods. **Supp. Fig. S52a** illustrates the wall time usage of each method as the number of CPU cores/threads increased. SpaMosaic consistently achieved the lowest wall time, followed by StabMap and scMoMaT. In contrast, MultiVI and CLUE exhibited the highest wall times across all datasets. As the number of CPU cores/threads increased, the wall time for SpaMosaic, scMoMaT, and StabMap generally decreased, indicating effective parallelization and scalability. In comparison, the wall time of the remaining methods either remained relatively unchanged or fluctuated with an increasing number of cores/threads available, suggesting limited multi-threading support**. Supp. Fig. S52b** presents the CPU time comparison across all methods. SpaMosaic and StabMap achieved the lowest CPU times, while MultiVI and CLUE remained among the most computationally expensive in terms of CPU usage. For GPU time usage, the trends largely mirrored those observed for wall time (**Supp. Fig. S52c**). Notably, SpaMosaic showed the lowest GPU runtime, indicating that most of its wall time was spent on preprocessing prior to model training. On RAM usage, StabMap and SpaMosaic once again had the lowest memory consumption across all datasets (**Supp. Fig. S52d**). On the other hand, scMoMaT exhibited the highest RAM usage on the embryonic mouse brain (Misar) and mouse embryo datasets, while UINMF consumed the highest amount of RAM on the embryonic mouse brain (Misar+Stereo) dataset. As for GPU memory consumption, CLUE consistently ranked lowest across all three datasets (**Supp. Fig. S52e**). scMoMaT also had the highest GPU memory usage, while SpaMosaic maintained moderate usage close to CLUE's.

To investigate how SpaMosaic scales with the number of features, modalities, and samples, we conducted additional experiments with the embryonic mouse brain (Misar) dataset under a maximum CPU core/thread limit of 16. Specifically, to evaluate the impact of varying feature numbers on SpaMosaic's computational performance, we randomly sampled a range of features from the RNA and ATAC modalities. For RNA, the number of sampled features employed was 1,000; 2,000; 4,000; 8,000; 16,000; 32,000. For ATAC, it was 1,000; 3,000; 9,000; 27,000; 81,000; 191,034. Each RNA-ATAC feature pair was used to test the performance of SpaMosaic, as shown in **Supp. Fig. S53.** Overall, as the number of features increased, both wall time and CPU time showed a logarithmic growth, though wall time remained under 60 seconds across all settings. GPU time remained under 4 seconds. This is expected because increasing the number of input features primarily affects the preprocessing stage which entails little GPU processing, while the GPU-intensive neural network training is performed on the fixed-size low-dimensional representations and is thus largely unaffected by the original feature count. As for RAM usage, it increased linearly with the number of features, whereas GPU memory usage showed no clear change.

We next evaluated the impact of the number of modalities on SpaMosaic's computational performance by using the embryonic mouse brain (Misar) to create simulated datasets with different modality counts. The embryonic mouse brain (Misar) originally contains two modalities, RNA and ATAC, and three batches where one batch contains both RNA and ATAC, while the other two contain only RNA and only ATAC, respectively. To simulate datasets with more modalities, we replicated each original modality per batch by a given multiplication factor and treated each copy as a distinct modality. For example, with a factor of 2: the first batch originally has RNA and ATAC; it became {RNA-1, RNA-2, ATAC-1, ATAC-2}; the second batch (originally RNA-only) became {RNA-1, RNA-2}; the third batch (originally ATAC-only) became {ATAC-1, ATAC-2}. This resulted in four distinct modalities. We employed multiplication factors from 1 to 5, corresponding to 2, 4, 6, 8, and 10 modalities.

The results obtained are shown in **Supp. Fig. S54**, where we see that the wall time and CPU time increased approximately linearly with the number of modalities. However, even at the highest setting with ten modalities, wall time remained under 4 minutes. While the curve for GPU time was steeper than that of the wall time's, the maximum time required was around 30 seconds, contributing minimally to the overall runtime. In contrast, RAM and especially GPU memory usage increased more sharply, from approximately 2 GB to 12 GB. This steep growth was primarily due to SpaMosaic's design, which required pairwise alignment between modalities and thus a quadratic increase in the number of modality pairs (**Supp. Fig. S54**). Currently, datasets with four or more modalities are rare and therefore it is not a major limitation presently. For datasets with three or fewer modalities, GPU memory usage remained below 4 GB.

We also evaluated the impact of increasing the number of sections using simulated data created from the embryonic mouse brain (Misar) dataset. Specifically, we replicated the original three sections with different multiplication factors to increase the number of sections. For example, with a factor of 2, each original slice and its associated modalities were duplicated once, resulting in a dataset with two multimodal sections, two RNA-only sections, and two ATAC-only sections. We employed several replication settings that corresponded to total section counts of 3, 6, 9, 15, 24, 45, 69, and 100. For all cases except the 100-section setting, the three original sections were replicated equally. The 100-section dataset was created by replicating the first two sections 33 times and the third section 34 times.

As shown in **Supp. Fig. S55**, SpaMosaic, CLUE, and Cobolt (GPU-based integration methods) scaled up to 100 sections with reasonable runtimes (<10,000s). While MultiVI was able to scale up to 100 sections, it required much longer runtime (>50,000s) than others. For scMoMaT, it was unable to handle 100 sections due to high GPU memory usage. Regarding specific resource usage, SpaMosaic achieved the lowest wall time, CPU time, and GPU time consumption. For CPU memory usage, all methods showed a linear increase with the number of sections, with MultiVI exhibiting the slowest growth and the others displaying similar rates. For GPU memory usage, CLUE, Cobolt, and MultiVI showed the slowest increase, likely due to their mini-batch training strategies. scMoMaT exhibited the highest memory growth, exceeding 23 GB at 69 sections, while SpaMosaic followed, though with a substantially lower growth rate.

Finally, we evaluated the maximum number of spots per section that SpaMosaic can handle. Using the embryonic mouse brain (Misar) dataset as template, we simulated data of multiple sets of three sections: one multimodal section (RNA+ATAC) and two single-modality sections (RNA or ATAC), with different numbers of spots. All three sections per set had the same number of spots. The spatial distribution of spots in each section is shown in **Supp. Fig. S56**. The RNA modality contained 10,000 genes with expression values drawn from a Poisson distribution with parameter $\lambda = 1$,

while the ATAC modality consisted of 100,000 accessibility peaks encoded as binary values (0 or 1) with 99.9% sparsity. We generated multiple dataset configurations in which each section contained 50,000, 100,000, 200,000, 400,000, or 800,000 spots. In the computation resource consumption testing, SpaMosaic was able to scale up to 800,000 spots per section (an approximately 895×895 spot grid) on the test server (**Supp. Fig. S57)**. Our future work will focus on optimizing the model implementation to enable mini-batch training, with the goal of further reducing GPU memory usage and thus further improving scalability.

In summary, we benchmarked SpaMosaic against existing mosaic integration methods across three distinct integration scenarios and found that SpaMosaic consistently achieved lower wall time, CPU time, and GPU time, along with relatively low GPU memory and RAM usage. Even on the largest dataset, embryonic mouse brain (Misar+Stereo) (with 50,000 spots across six sections), SpaMosaic completed the integration within one minute, while the slowest method, MultiVI, required over 100 minutes. Using a fixed dataset, increasing the number of input features primarily affected preprocessing time and RAM usage, but had minimal impact on model training and inference speed. In contrast, increasing the number of modalities led to a quadratic increase in GPU time and memory usage, as SpaMosaic performs pairwise alignment across all modality combinations. Nevertheless, even with ten modalities, SpaMosaic's GPU runtime remained below 30 seconds. For increases in the number of sections, it entailed a linear rise in GPU time and memory usage by SpaMosaic. However, on the test server, SpaMosaic could handle up to a hundred sections or more. Finally, with a mosaic dataset (one RNA+ATAC section, one RNA section, and one ATAC section), SpaMosaic successfully processed data with up to 800,000 spots per section.

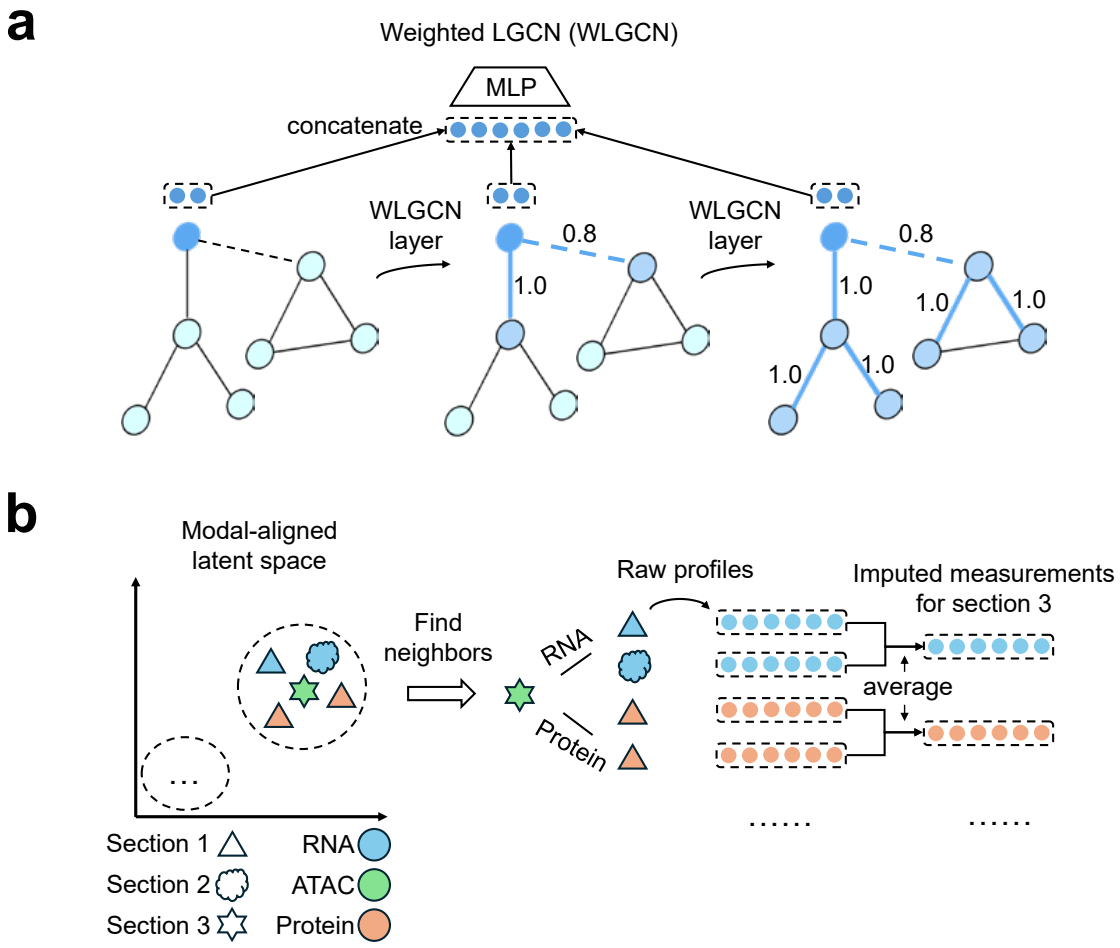
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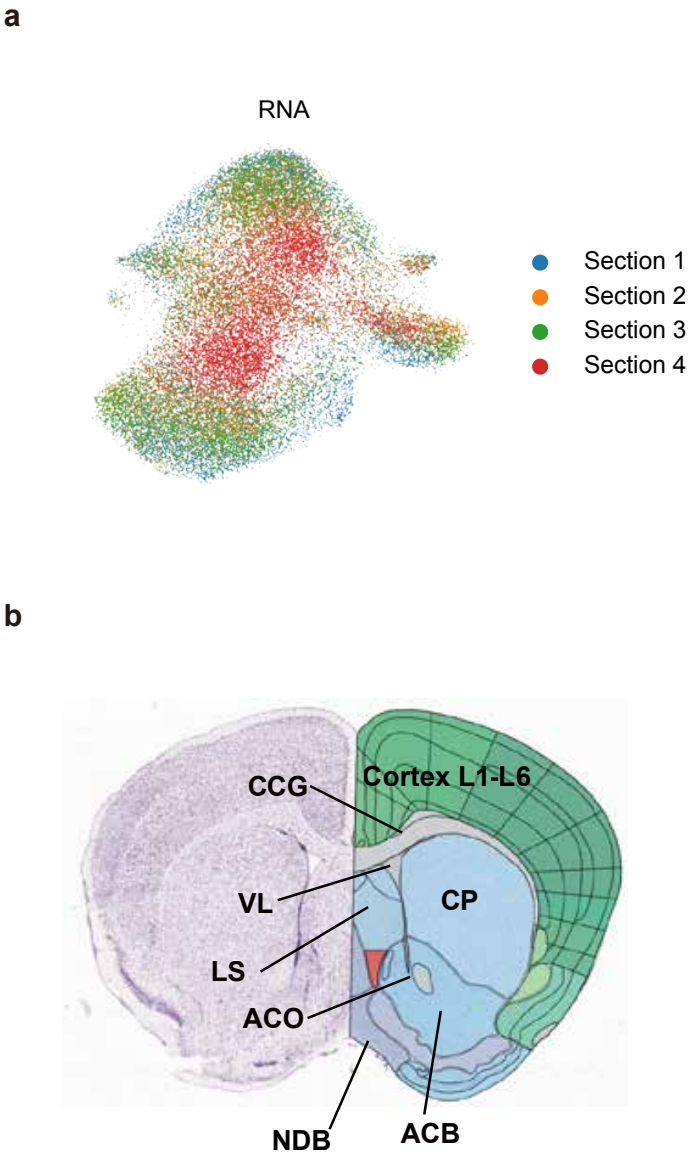
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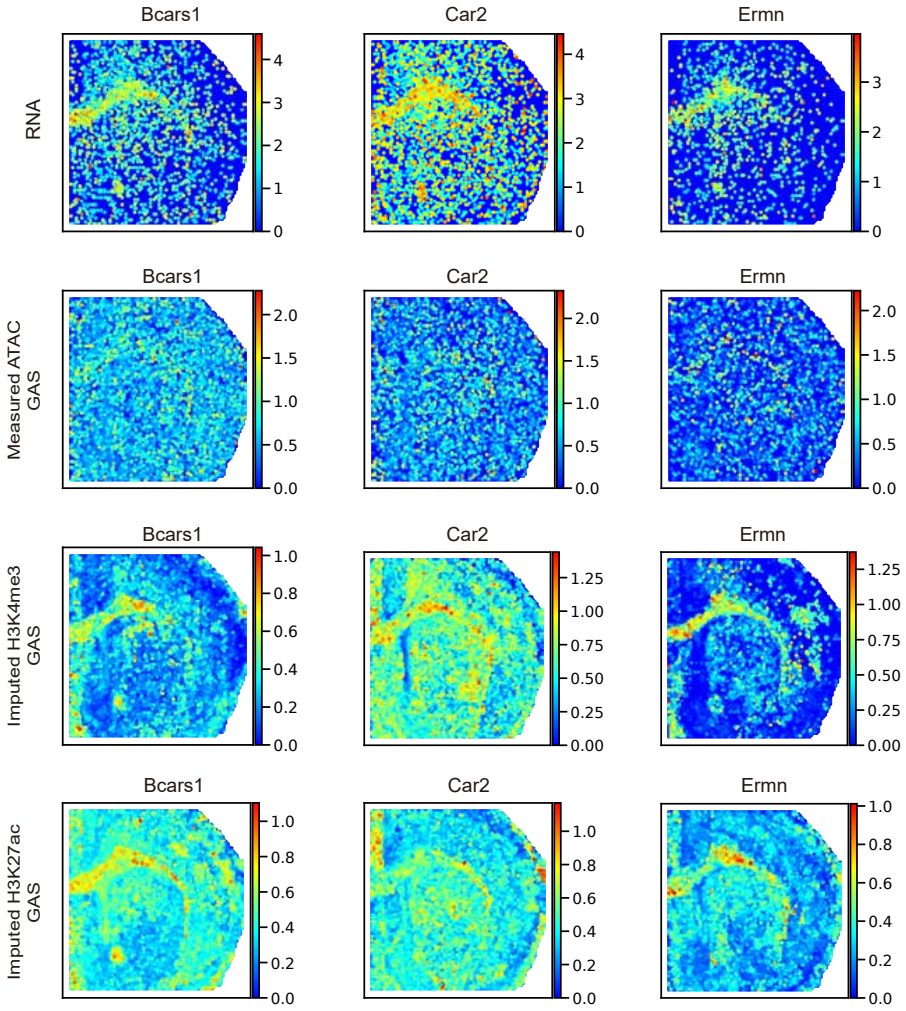
Supplementary Figures



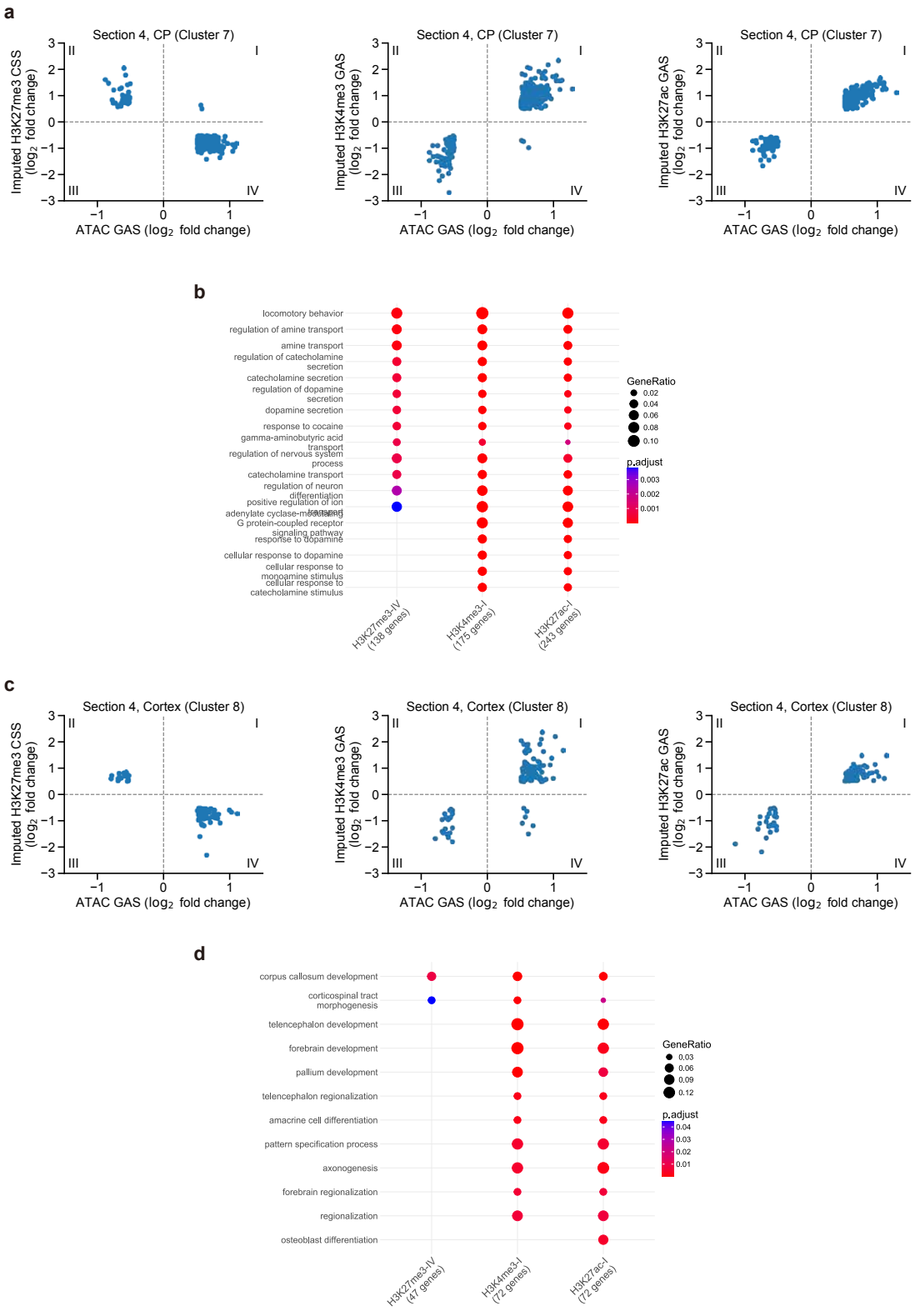
Supplementary Figure S1 Graph neural network architecture and imputation strategy. a Aggregation and propagation processes of weighted LGCN. The weight for spatial neighbors is set to 1 while the weight for expression neighbors is set to 0.8. In each layer, each node aggregates the information from its neighbors using the weighted average. After two layers, the information from each node is concatenated across three output layers and then processed by a multiple layer perceptron. **b** Imputation process. Taking section 3 as an example, which only has ATAC modality, we aim to impute its RNA and protein profiles. To impute the RNA profile of each spot in section 3, we find its RNA-specific neighbors from the other sections and then compute the average target expression profiles of these neighbors. The same approach is also taken for other modalities.



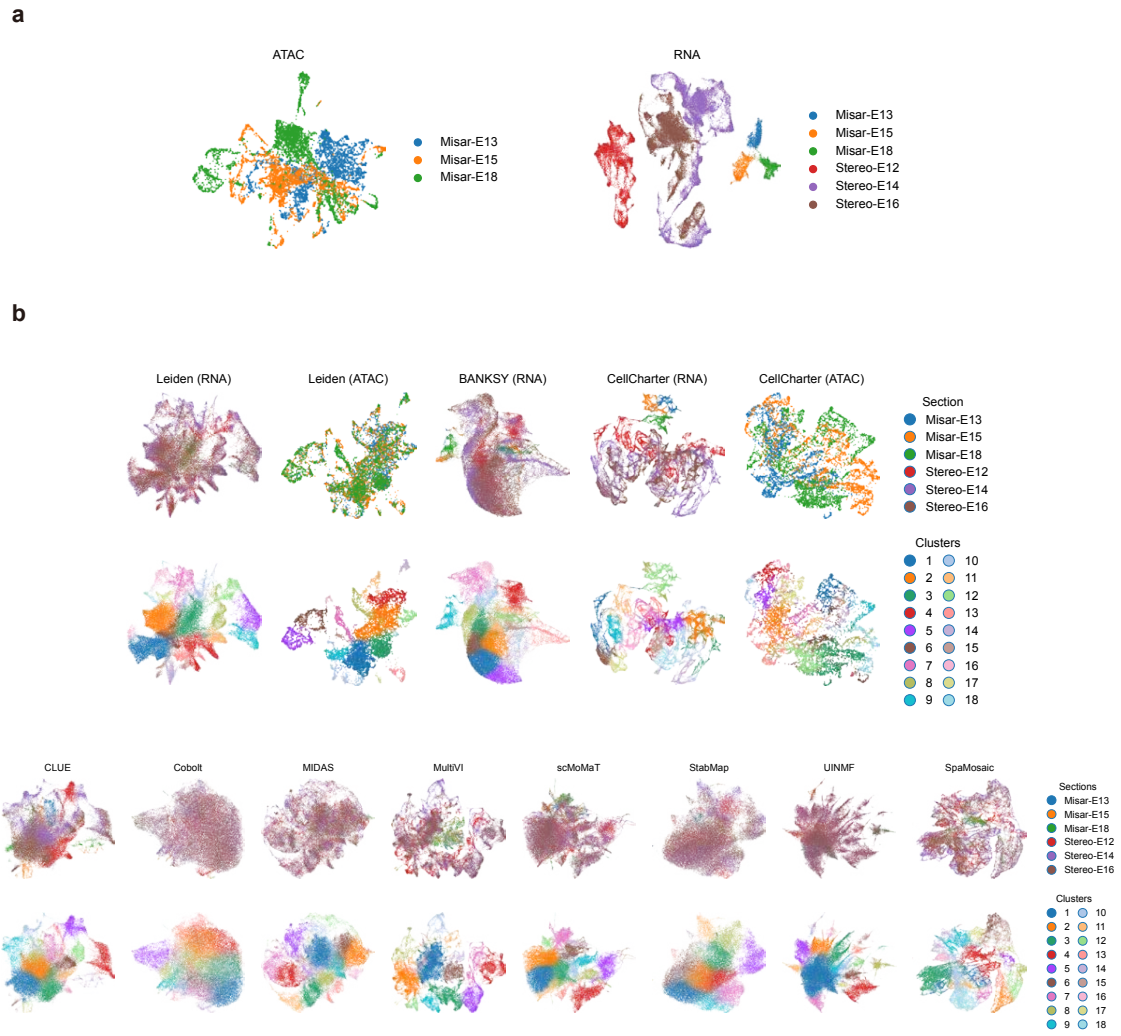
Supplementary Figure S2 UMAP plot and reference atlas of postnatal mouse brain (spRA+spRH) dataset. a UMAP plots of raw RNA profiles. Spots are colored by section labels. **b** Reference atlas of postnatal mouse brain structures. Images were from the Allen Reference Atlas (https://atlas.brain-map.org/).



Supplementary Figure S3 Spatial heatmap of measured and imputed gene expression/gene activity scores. Spatial heatmap of measured RNA, measured ATAC GAS, imputed H3K4me3 GAS and imputed H3K27ac GAS for CCG-specific genes *Bcas1*, *Car2* and *Ermn*.

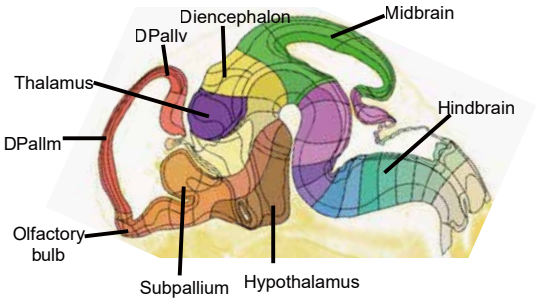


Supplementary Figure S4 Relationship between ATAC GAS and histone modification GAS/CSS in the CP and cortex regions. **a** Relationship between imputed GAS/CSS and measured ATAC GAS for CP-specific genes on the fourth section. **b** Enriched GO terms for CP-specific genes in quadrants I and IV of panel **a**. Enrichment analysis was performed using a one-sided hypergeometric test, and P values were adjusted for multiple comparisons using the Benjamini-Hochberg (BH) method. **c** Relationship between imputed GAS/CSS and measured ATAC GAS for cortex-specific genes on the fourth section. **d** Enriched GO terms for cortex-specific genes in quadrants I and IV in panel **c**. Enrichment analysis was performed using a one-sided hypergeometric test, and P values were adjusted for multiple comparisons using the Benjamini-Hochberg (BH) method.

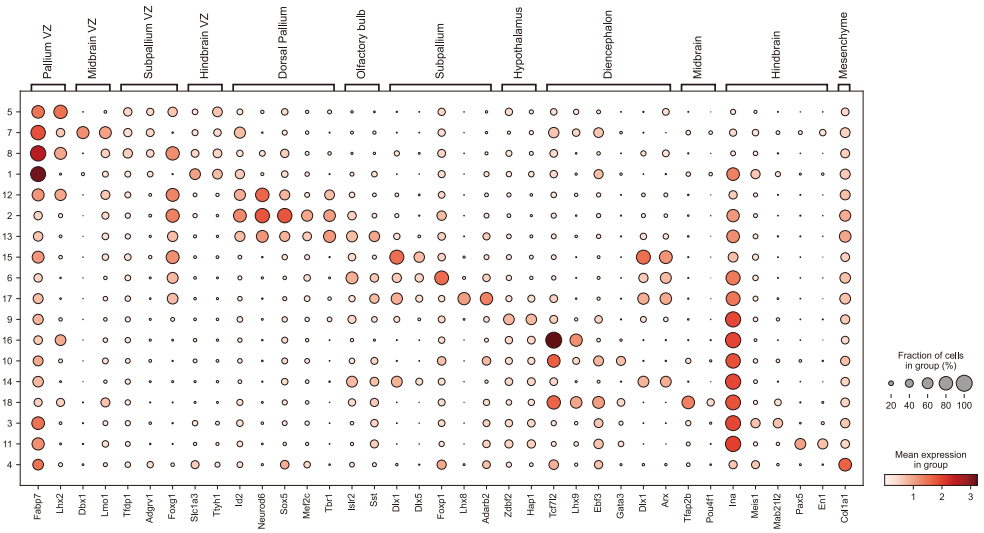


Supplementary Figure S5 UMAP plots of embryonic mouse brain dataset (Misar+Stereo) integration analysis. a UMAP plots of the unintegrated ATAC and RNA profiles. Spots are colored by section labels. **b** UMAP plots of the embeddings from baseline methods and mosaic integration methods.

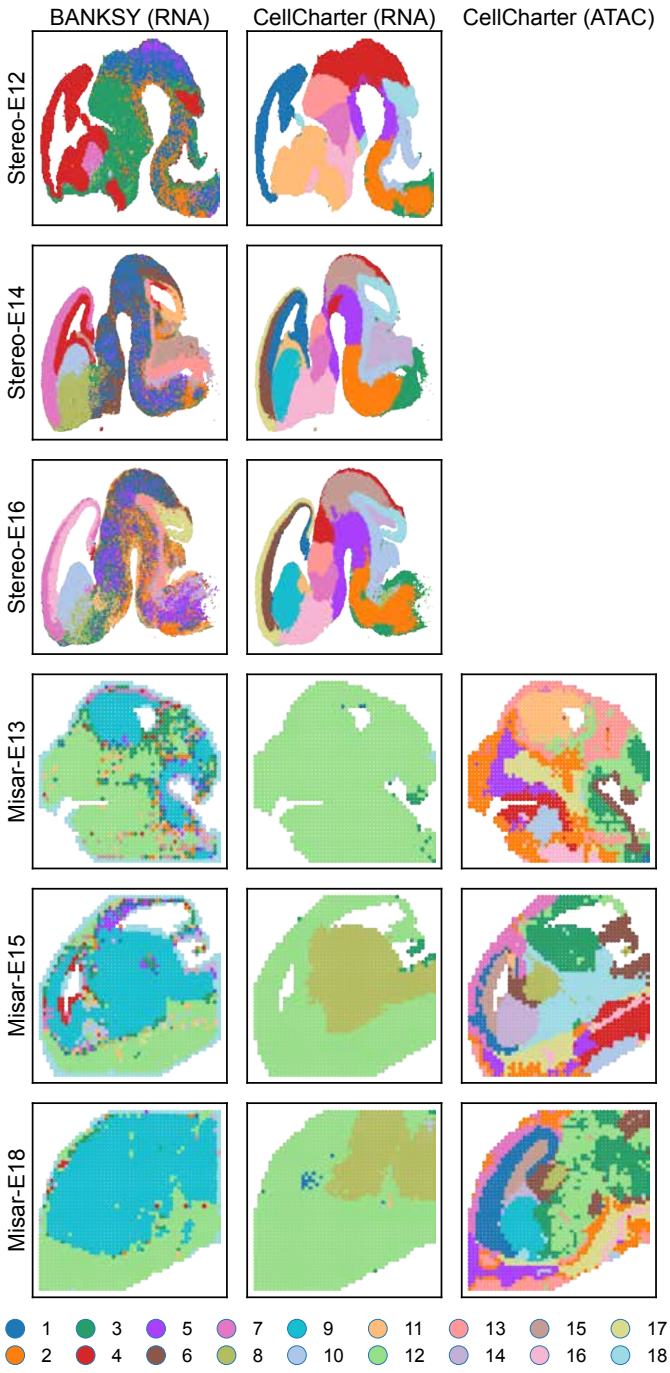
a



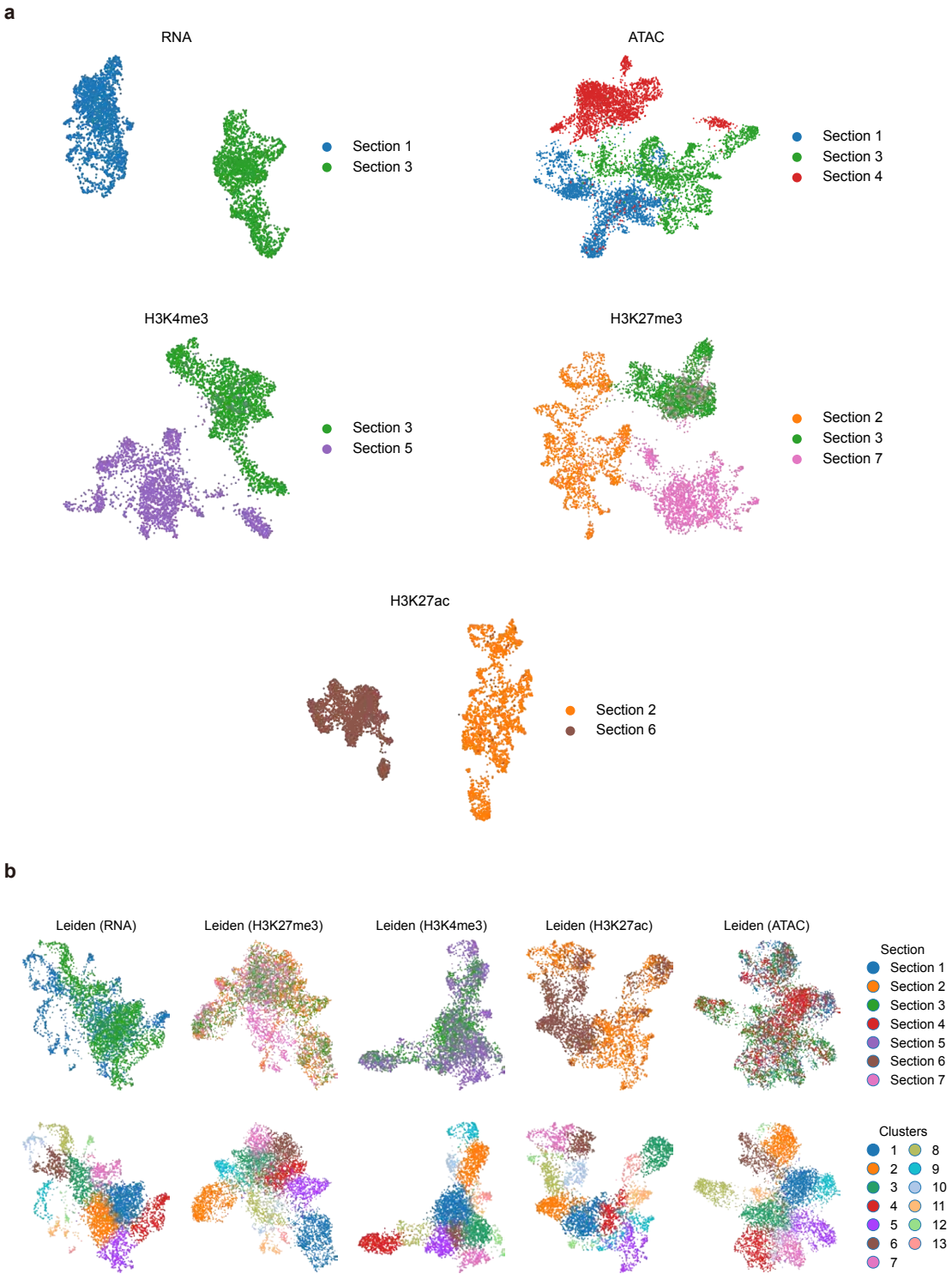
b



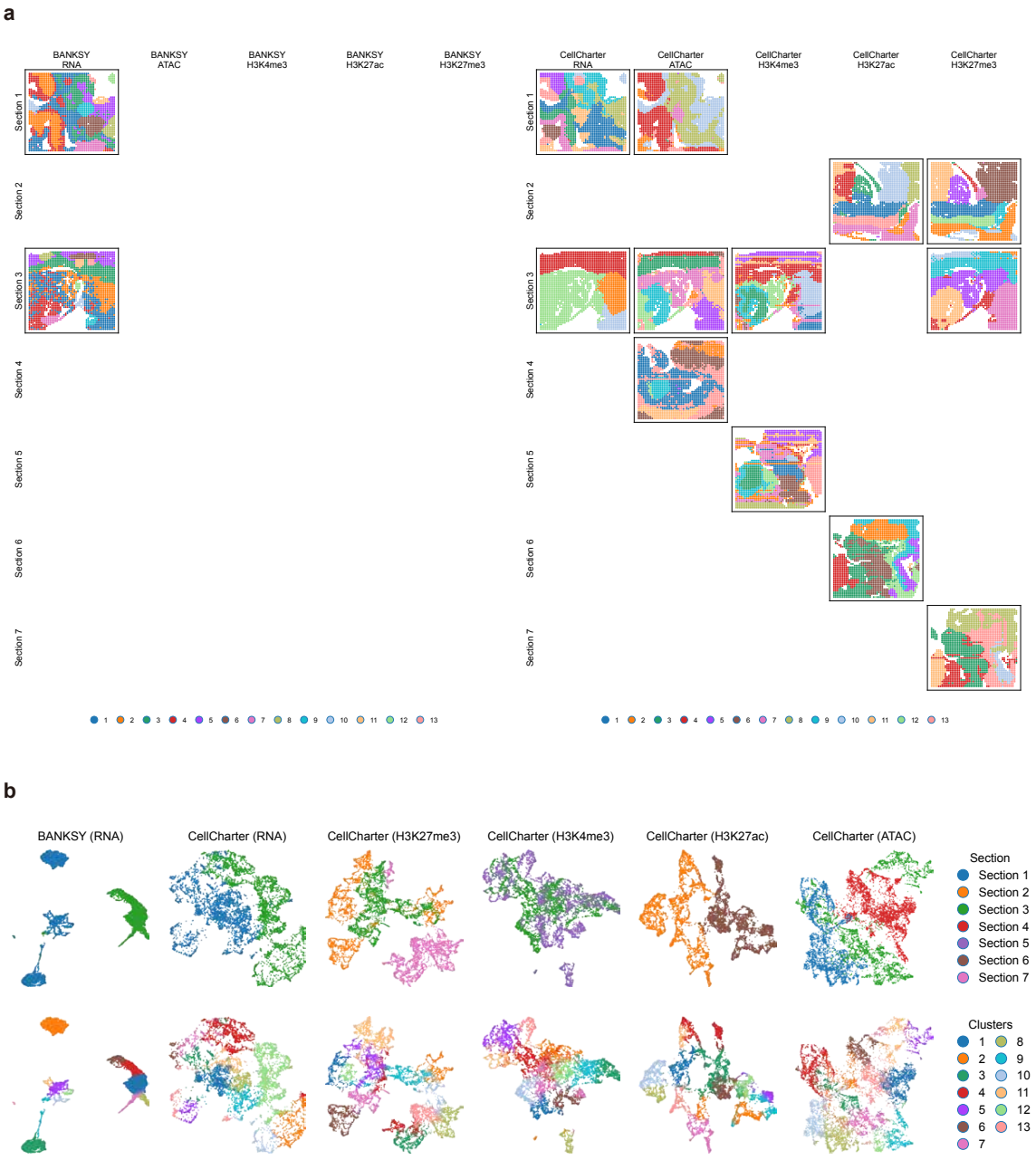
Supplementary Figure S6 Reference structure and marker gene expression validation of SpaMosaic's identified clusters in embryonic mouse brain (Misar+Stereo) dataset. **a** Reference atlas of developing mouse brain structures at the E13.5 stage. Image was taken from the Allen Reference Atlas (https://atlas.brain-map.org/). **b** Dot plot of region-specific marker gene RNA expression in SpaMosaic's clusters.



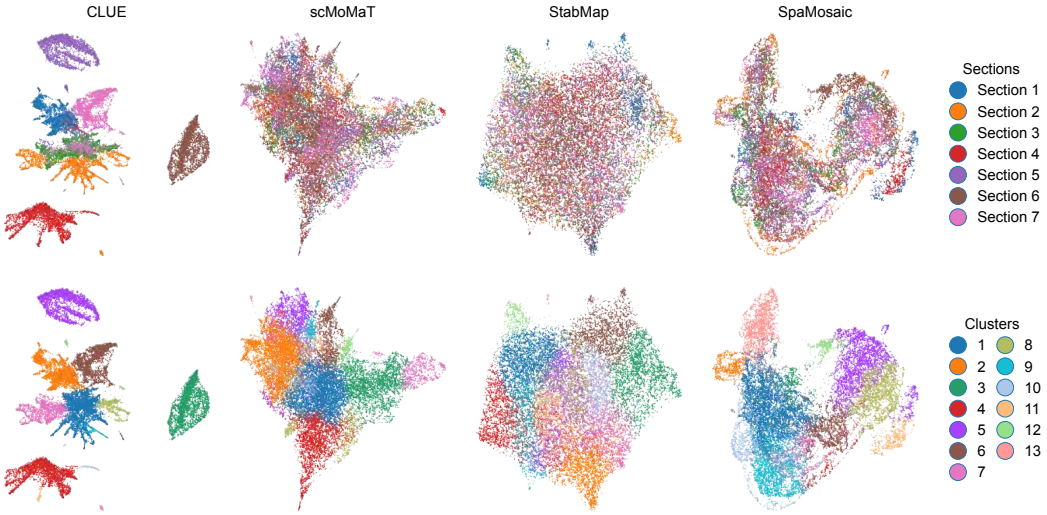
Supplementary Figure S7 Spatial plots of clustering results from BANKSY and CellCharter on embryonic mouse brain (Misar+Stereo) dataset.



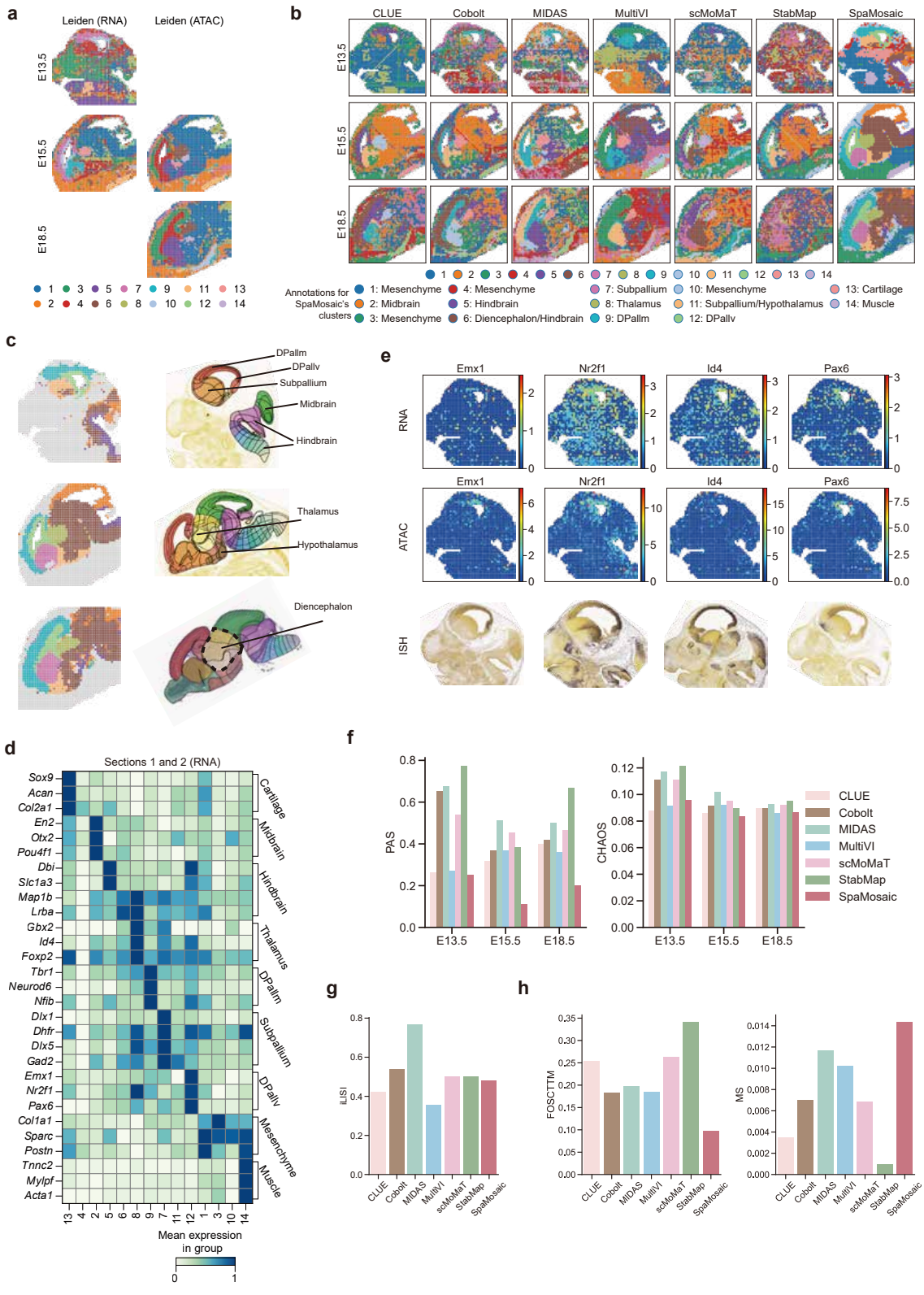
Supplementary Figure S8 UMAP plots of mouse embryo dataset integration analysis. a UMAP plots of unintegrated RNA, ATAC, H3K4me3, H3K27me3, and H3K27ac histone modification profiles. Spots are colored by section labels. **b** UMAP plots of embeddings from baseline methods. Spots are colored by section labels in the first row and the second row by cluster labels.



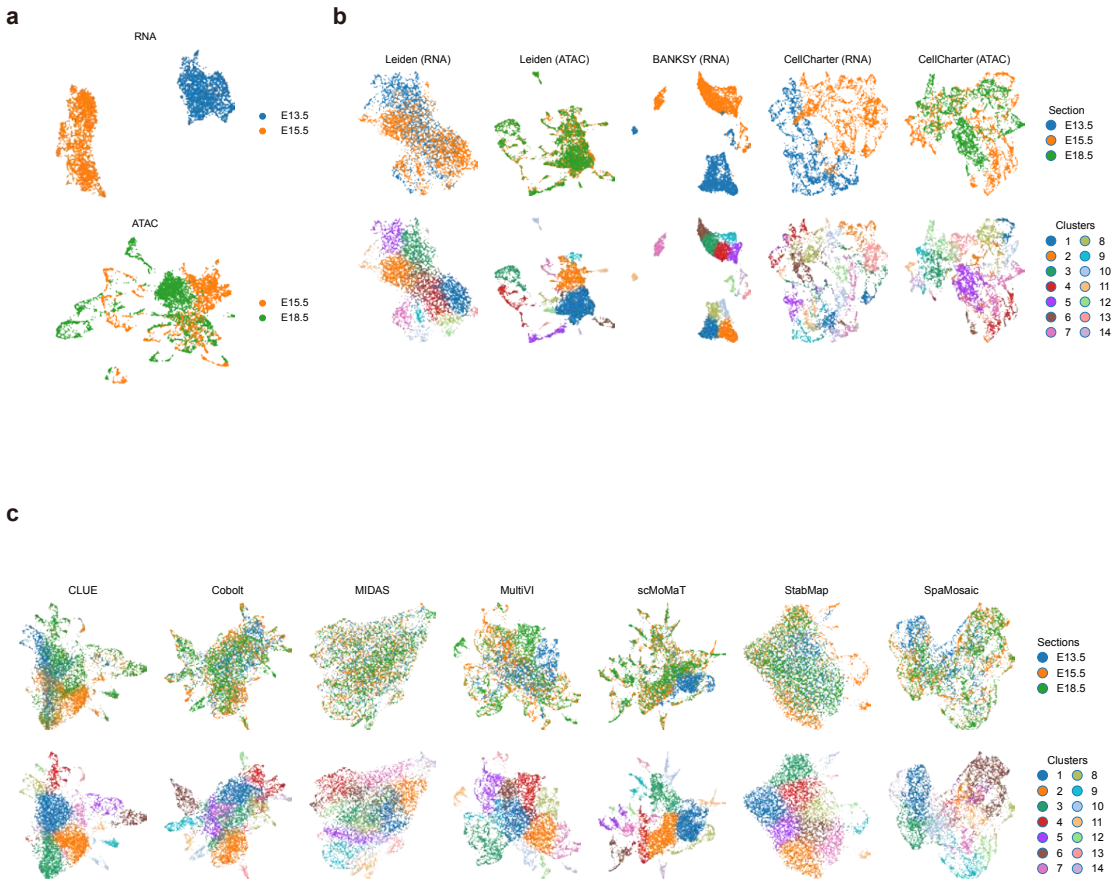
Supplementary Figure S9 Spatial clustering plots and UMAP plots of baseline methods. **a** Spatial plots of clustering results from the baseline methods. **b** UMAP plots of embeddings from the baseline methods. Spots are colored by section labels in the first row and the second row by cluster labels.



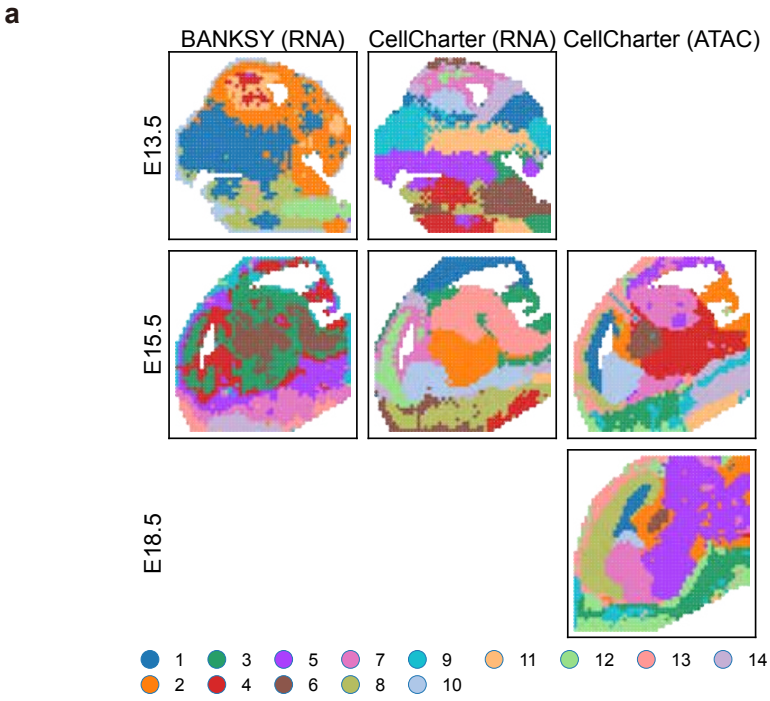
Supplementary Figure S10 UMAP plots of embeddings from mosaic integration methods. Spots are colored by section labels in the first row and the second row by cluster labels.

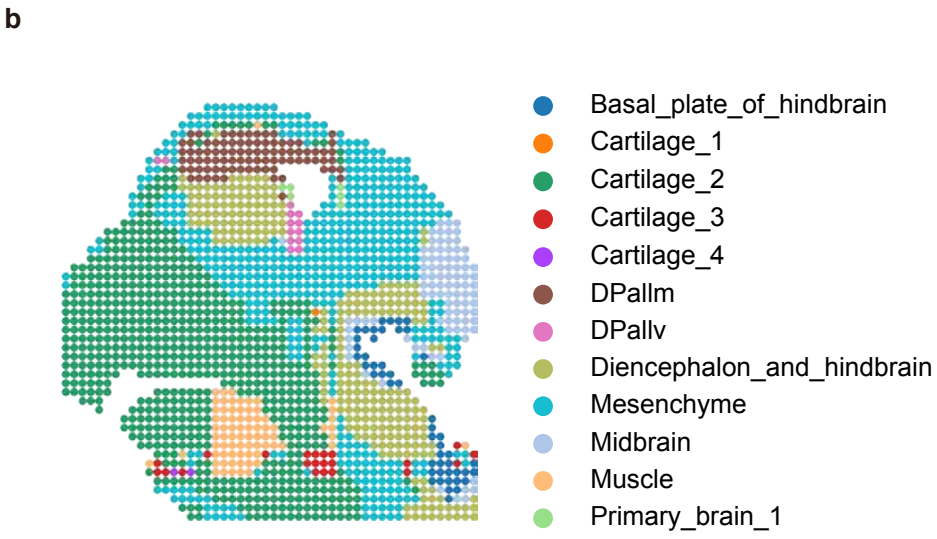

a
Leiden (RNA)
Leiden (ATAC)
E13.5
E15.5
E18.5
b
CLUE
Cobolt
MIDAS
MultiVI
scMoMaT
StabMap
SpaMosaic
Annotations for SpaMosaic's clusters
1: Mesenchyme
2: Midbrain
3: Mesenchyme
4: Mesenchyme
5: Hindbrain
6: Diencephalon/Hindbrain
7: Subpallium
8: Thalamus
9: DPallm
10: Mesenchyme
11: Subpallium/Hypothalamus
12: DPallv
13: Cartilage
14: Muscle
c
DPallm
DPallv
Subpallium
Midbrain
Hindbrain
Thalamus
Hypothalamus
Diencephalon
d
Sections 1 and 2 (RNA)
Sox9
Acan
Col2a1
En2
Otx2
Pou4f1
Dbi
Slc1a3
Map1b
Lrba
Gbx2
Id4
Foxp2
Tbr1
Neurod6
Nfib
Dlx1
Dhfr
Dlx5
Gad2
Emx1
Nr2f1
Pax6
Col1a1
Sparc
Postn
Tnnc2
Mylpf
Acta1
Cartilage
Midbrain
Hindbrain
Thalamus
DPallm
Subpallium
DPallv
Mesenchyme
Muscle
Mean expression in group
e
Emx1
Nr2f1
Id4
Pax6
RNA
ATAC
ISH
f
PAS
CHAOS
E13.5
E15.5
E18.5
CLUE
Cobolt
MIDAS
MultiVI
scMoMaT
StabMap
SpaMosaic
g
iLISI
h
FOSCTTM
MS

Supplementary Figure S11 Downstream analysis results obtained with SpaMosaic and quantitative benchmarking on an embryonic mouse brain (Misar) dataset. **a** Mosaic dataset and clustering patterns for all three sections: E13.5 without the ATAC modality, E15.5 with both RNA and ATAC expression profiles, and E18.5 without the RNA modality. Modality-specific clustering was jointly performed across relevant sections using the Leiden algorithm on batch-corrected representations. **b** Spatial plots of the clusters obtained from mosaic integration methods for the three sections. Manual annotation was performed to identify SpaMosaic's clusters. **c** Comparison of SpaMosaic clustering results with reference annotations from the Allen Reference Atlas for the corresponding three developmental stages. Reference images were cropped from the reference atlas (https://atlas.brain-map.org/). **d** RNA expression of marker genes specific to the brain regions captured by SpaMosaic's clusters. **e** Gene activity scores of DPallv-specific marker genes (*Emx1*, *Emx2*, *Id4*, *Nr2f1*, and *Pax6*) in SpaMosaic's clusters. The bottom row shows the ISH images from the Allen Brain Atlas for the corresponding markers. Reference images were cropped from the Allen Brain Atlas (https://developingmouse.brain-map.org/). **f** PAS and CHAOS scores of benchmarked methods. Bars represent metric values calculated individually for each embryonic time point (n=1). **g** iLISI scores of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1). **h** Modality alignment scores (FOSCTTM and MS) of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1).

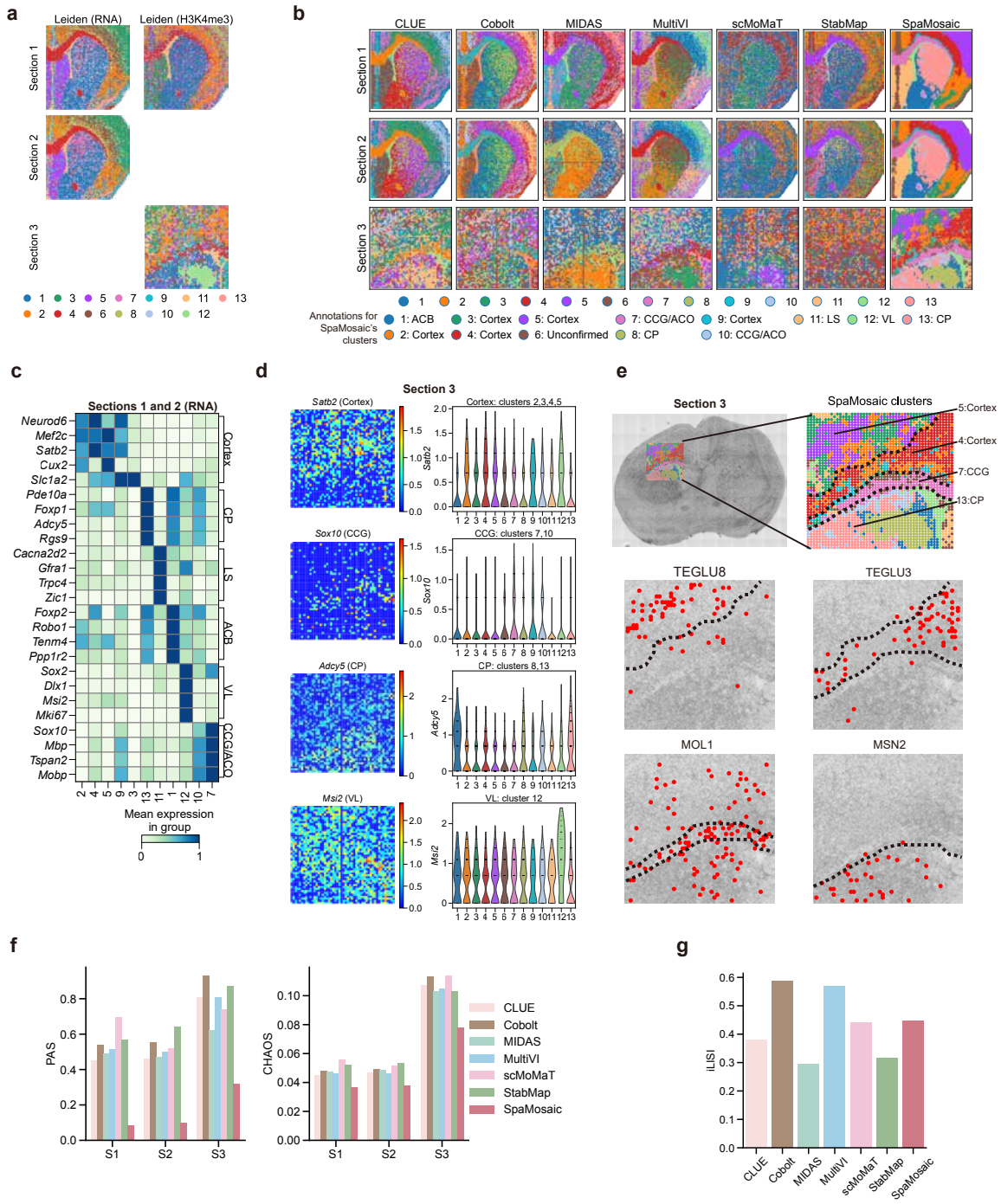


Supplementary Figure S12 UMAP plots of embryonic mouse brain (Misar) dataset integration analysis. **a** UMAP plots of the unintegrated RNA and ATAC profiles. Spots are colored by section labels. **b** UMAP plots of the embeddings from baseline methods. Spots in the first row are colored by section and the second row by cluster. **c** UMAP plots of the embeddings from benchmarked methods.

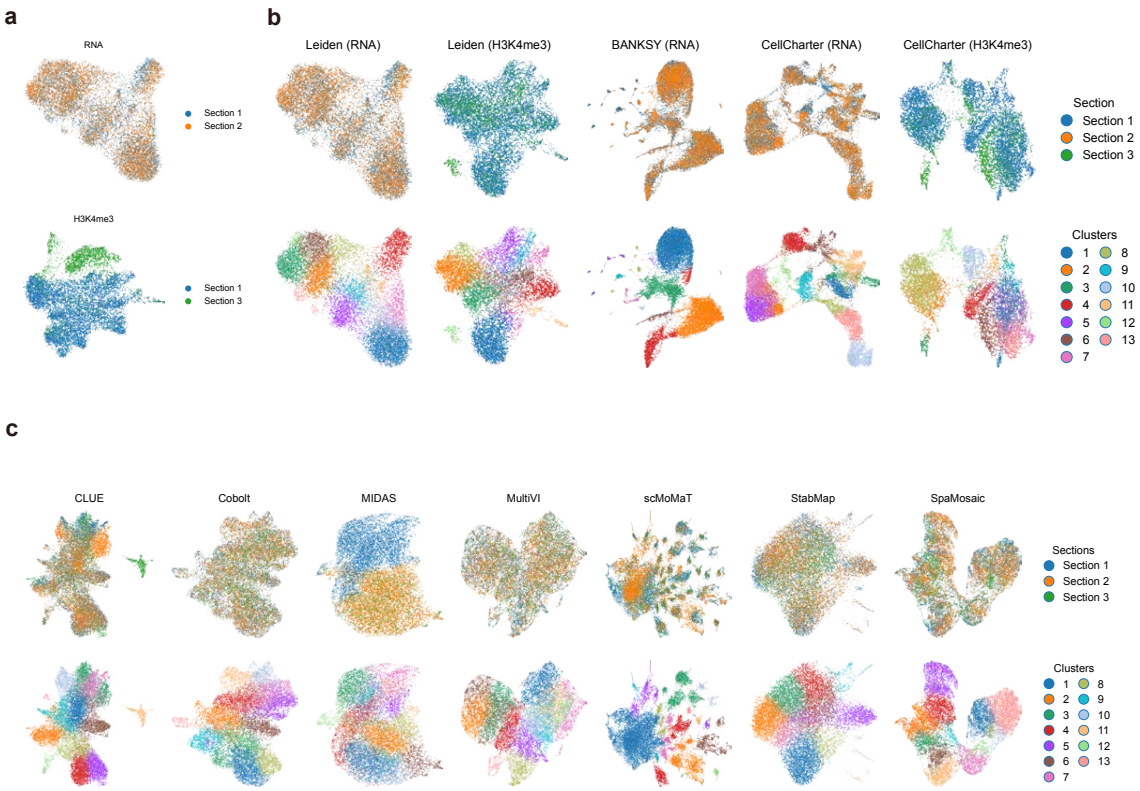




Supplementary Figure S13 Spatial plots of embryonic mouse brain (Misar) dataset integration analysis. **a** Spatial plots of clustering results from baseline methods. **b** Reference annotation for the E13.5 section from the original publication.

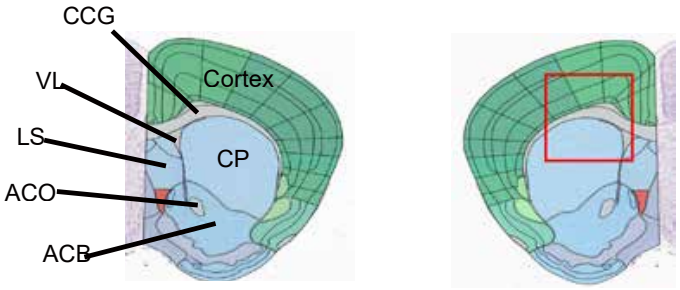


Supplementary Figure S14 Downstream analysis results obtained with SpaMosaic and quantitative benchmarking on a postnatal mouse brain (spRH+spC&T) dataset. **a** Mosaic dataset configuration and clustering patterns obtained for all three sections. Modality-specific clustering was jointly performed across relevant sections using Leiden clustering on the batch-corrected representations. **b** Spatial plots of the clusters obtained from benchmarked methods for all three sections. Manual annotation was performed to identify SpaMosaic's clusters. **c** RNA expression of marker genes specific to the six brain regions captured by SpaMosaic's clusters. **d** Spatial heatmap and violin plots of gene activity scores (GAS) for region-specific markers on the third section and in SpaMosaic's clusters, respectively. **e** SpaMosaic's clusters overlaid on the image of the third section and cell types identified by label transfer. **f** PAS and CHAOS scores of benchmarked methods across the three sections. Bars represent metric values calculated individually for each section (n=1). **g** iLISI scores of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1).

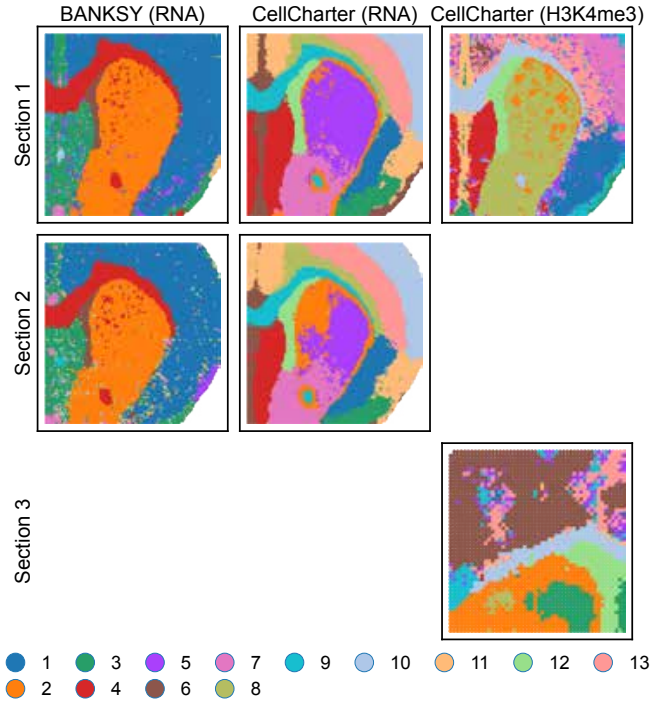


Supplementary Figure S15 UMAP plots of postnatal mouse brain dataset (spRH+spC&T) integration analysis. a UMAP plots of the unintegrated RNA and H3K4me3 histone modification profiles. Spots are colored by section labels. **b** UMAP plots of the embeddings from baseline methods. Spots in the first row are colored by section and the second row by cluster. **c** UMAP plots of the embeddings from benchmarked methods.

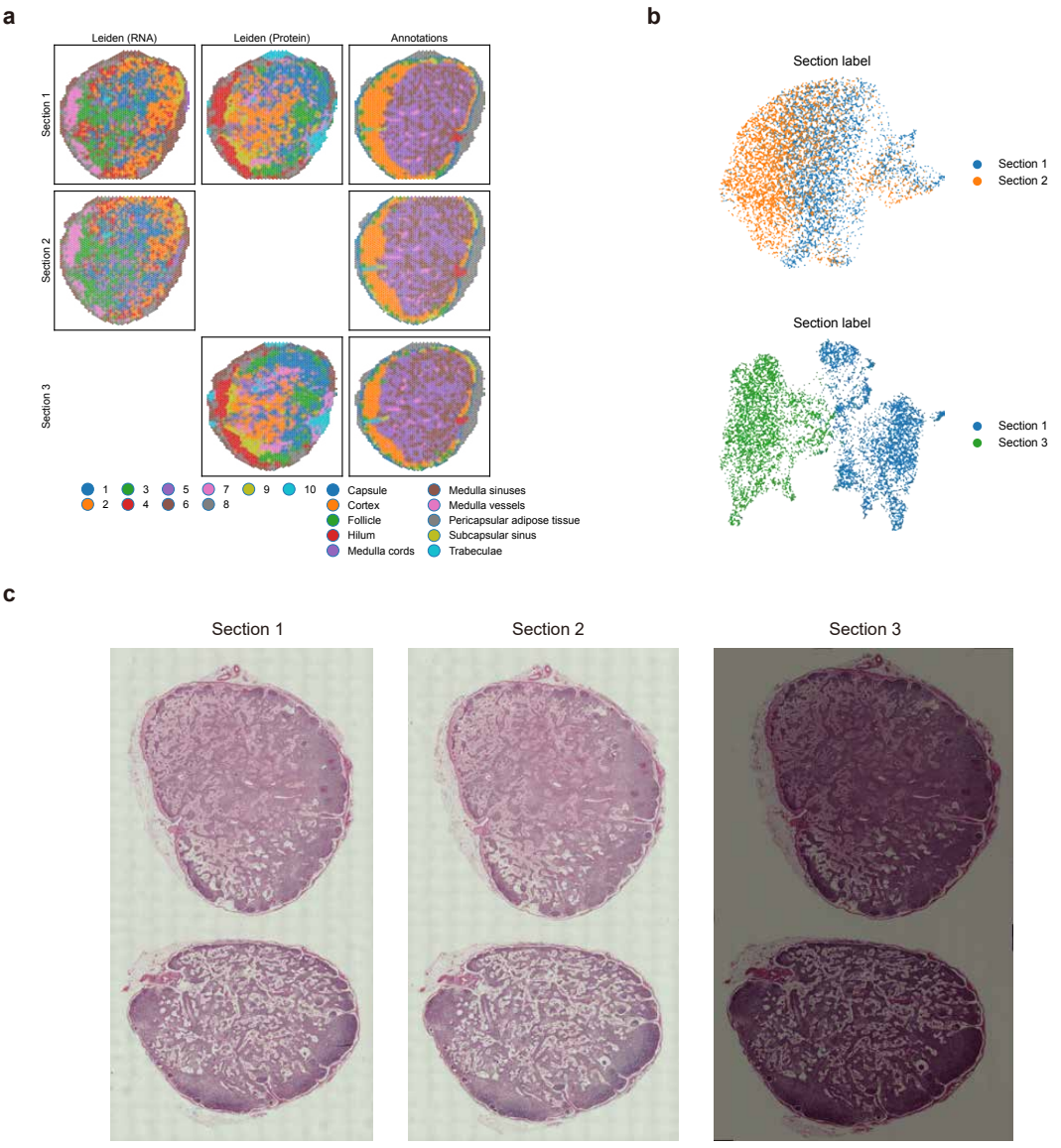
a



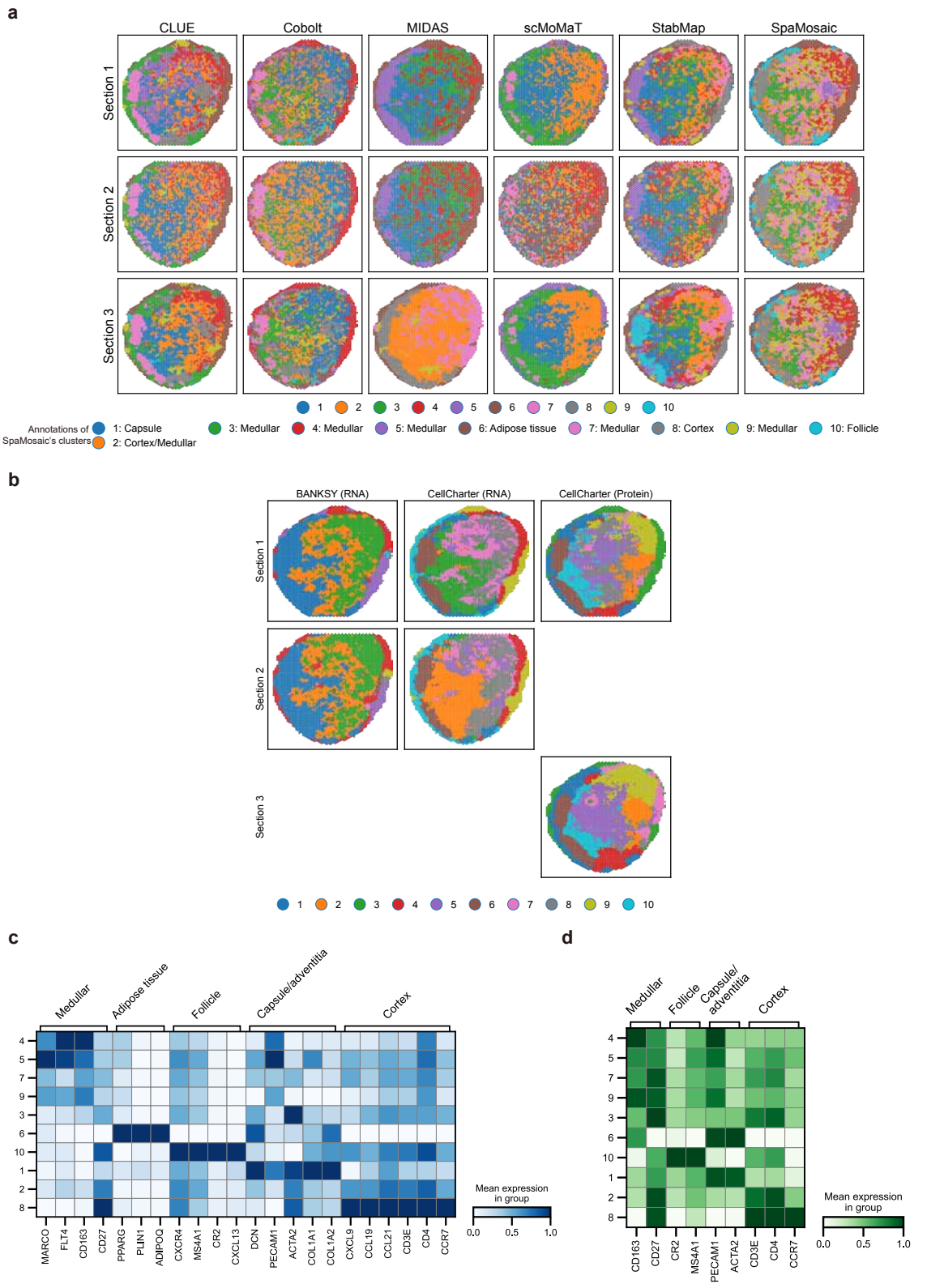
b



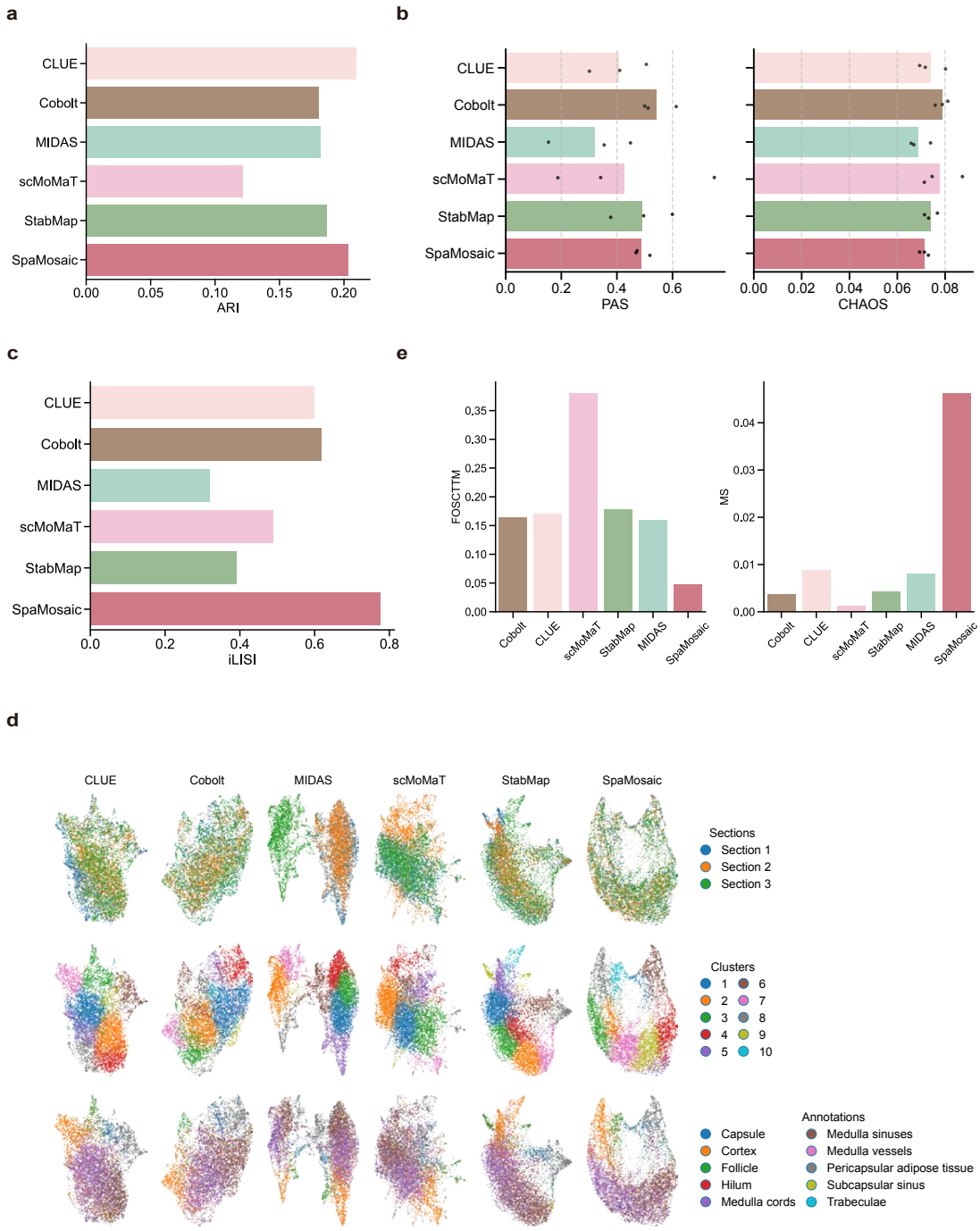
Supplementary Figure S16 Reference atlas and spatial plots of clustering results from baseline methods on the postnatal mouse brain (spRH+spC&T) dataset. a Postnatal mouse brain structure reference atlas. Images were taken from the Allen Reference Atlas (https://atlas.brain-map.org/). **b** Spatial plots of clustering results from baseline methods.



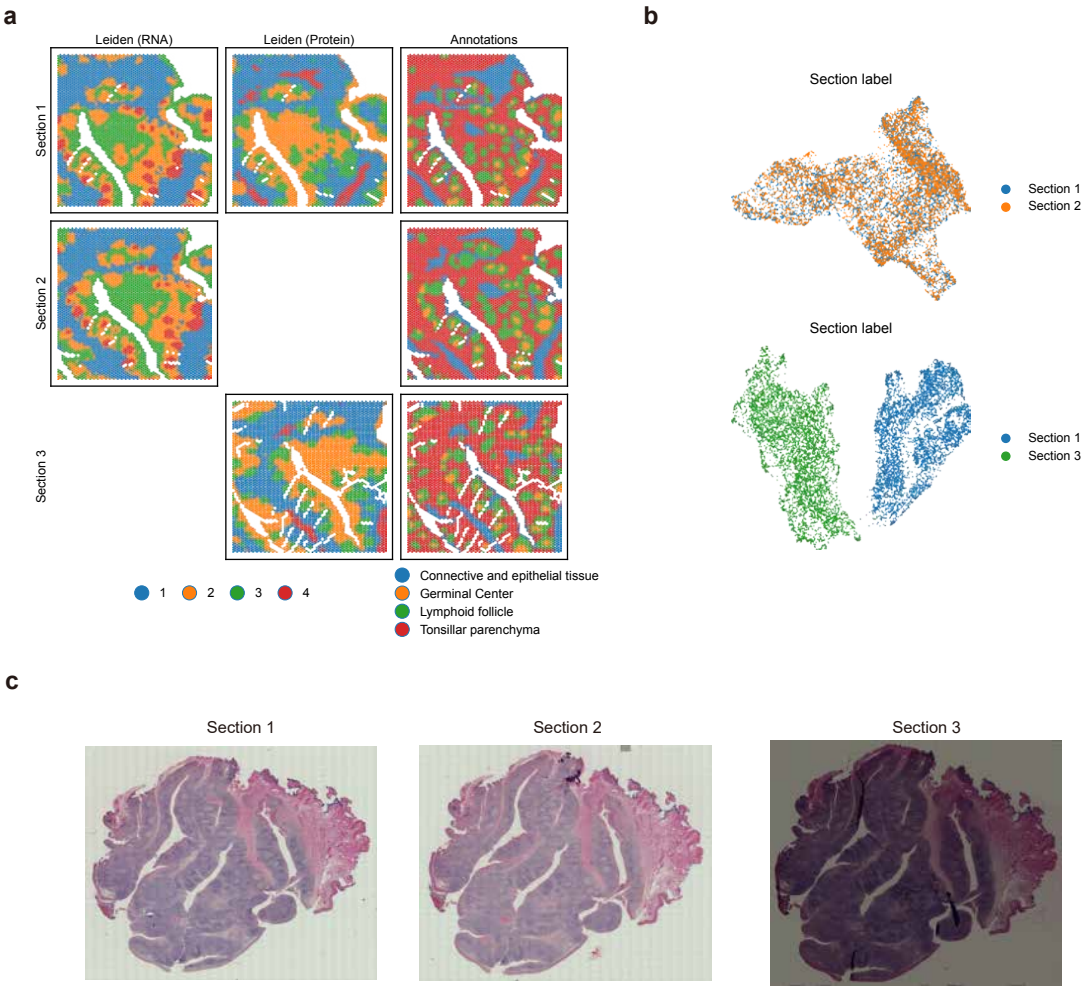
Supplementary Figure S17 Human lymph node dataset setup and visualization. **a** Mosaic dataset configuration and clustering patterns for all three sections: section 1 with both RNA and protein expression profiles, section 2 without the protein expression modality, and section 3 without the RNA modality. Modality-specific clustering was performed jointly across relevant sections using the Leiden algorithm on batch-corrected representations. **b** UMAP plots of uncorrected RNA and protein profiles. **c** H&E staining images of the three sections.



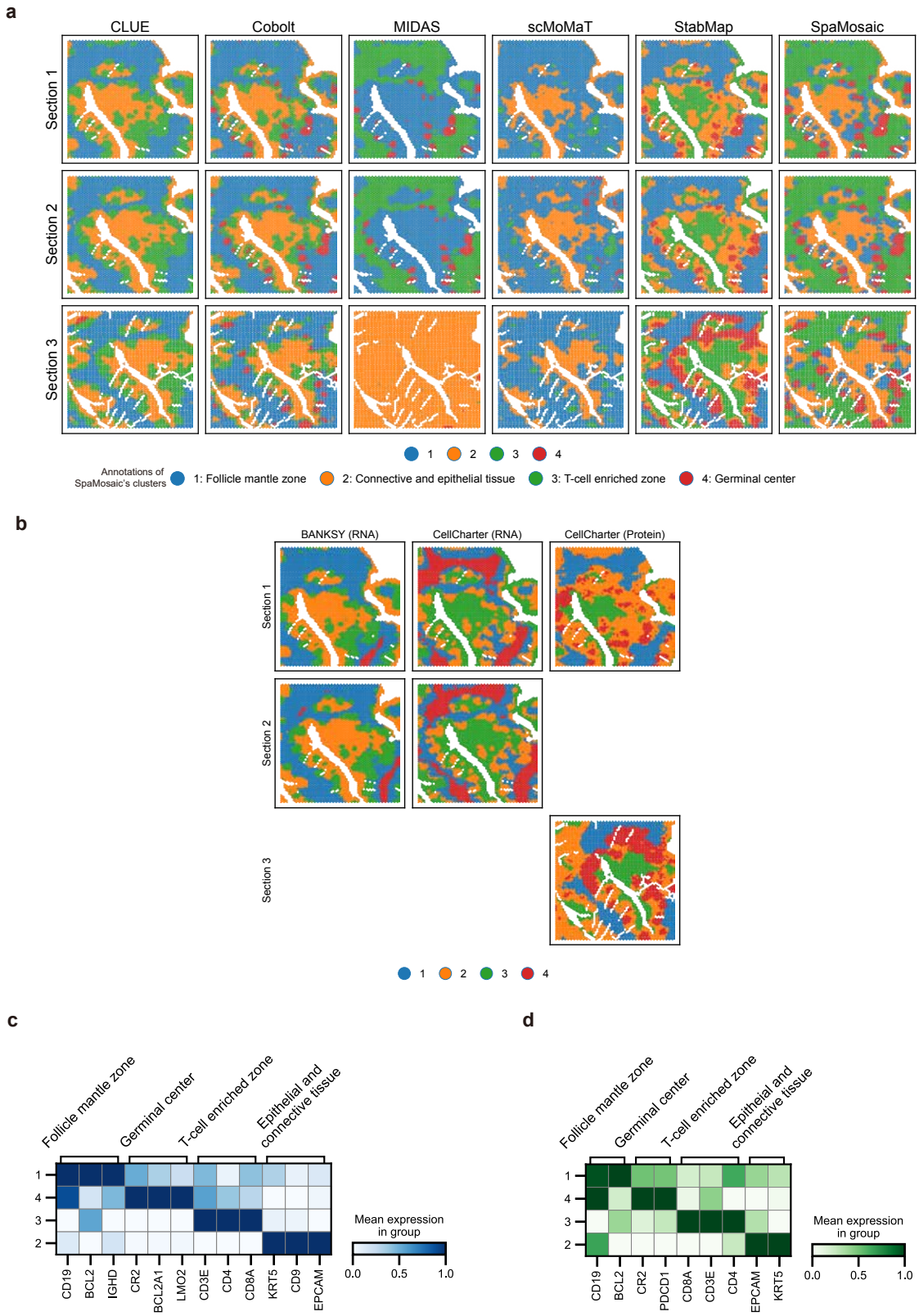
Supplementary Figure S18 Spatial clustering plots and marker analysis on human lymph node dataset. **a** Spatial plots of clustering results from benchmarked methods. **b** Spatial plots of clustering results from BANKSY and CellCharter. **c** RNA expression of region-specific marker genes in SpaMosaic's clusters. **d** Protein expression of region-specific markers in SpaMosaic's clusters.



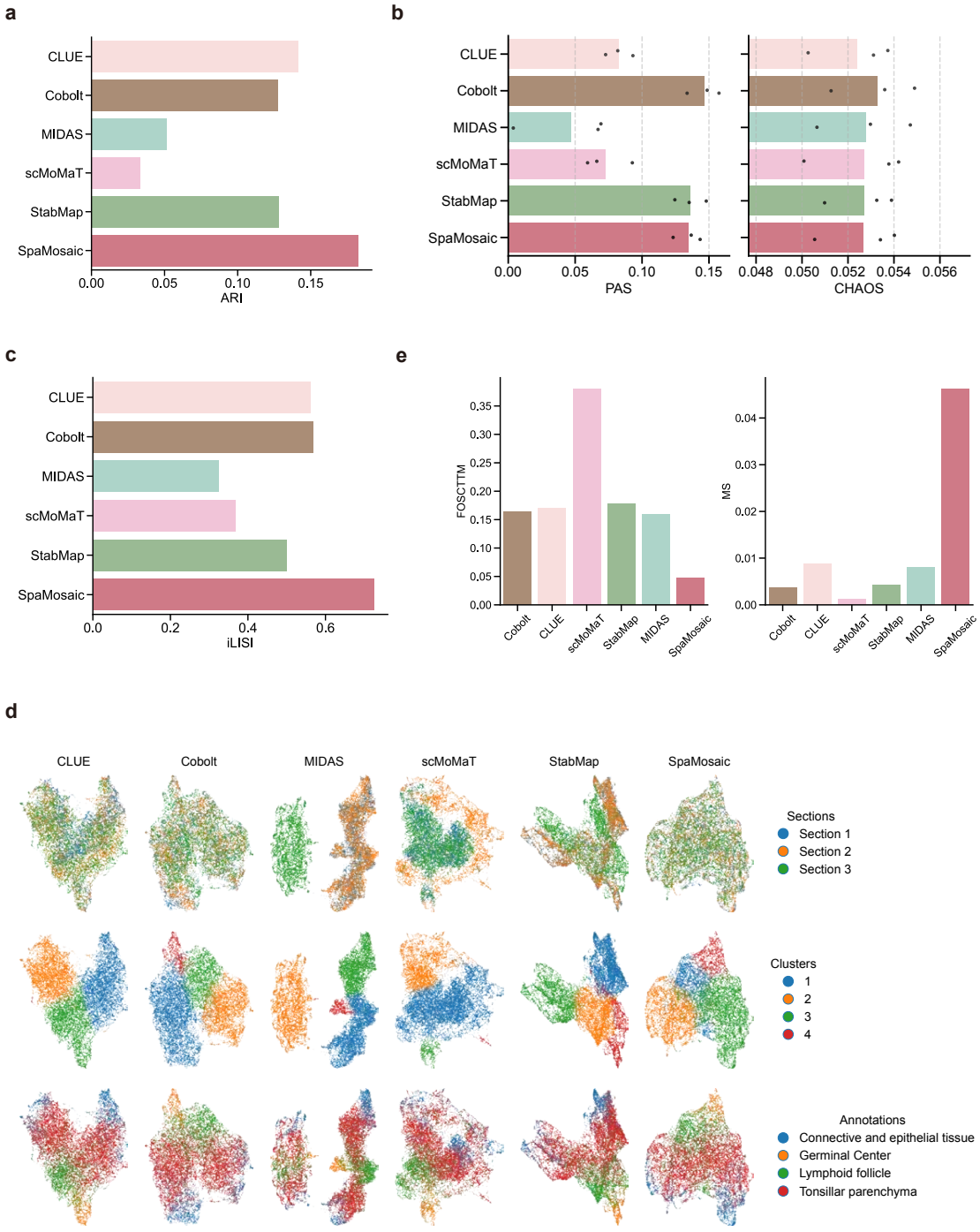
Supplementary Figure S19 Quantitative benchmarking and UMAP plots of human lymph node dataset. **a** ARI scores of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1). **b** PAS and CHAOS scores of benchmarked methods. Bars represent the mean values, with overlaid dots indicating individual sections (n = 3). **c** iLISI scores of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1). **d** UMAP plots of embeddings from benchmarked methods. From top to bottom, spots are colored by section, cluster, and annotation, respectively. **e** FOSCTTM and MS scores of the benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1).



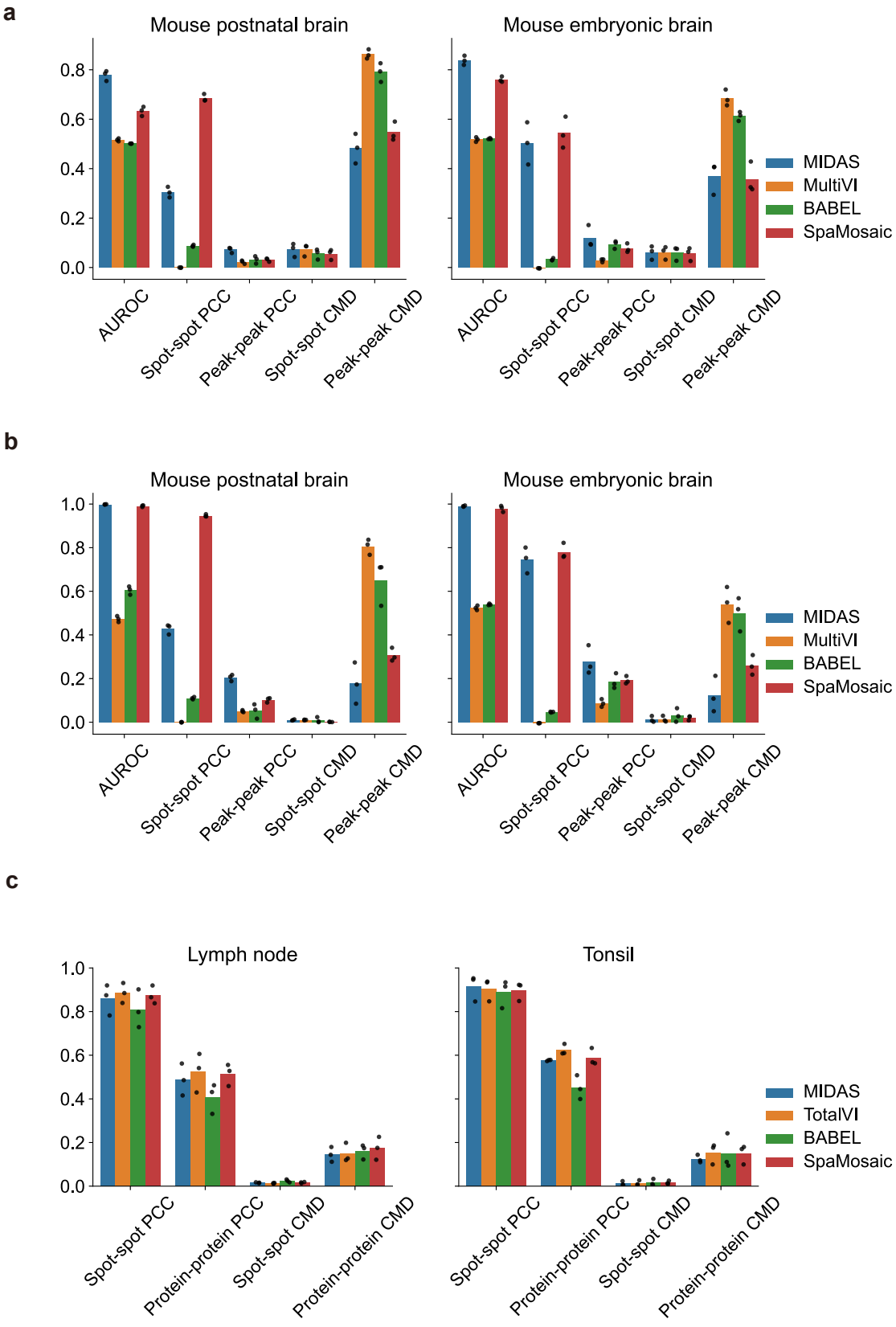
Supplementary Figure S20 Human tonsil dataset setup and visualization. **a** Mosaic dataset configuration and clustering patterns for all three sections: section 1 with both RNA and protein expression profiles, section 2 without the protein expression modality, and section 3 without the RNA modality. Modality-specific clustering was performed jointly across relevant sections using the Leiden algorithm on batch-corrected representations. **b** UMAP plots of unintegrated RNA and protein profiles. **c** H&E staining images of the three sections.



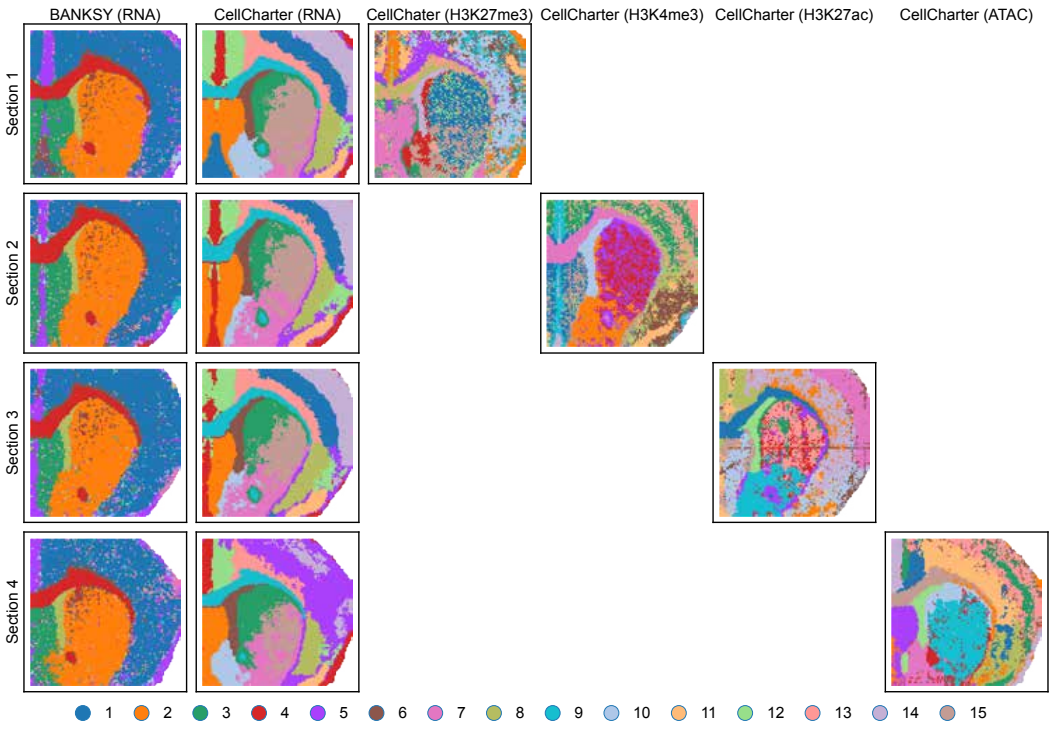
Supplementary Figure S21 Spatial clustering plots and expression heatmap for human tonsil dataset. a Spatial plots of clustering results from benchmarked methods. **b** Spatial plots of clustering results from BANKSY and CellCharter. **c** RNA expression of region-specific marker genes in SpaMosaic's clusters. **d** Protein expression of region-specific markers in SpaMosaic's clusters.



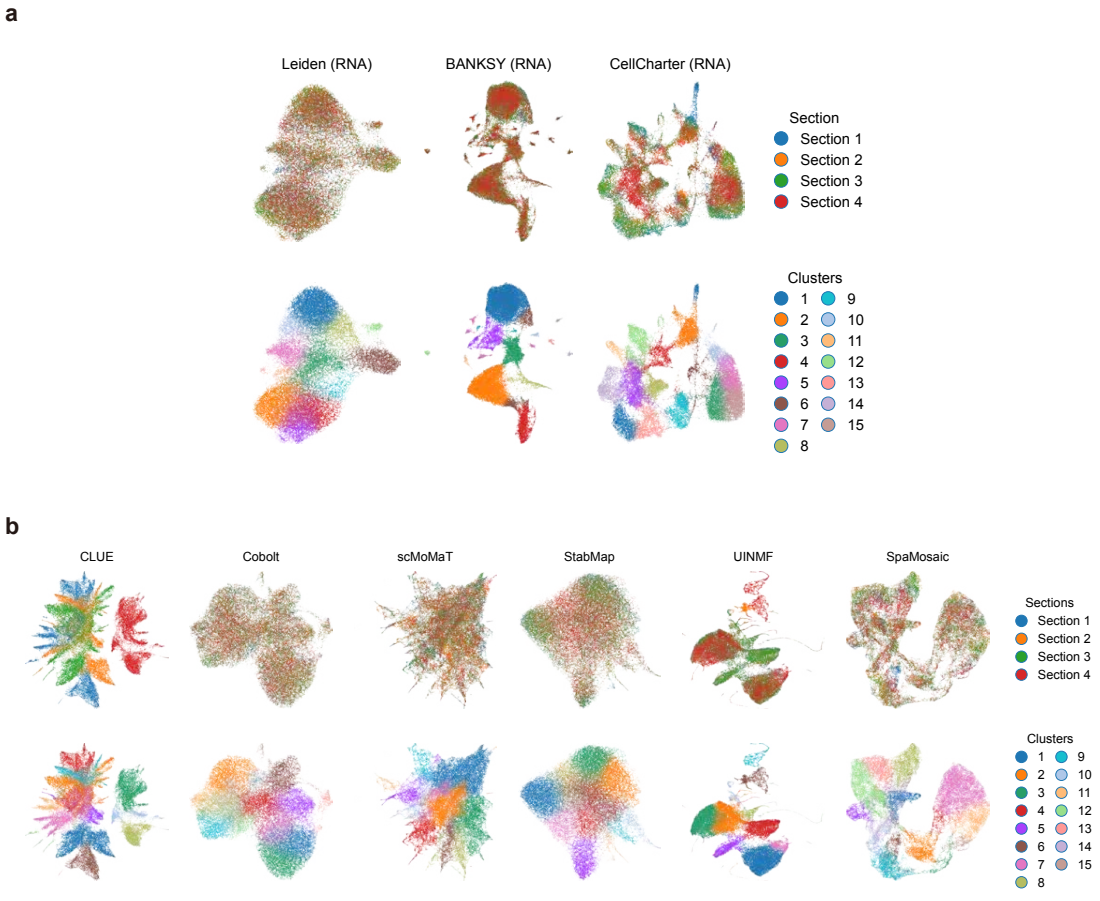
Supplementary Figure S22 Quantitative benchmarking and UMAP plots on human tonsil dataset. **a** ARI scores of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1). **b** PAS and CHAOS scores of benchmarked methods. Bars represent the mean values, with overlaid dots indicating individual sections (n = 3). **c** iLISI scores of benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1). **d** UMAP plots of embeddings from benchmarked methods. From top to bottom, spots are colored by section, cluster, and annotation, respectively. **e** FOSCTTM and MS scores of the benchmarked methods. Bars represent global scores calculated for the integrated dataset (n=1).



Supplementary Figure S23 Quantitative benchmarking on imputation performance. **a** Benchmarking of ATAC imputation on the embryonic mouse brain (Misar) and postnatal mouse brain datasets. Bars represent mean values, with overlaid dots indicating scores from a 3-fold cross-validation (n=3). **b** Benchmarking of ATAC imputation with smoothed ground truth peak count matrices. Bars represent mean values, with overlaid dots indicating scores from a 3-fold cross-validation (n=3) **c** Benchmarking of protein imputation on the human lymph node and tonsil datasets. Bars represent mean values, with overlaid dots indicating scores from a 3-fold cross-validation (n=3)

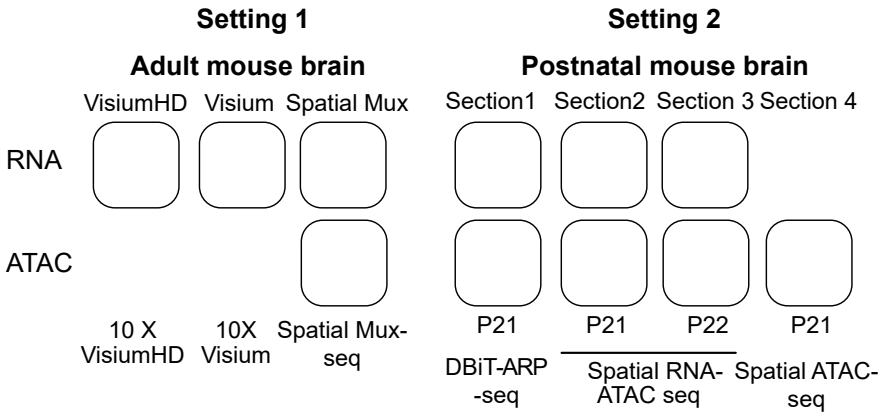


Supplementary Figure S24 Spatial plots of clustering results from baseline methods on the postnatal mouse brain (spRA+spRH) dataset.



Supplementary Figure S25 UMAP plots of postnatal mouse brain dataset (spRA+spRH) integration analysis. a UMAP plots of the embeddings from baseline methods. Spots in the first row are colored by section and the second row by cluster. **b** UMAP plots of the embeddings from benchmarked methods.

a



b

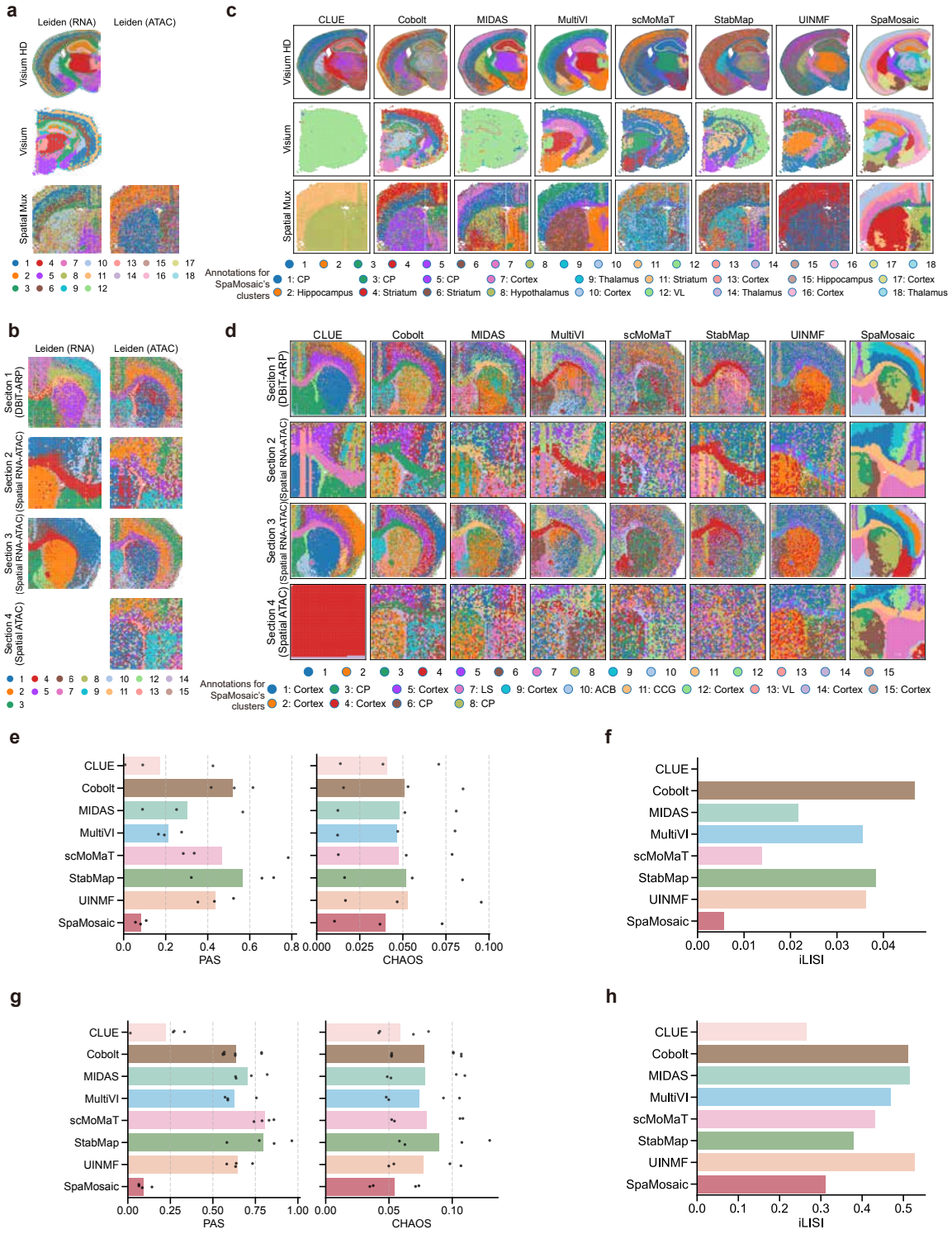
Setting 1

	Resolution	#Cell (Spot)
VisiumHD	16 (8×2) μm	98917
Visium	55 μm	9686
Spatial Mux	50 μm	2707

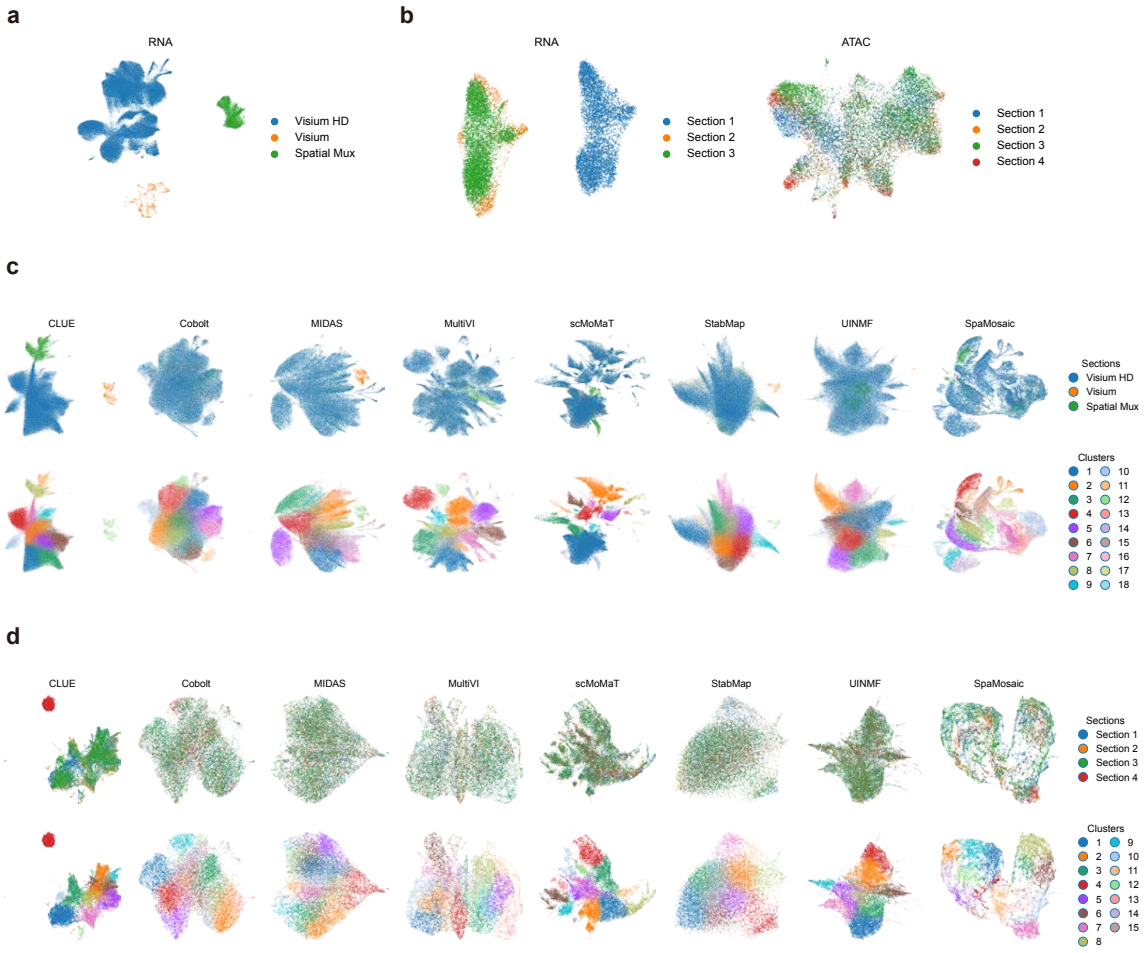
Setting 2

	Resolution	#Cell (Spot)	Technology
Section 1	20 μm	9787	DBiT-ARP-seq
Section 2	20 μm	2497	Spatial RNA-ATAC-seq
Section 3	20 μm	9215	
Section 4	20 μm	2499	Spatial ATAC-seq

Supplementary Figure S26 Dataset overview of scenarios 1 and 2 in cross-technology integration. **a** Configuration of data modalities for both integration scenarios. **b** Metadata for each section in the two scenarios, including resolutions and spot counts.

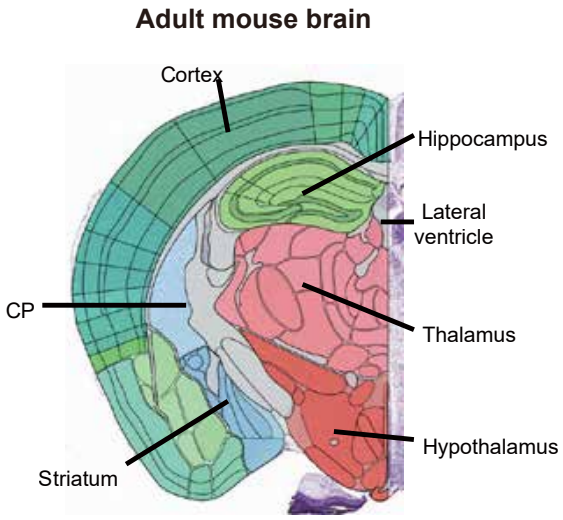

a
Leiden (RNA)
Leiden (ATAC)
Visium HD
Visium
Spatial Mux
c
CLUE
Cobolt
MIDAS
MultiVI
scMoMaT
StabMap
UINMF
SpaMosaic
Annotations for SpaMosaic's clusters
1: CP
2: Hippocampus
3: CP
4: Striatum
5: CP
6: Striatum
7: Cortex
8: Hypothalamus
9: Thalamus
10: Cortex
11: Striatum
12: VL
13: Cortex
14: Thalamus
15: Hippocampus
16: Cortex
17: Cortex
18: Thalamus
b
Leiden (RNA)
Leiden (ATAC)
Section 1 (DBiT-ARP)
Section 2 (Spatial RNA-ATAC)
Section 3 (Spatial RNA-ATAC)
Section 4 (Spatial ATAC)
d
CLUE
Cobolt
MIDAS
MultiVI
scMoMaT
StabMap
UINMF
SpaMosaic
Seciton 1 (DBiT-ARP)
Section 2 (Spatial RNA-ATAC)
Section 3 (Spatial RNA-ATAC)
Section 4 (Spatial ATAC)
Annotations for SpaMosaic's clusters
1: Cortex
2: Cortex
3: CP
4: Cortex
5: Cortex
6: CP
7: LS
8: CP
9: Cortex
10: ACB
11: CCG
12: Cortex
13: VL
14: Cortex
15: Cortex
e
CLUE
Cobolt
MIDAS
MultiVI
scMoMaT
StabMap
UINMF
SpaMosaic
0.0 0.2 0.4 0.6 0.8
PAS
0.000 0.025 0.050 0.075 0.100
CHAOS
f
0.00 0.01 0.02 0.03 0.04
iLISI
g
0.00 0.25 0.50 0.75 1.00
PAS
0.00 0.05 0.10
CHAOS
h
0.0 0.1 0.2 0.3 0.4 0.5
iLISI

Supplementary Figure S27 Cross-technology integration of mouse brain data across heterogeneous sizes and resolutions. **a** Mosaic dataset configuration and clustering patterns obtained for all three sections in scenario 1. Modality-specific clustering was performed jointly across relevant sections using the Leiden algorithm on batch-corrected representations. **b** Spatial plots of the clusters obtained from benchmarked methods for all three sections in scenario 1. Manual annotation was performed to identify SpaMosaic's clusters. **c** Mosaic dataset configuration and clustering patterns obtained for all four sections in scenario 2. Modality-specific clustering was performed jointly across relevant sections using the Leiden algorithm on batch-corrected representations. **d** Spatial plots of the clusters obtained from benchmarked methods for all four sections in scenario 2. Manual annotation was performed to identify SpaMosaic's clusters. **e** PAS and CHAOS scores of benchmarked methods in scenario 1. Bars represent the mean values, with overlaid dots indicating individual sections (n = 3). **f** iLISI scores in scenario 1. Bars represent global scores calculated for the integrated dataset (n=1). **g** PAS and CHAOS scores of benchmarked methods in scenario 2. Bars represent the mean values, with overlaid dots indicating individual sections (n = 4). **h** iLISI scores in scenario 2. Bars represent global scores calculated for the integrated dataset (n=1).

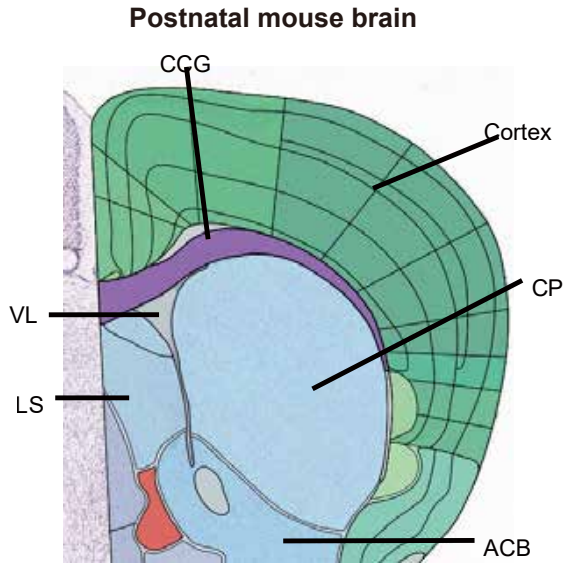


Supplementary Figure S28 UMAP plots of cross-technology integration analysis. **a** UMAP plots of the unintegrated RNA profiles in scenario 1. Spots are colored by section labels. **b** UMAP plots of the unintegrated RNA and ATAC profiles in scenario 2. **c** UMAP plots of the embeddings from all mosaic integration methods in scenario 1. **d** UMAP plots of the embeddings from all mosaic integration methods in scenario 2.

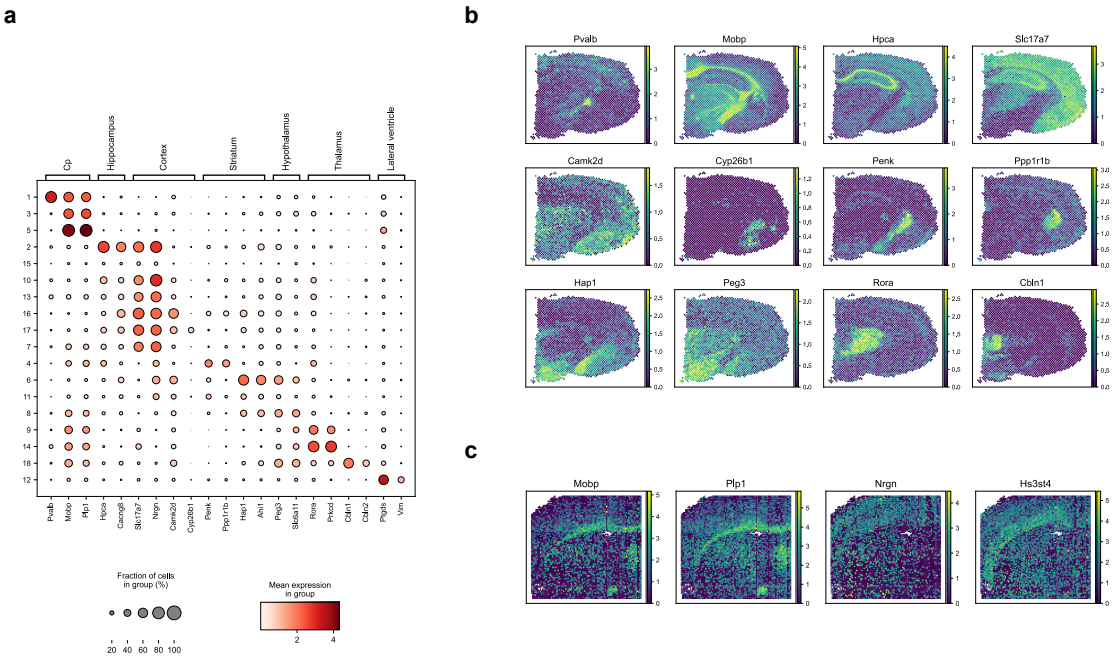
a



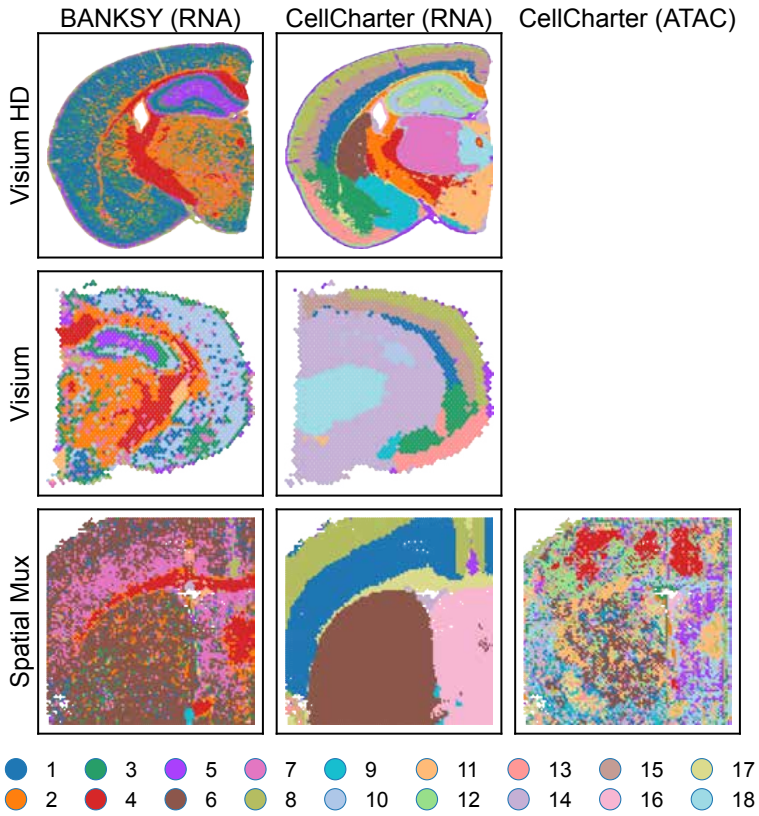
b



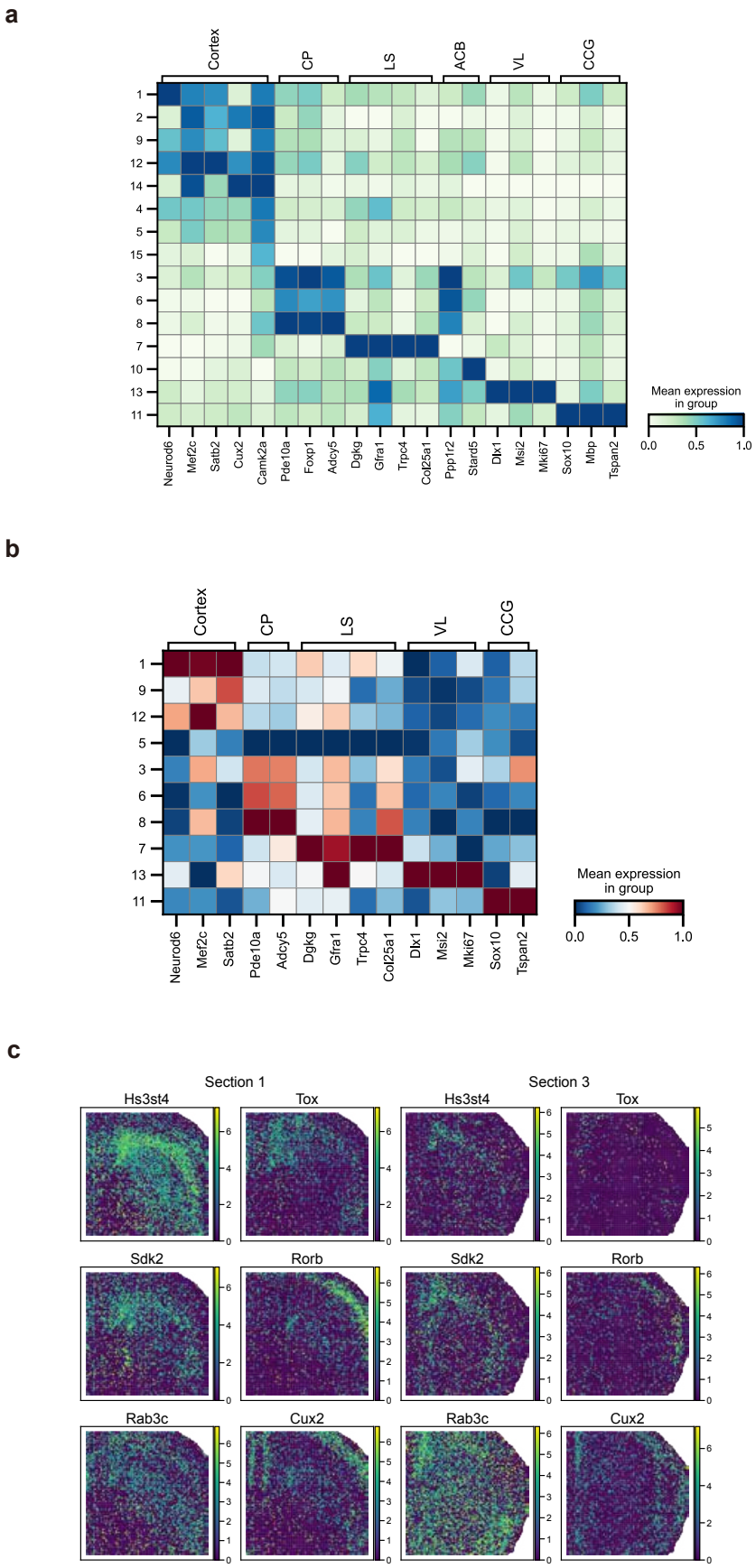
Supplementary Figure S29 Reference atlas of mouse brain for cross-technology integration analysis. **a** Reference atlas of adult mouse brain structures. **b** Reference atlas of postnatal mouse brain structures. Both images were from the Allen Reference Atlas (https://atlas.brain-map.org/)



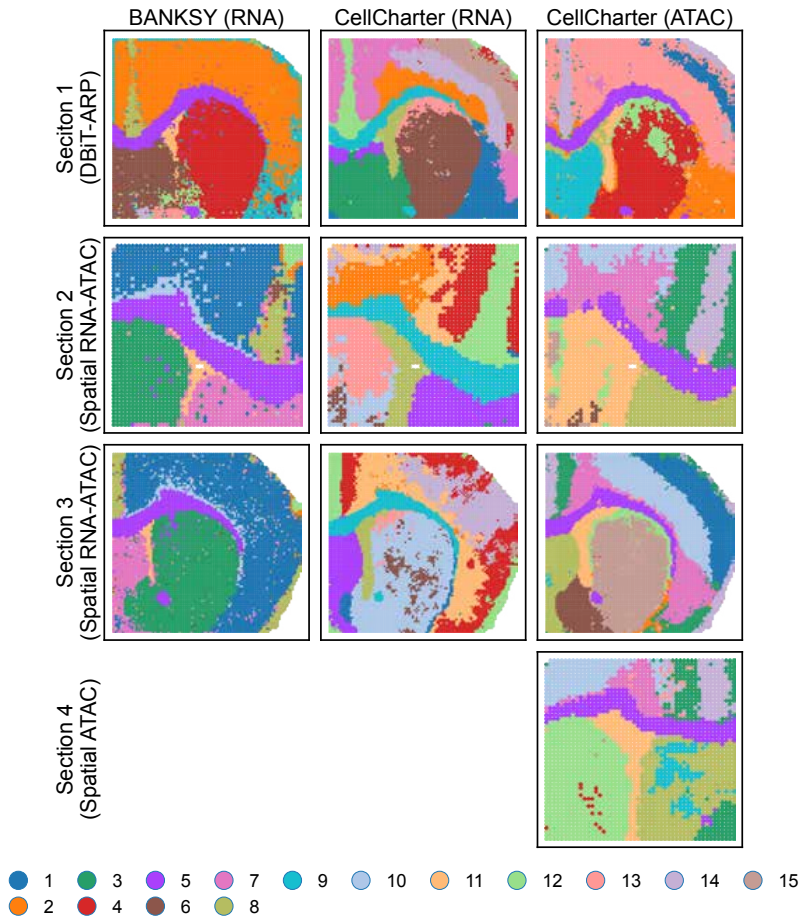
Supplementary Figure S30 Marker gene validation in cross-technology integration (Scenario 1). **a** Dot plot of region-specific marker gene expression in SpaMosaic's clusters. **b** Spatial heatmap of RNA expression for selected marker genes in the 10x Genomics Visium section. **c** Spatial plots of RNA expression for selected marker genes in the Spatial Mux-seq section.



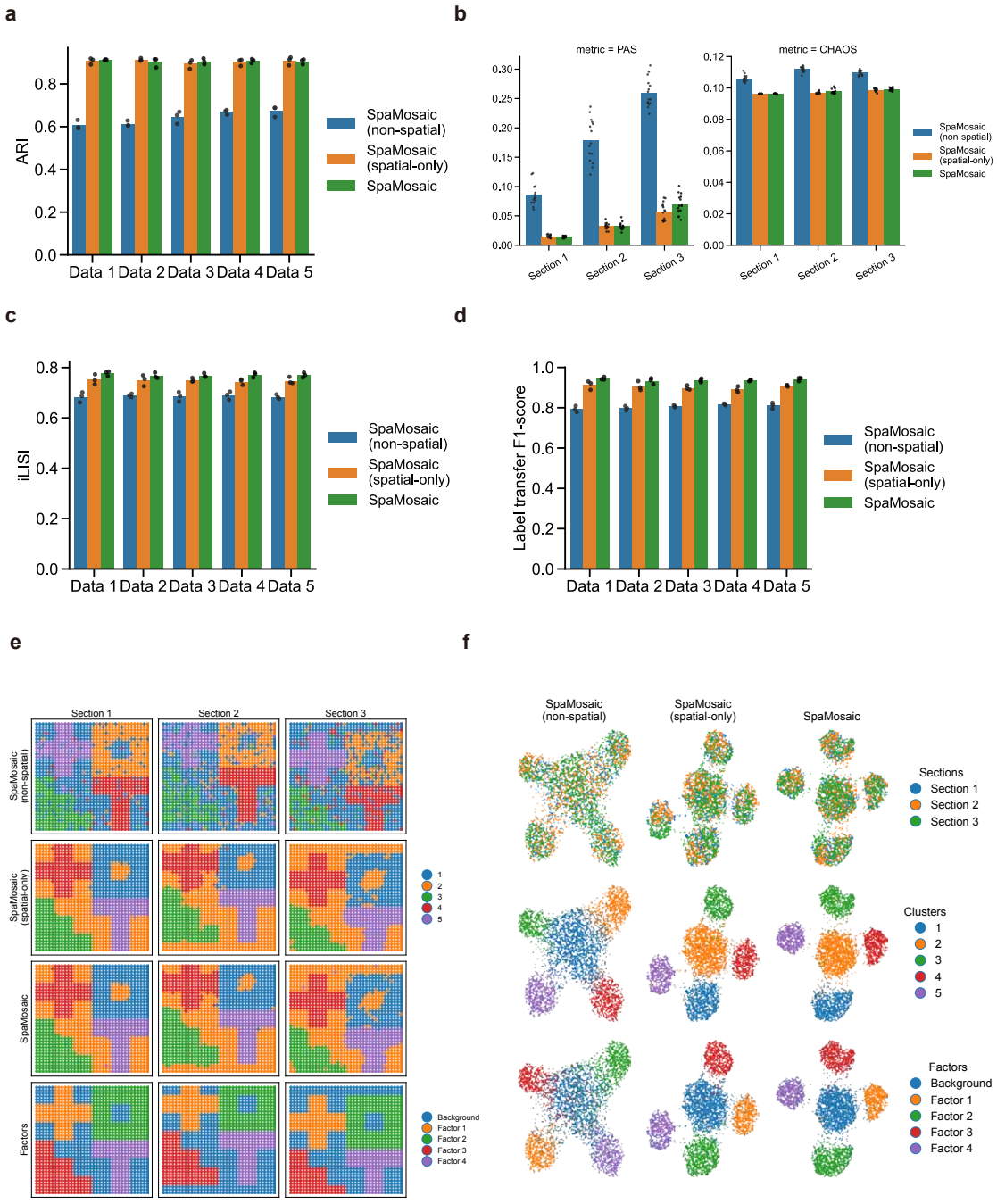
Supplementary Figure S31 Spatial plots of clustering results from BANKSY and CellCharter in cross-technology integration (Scenario 1).



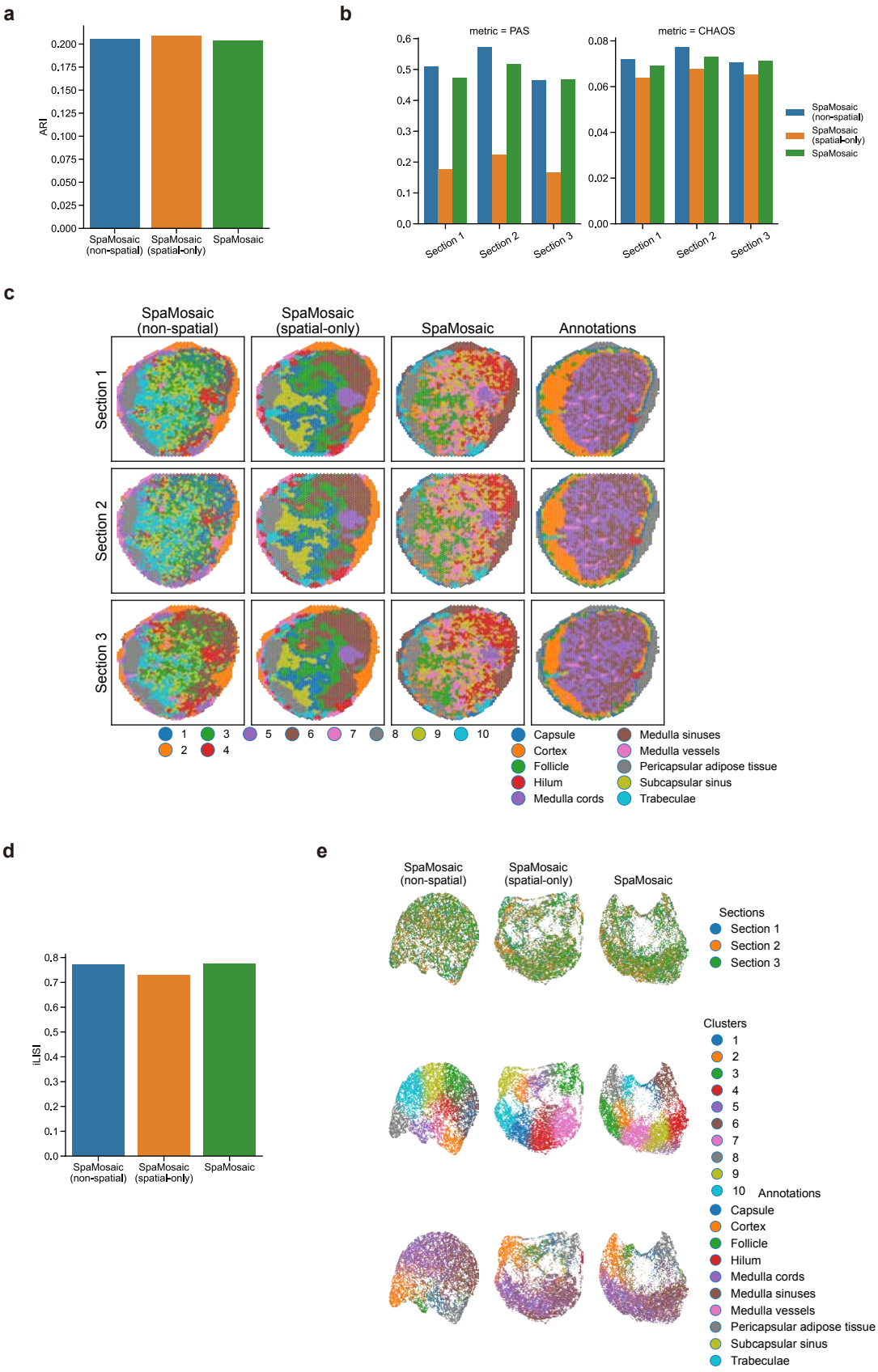
Supplementary Figure S32 Marker gene expression and gene activity validation in cross-technology integration (Scenario 2). **a** Average RNA expression of region-specific marker genes in SpaMosaic's clusters. **b** Average gene activity scores of region-specific marker genes in SpaMosaic's clusters. **c** Spatial plots of RNA expression for cortical layer-specific markers (*Hs3st4*, *Tox*, *Sdk2* for deeper layers; *Rorb*, *Rab3c*, *Cux2* for upper layers) in sections 1 and 3.



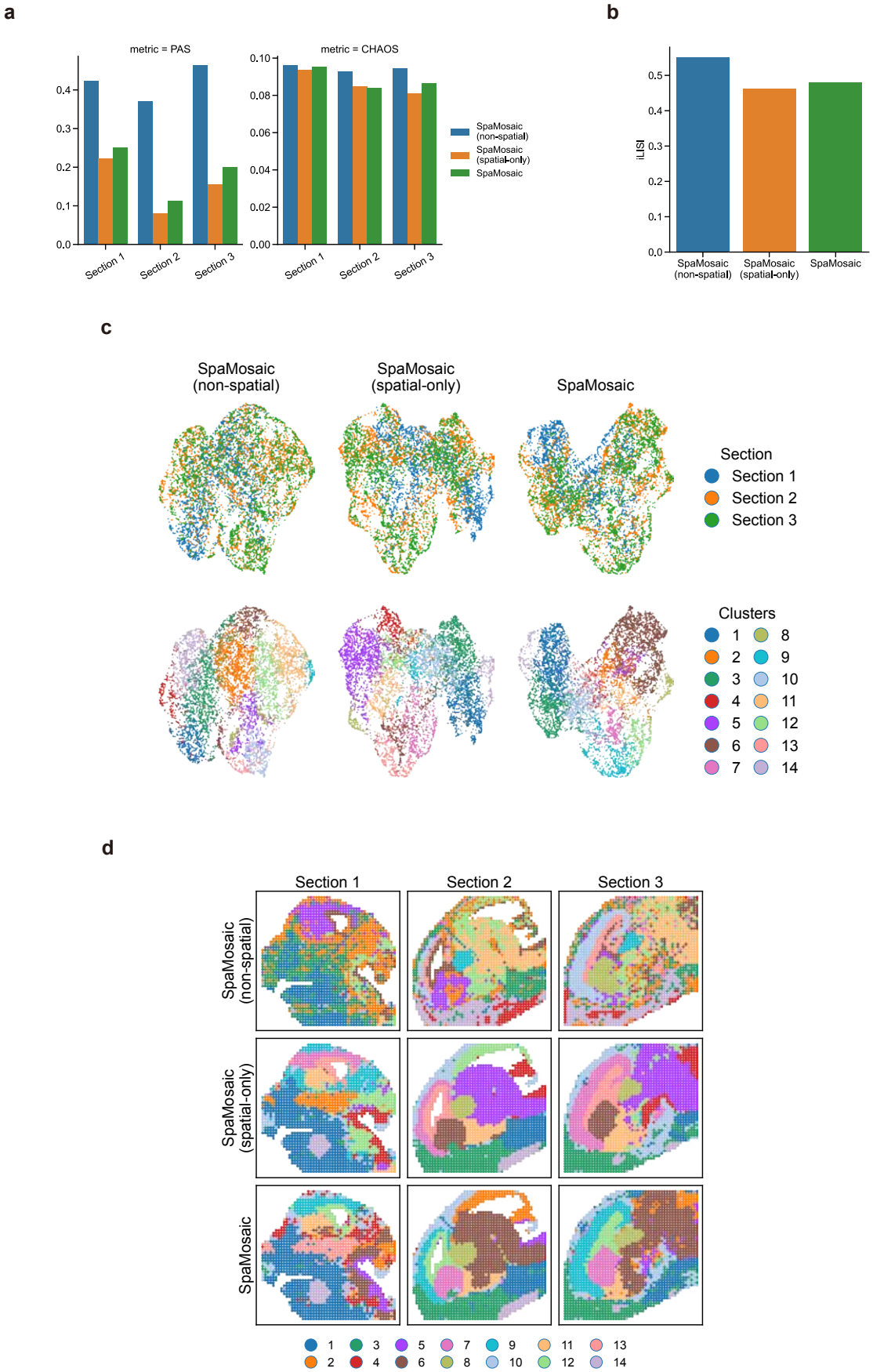
Supplementary Figure S33 Spatial plots of clustering results from BANKSY and CellCharter in cross-technology integration (Scenario 2).



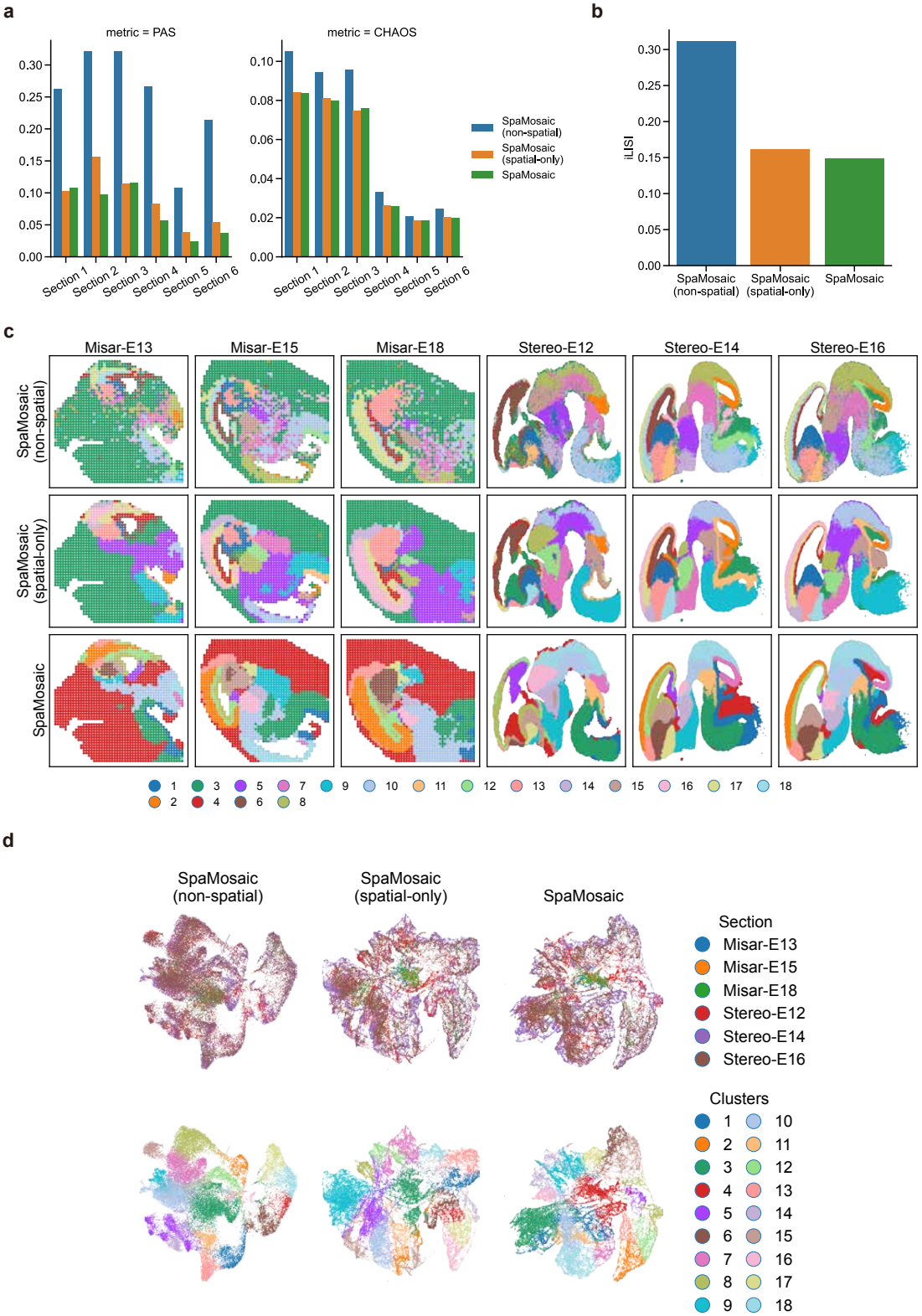
Supplementary Figure S34 Ablation study on simulation datasets. **a** Comparison of ARI scores among the three model variants on the simulation datasets. Bars represent the mean values derived from three replicates, with overlaid dots indicating the individual data points (n=3). **b** PAS and CHAOS scores of model variants. Bars represent mean scores for each section. Dots indicate individual values from five simulation cases with three independent replicates each (n=15 per bar). **c** iLISI scores of model variants. Bars represent the mean values derived from three replicates, with overlaid dots indicating the individual data points (n=3). **d** Label transfer F1-scores of model variants. Bars represent the mean values derived from three replicates, with overlaid dots indicating the individual data points (n=3). **e** Spatial plots of clustering results. The last row represents the ground truth of factor patterns. **f** UMAP plots of embeddings from the variants. From top to bottom, spots are colored by section, cluster, and factor labels.



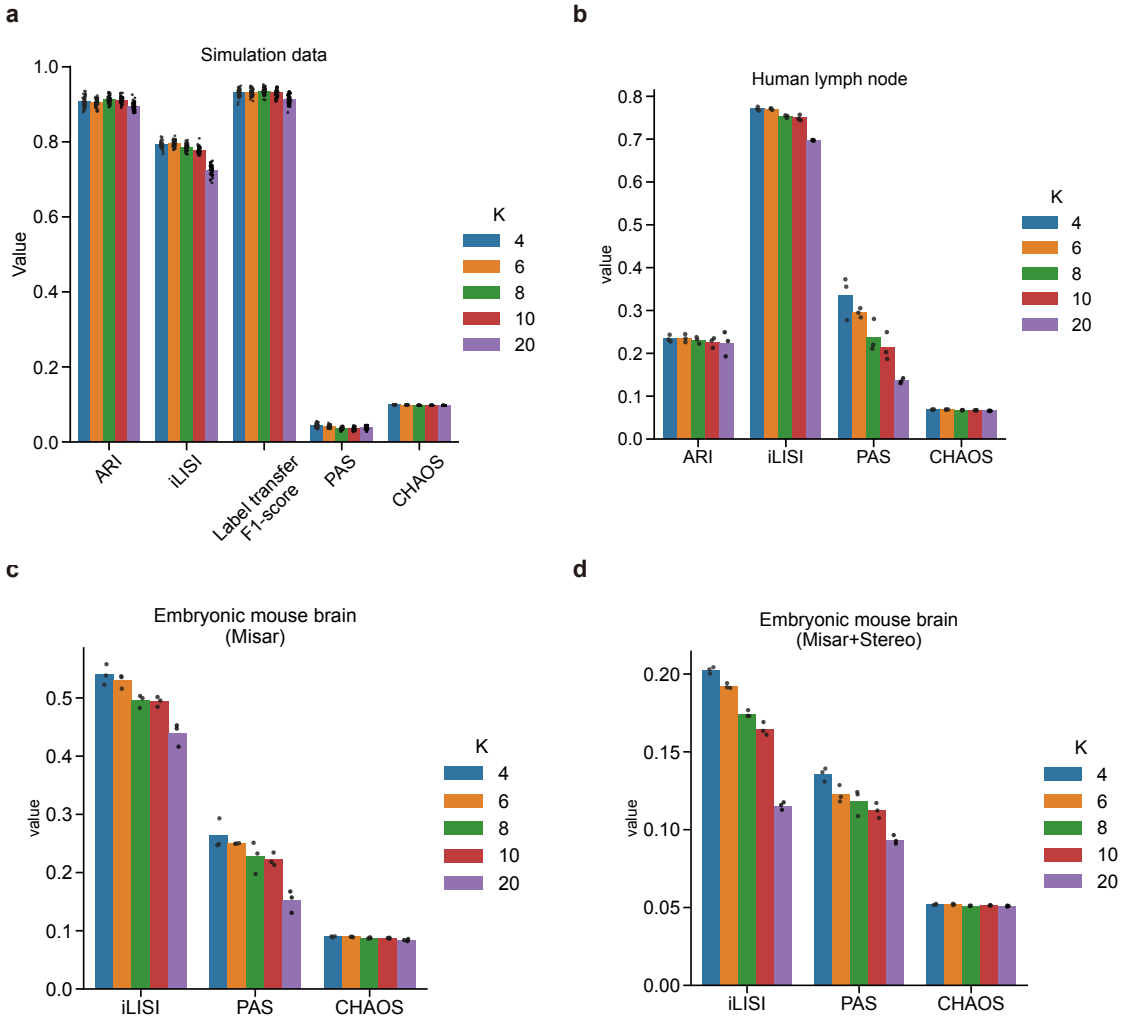
Supplementary Figure S35 Ablation study on the human lymph node dataset. **a** Comparison of ARI scores among the three model variants on the human lymph node dataset. Bars represent global scores calculated for the integrated dataset (n=1). **b** PAS and CHAOS scores of model variants. Bars represent metric values calculated individually for each section (n=1). **c** Spatial plots of clustering results. The last column depicts the manual annotation. **d** iLISI scores of model variants. Bars represent global scores calculated for the integrated dataset (n=1). **e** UMAP plots of embeddings from the variants. Spots are colored by section (top) and cluster (bottom).



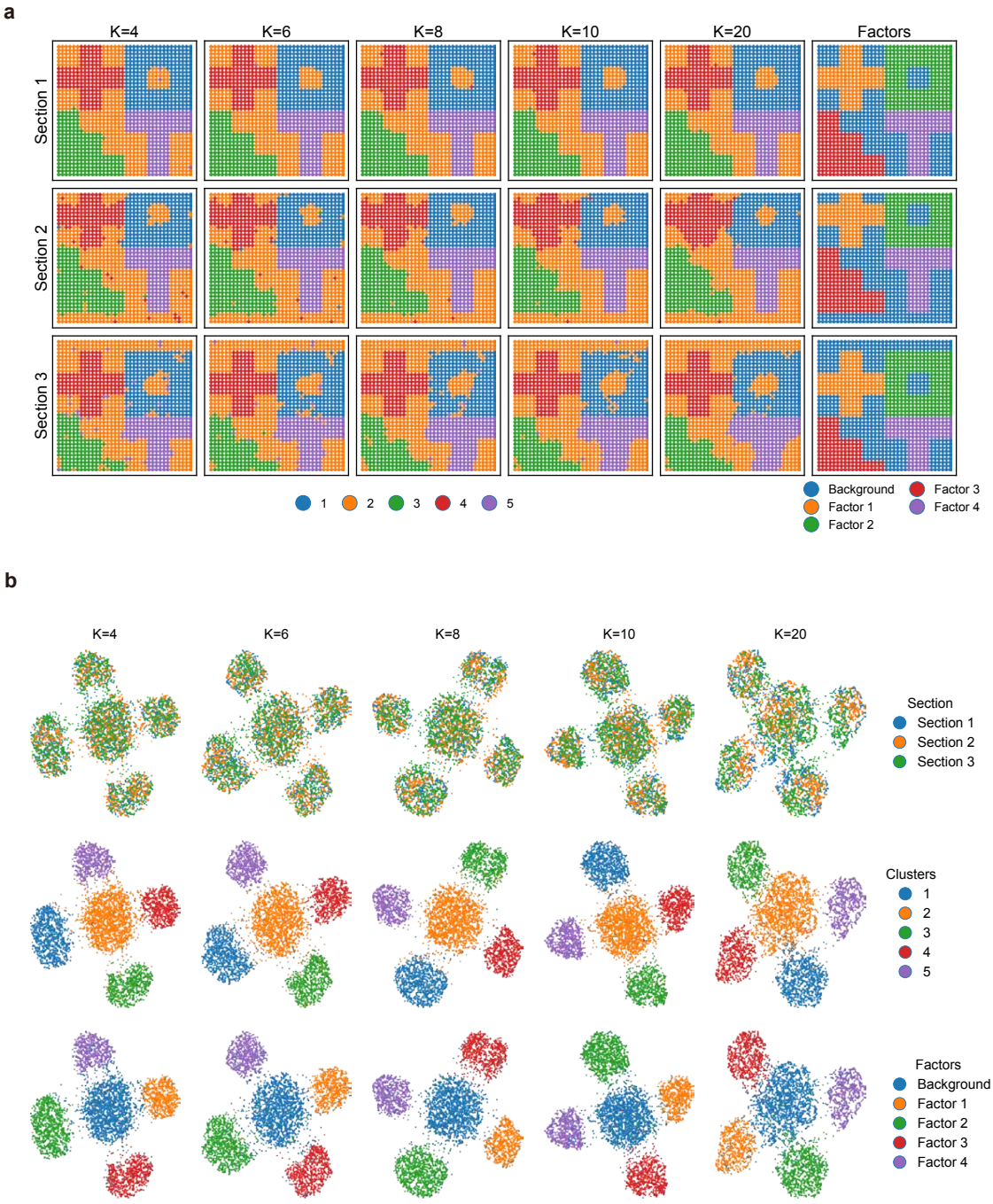
Supplementary Figure S36 Ablation study on the embryonic mouse brain dataset (Misar). **a** PAS and CHAOS scores of the three model variants on the embryonic mouse brain dataset (Misar). Bars represent metric values calculated individually for each section (n=1). **b** iLISI scores of model variants. Bars represent global scores calculated for the integrated dataset (n=1). **c** UMAP plots of embeddings from the variants. Spots are colored by section (top) and cluster (bottom). **d** Spatial plots of clustering results.



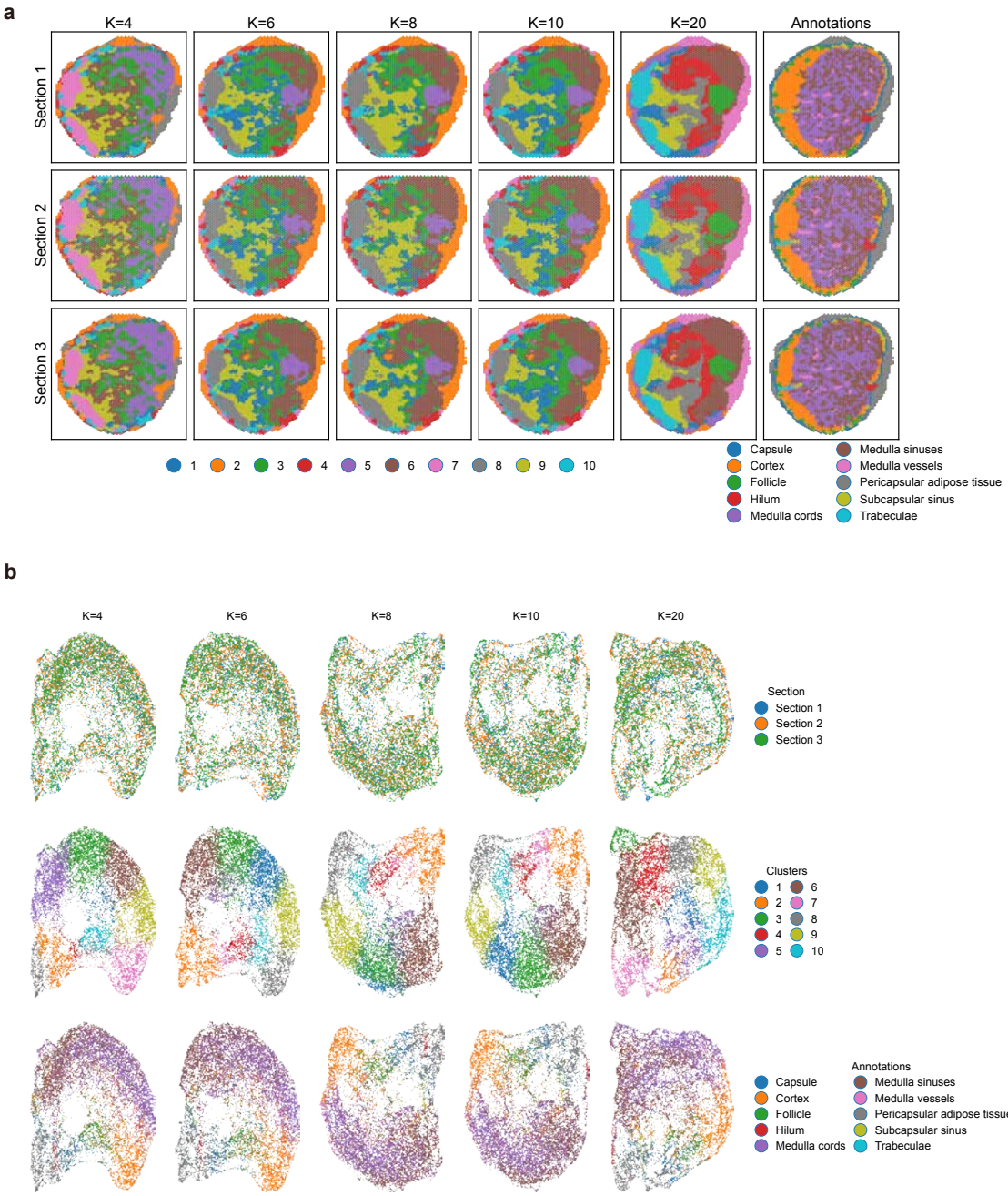
Supplementary Figure S37 Ablation study on the embryonic mouse brain dataset (Misar+Stereo). **a** PAS and CHAOS scores of the three model variants on the embryonic mouse brain dataset (Misar+Stereo). Bars represent metric values calculated individually for each section (n=1). **b** iLISI scores of model variants. Bars represent global scores calculated for the integrated dataset (n=1). **c** Spatial plots of clustering results. **d** UMAP plots of embeddings from the variants. Spots are colored by section (top) and cluster (bottom).



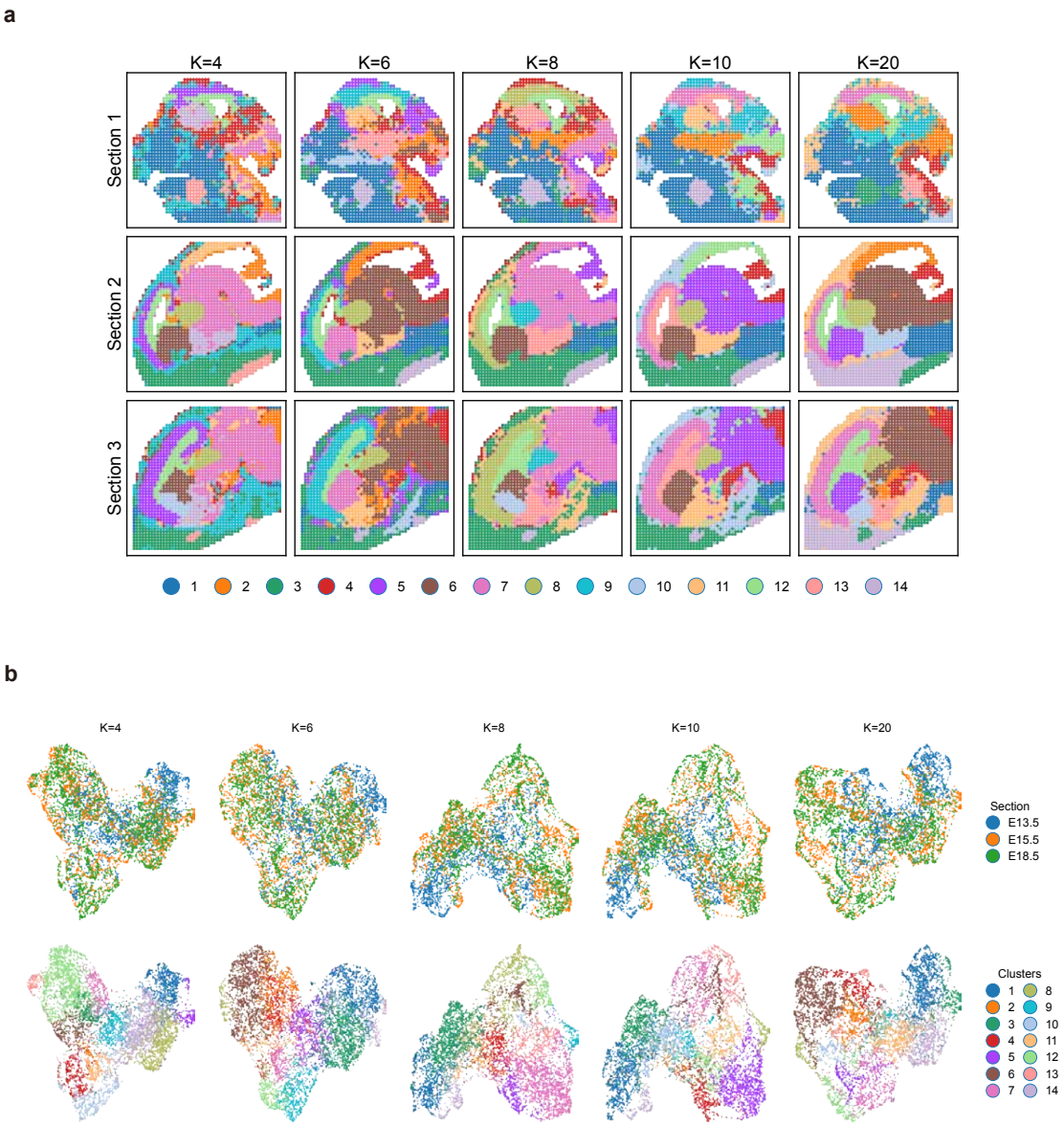
Supplementary Figure S38 Quantitative evaluation of parameter k sensitivity in spatial adjacency graph construction. **a** Benchmarking metrics computed on the simulation datasets across varying values of parameter K. Bars represent the mean values for ARI, iLISI, label transfer F1-score, PAS, and CHAOS. Overlaid dots indicate data points calculated from five simulation cases, each comprising three independent replicates with three random model initializations (total n=45 data points per bar). For PAS and CHAOS, each data point represents the average score calculated across the three slices. **b** Benchmarking metrics computed on the human lymph node dataset across varying values of parameter K. Bars represent the mean values for ARI, iLISI, PAS, and CHAOS. Overlaid dots indicate data points derived from three random model initializations (n=3). For PAS and CHAOS, each data point represents the average score calculated across the three slices. **c** Benchmarking metrics computed on the embryonic mouse brain dataset (Misar) across varying values of parameter K. Bars represent the mean values for iLISI, PAS, and CHAOS. Overlaid dots indicate data points derived from three random model initializations (n=3). For PAS and CHAOS, each data point represents the average score calculated across the three slices. **d** Benchmarking metrics computed on the embryonic mouse brain dataset (Misar+Stereo) across varying values of parameter K. Bars represent the mean values for iLISI, PAS, and CHAOS. Overlaid dots indicate data points derived from three random model initializations (n=3). For PAS and CHAOS, each data point represents the average score calculated across the six slices.



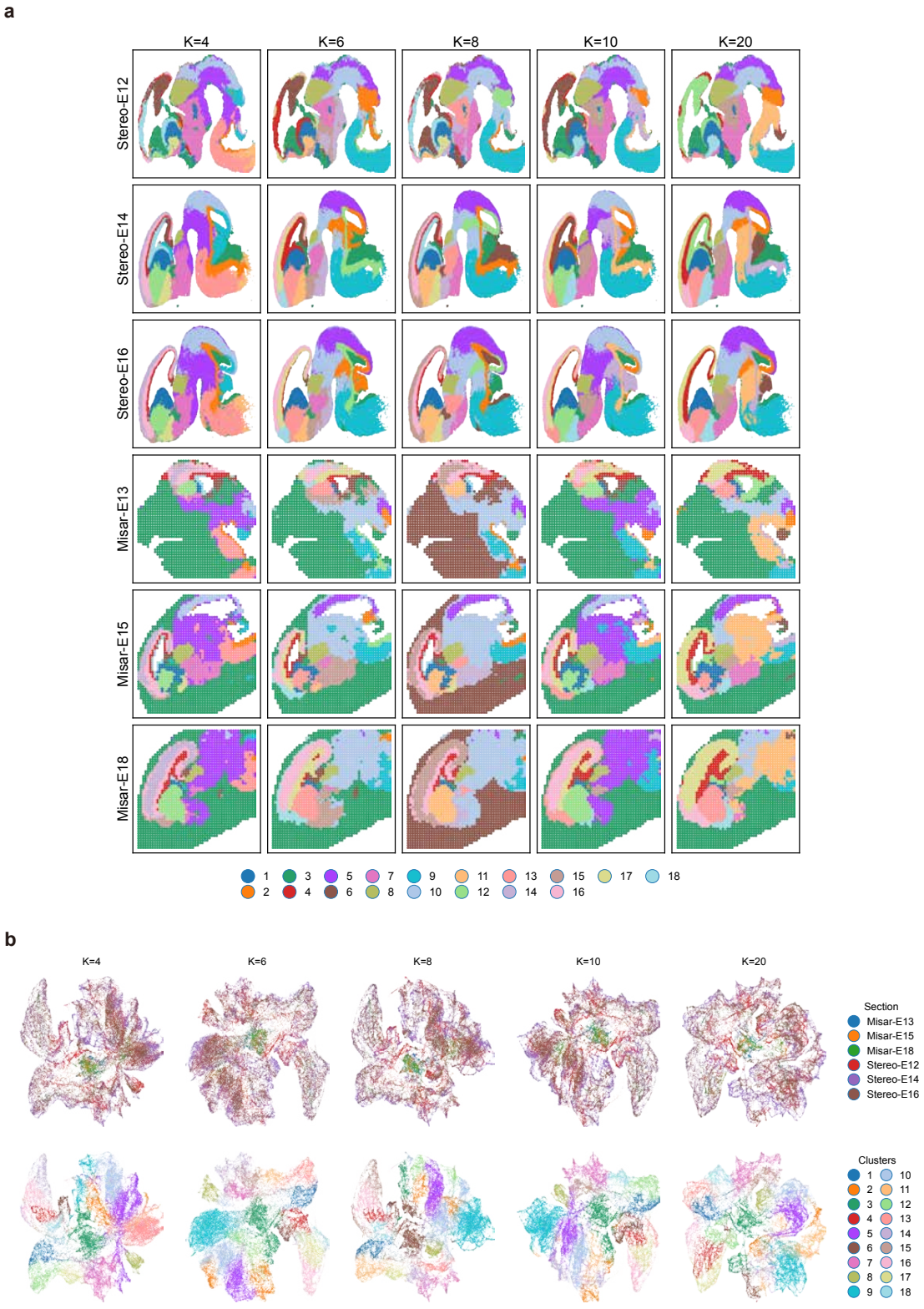
Supplementary Figure S39 Visualization of parameter k impact on the first replicate of simulated datasets. a Spatial plots of clustering results from SpaMosaic with different k values. The rightmost column shows the ground truth of factor patterns. **b** UMAP plots of embeddings from SpaMosaic with different k values. From top to bottom, spots are colored by section, cluster and factor labels.



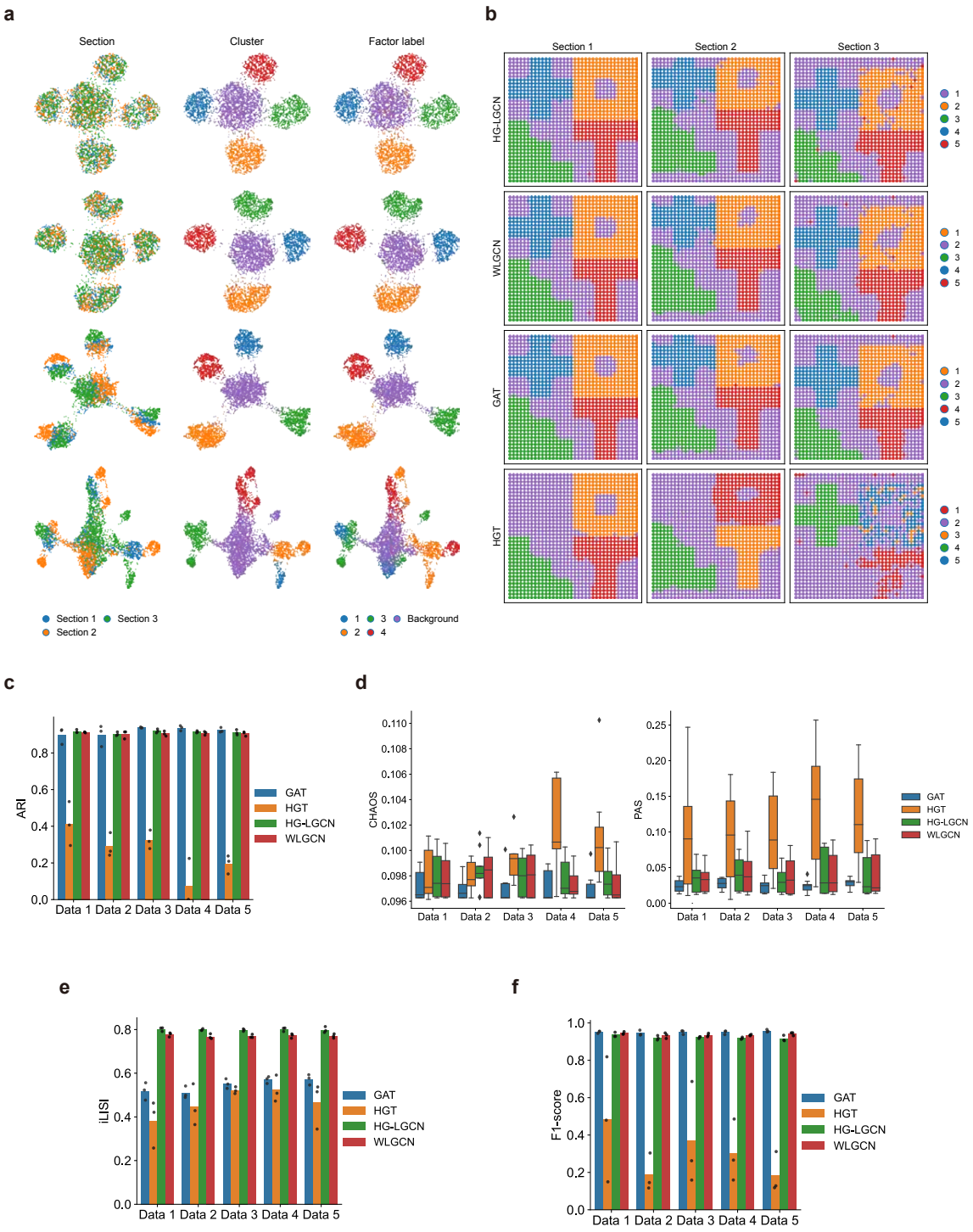
Supplementary Figure S40 Visualization of parameter k impact on the human lymph node dataset. a Spatial plots of clustering results from SpaMosaic with different k values. The rightmost column shows the expert manual annotations. **b** UMAP plots of embeddings from SpaMosaic with different k values. From top to bottom, spots are colored by section, cluster and annotation labels.



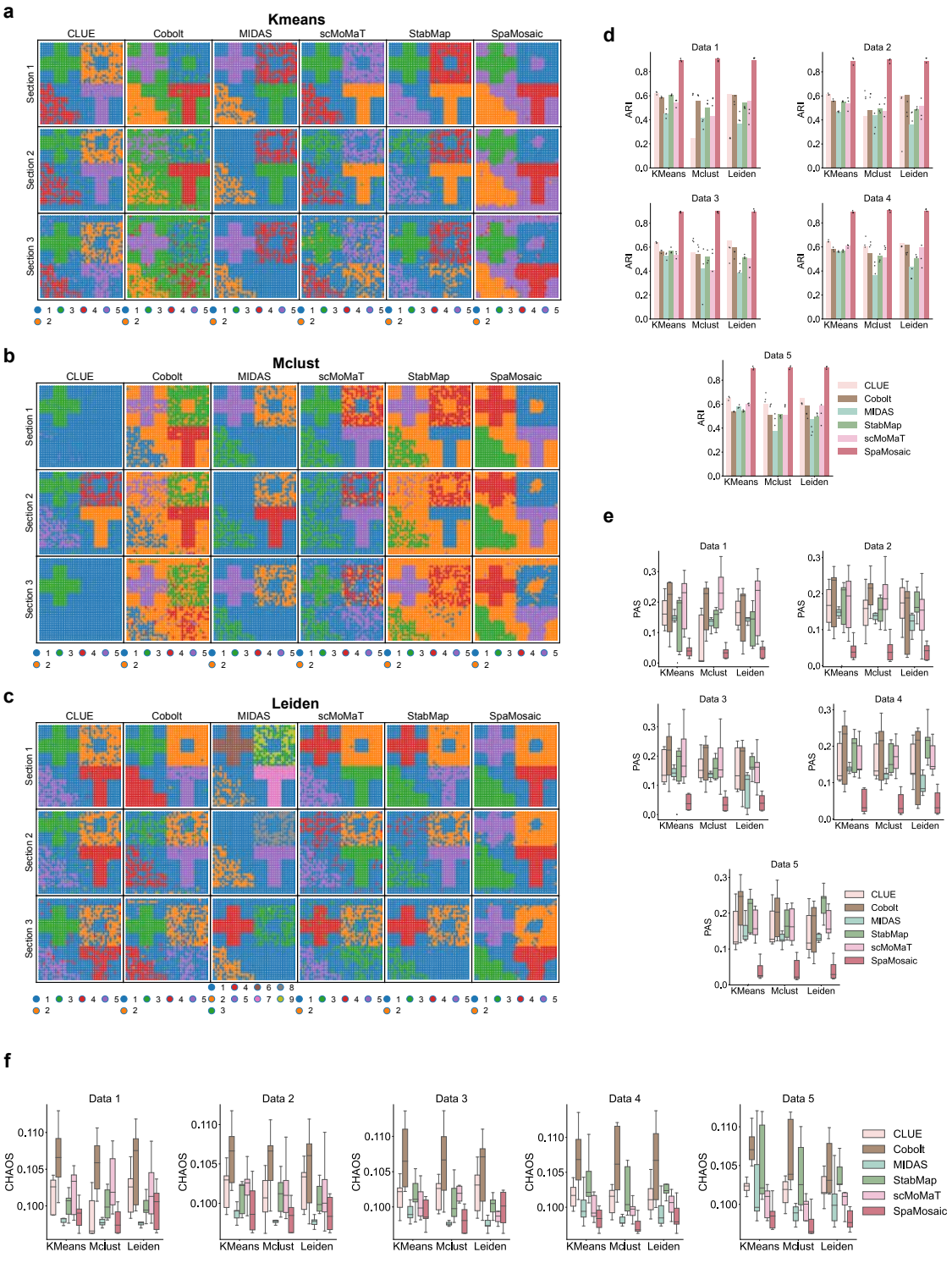
Supplementary Figure S41 Visualization of parameter k impact on the embryonic dataset (Misar). a Spatial plots of clustering results from SpaMosaic with different k values. **b** UMAP plots of embeddings from SpaMosaic with different k values. From top to bottom, spots are colored by section and cluster labels.



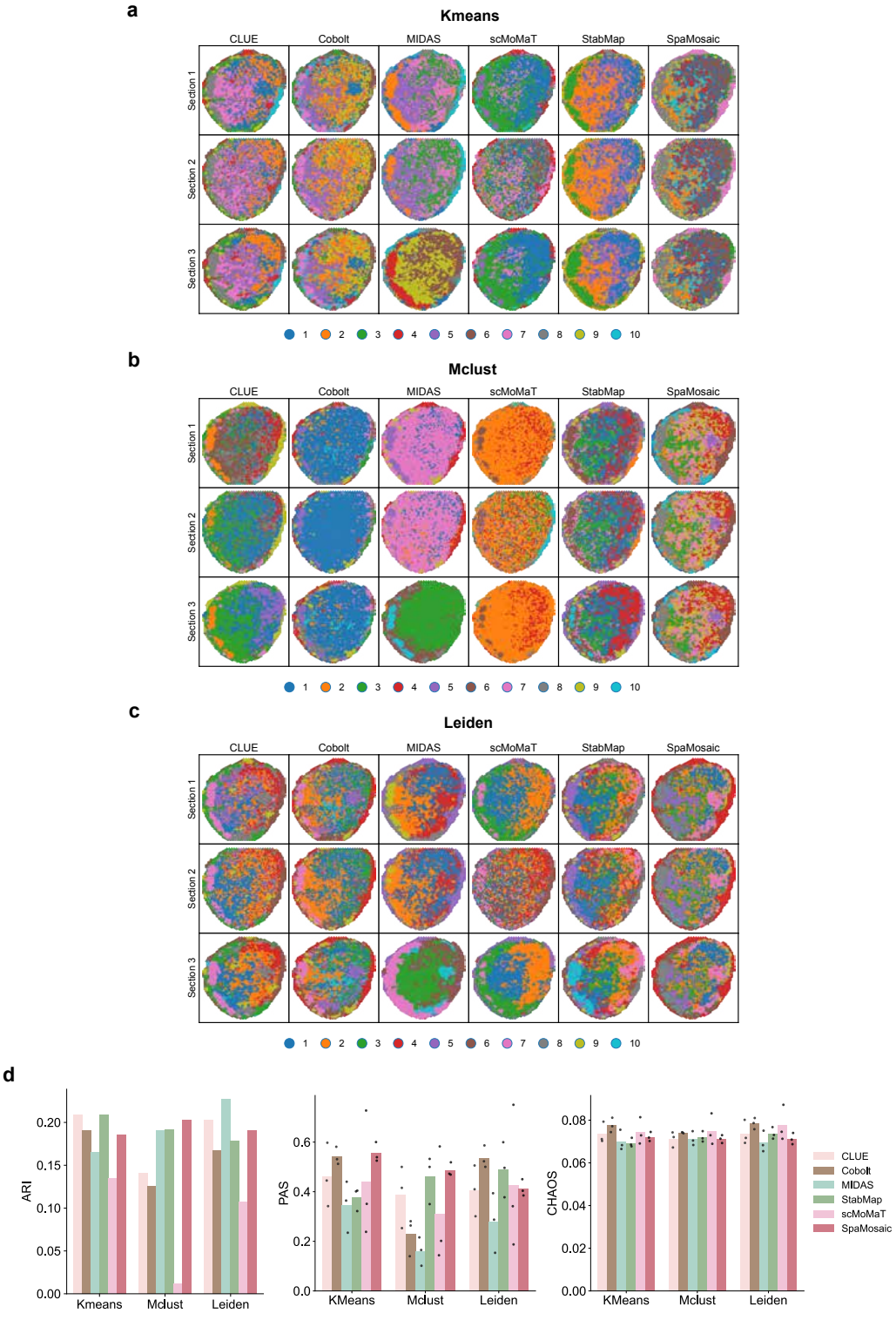
Supplementary Figure S42 Visualization of parameter k impact on the embryonic dataset (Misar+Stereo). a Spatial plots of clustering results from SpaMosaic with different k values. **b** UMAP plots of embeddings from SpaMosaic with different k values. From top to bottom, spots are colored by section and cluster labels.



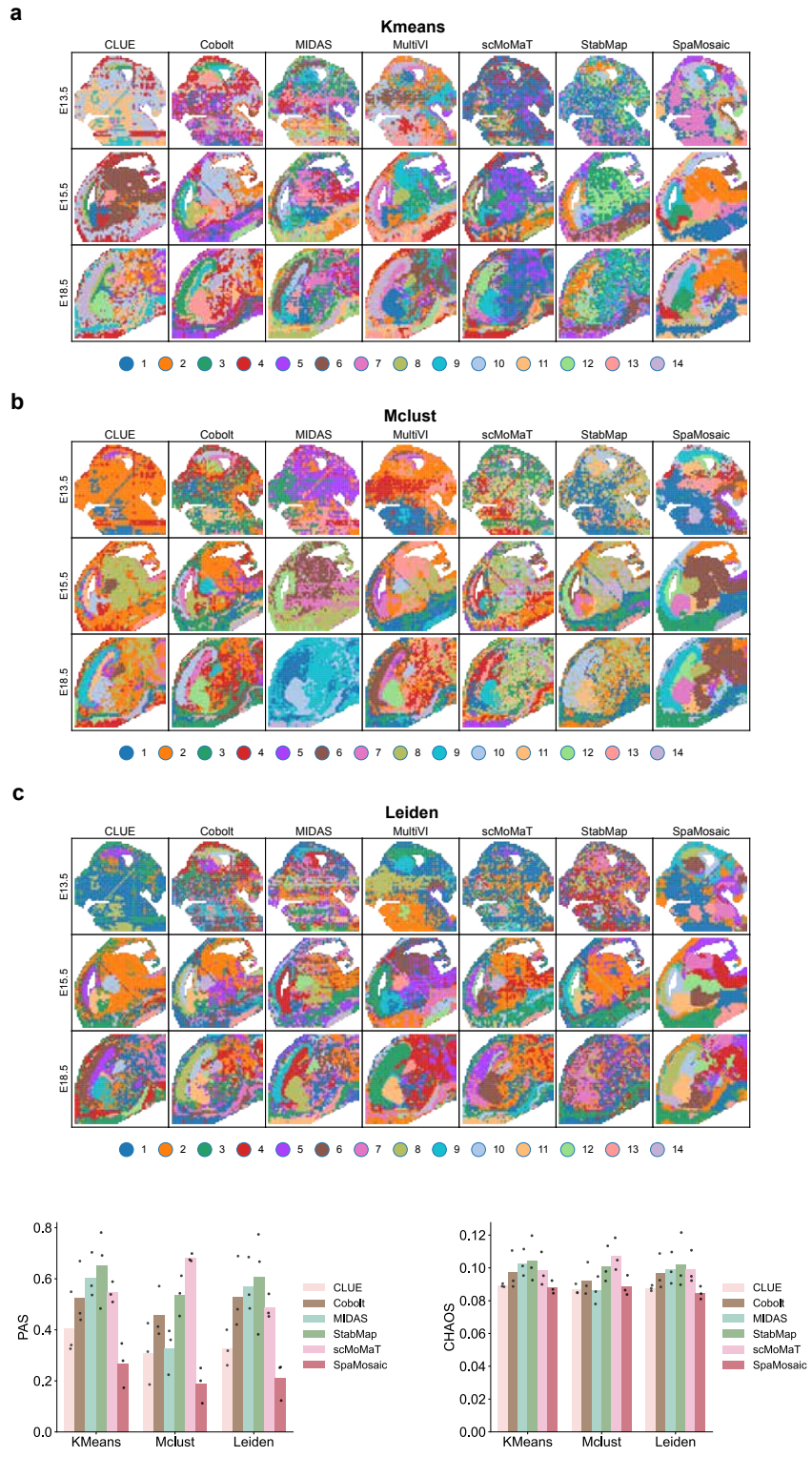
Supplementary Figure S43 Benchmarking different graph neural network architectures. **a** UMAP plots of embeddings generated with the different GNN architectures on the first replicate dataset of the first simulation setting. From left to right, the spots are colored by section, cluster and ground truth factor labels. **b** Spatial plots of clusters obtained with the different GNN architectures. The spot colors are matched with the cluster colors in **a**. **c** ARI scores of different GNN architectures. Bars represent the mean values, with overlaid dots indicating individual replicates (n = 3). **d** PAS and CHAOS scores of different GNN architectures across simulation datasets. Box plots display the distribution of scores derived from three independent replicates, each consisting of three slices (total n=9 data points per box). Center lines indicate the median; box limits represent the interquartile range (IQR); whiskers extend to 1.5× IQR; dots indicate outliers beyond 1.5×IQR. **e** iLISI scores of different GNN architectures. Bars represent the mean values, with overlaid dots indicating individual replicates (n = 3). **f** Label transfer F1-scores of different GNN architectures across five simulation datasets. Bars represent the mean values, with overlaid dots indicating individual replicates (n = 3). Each replicate data point is calculated as the average of reciprocal transfers scores between two single-modal slices.



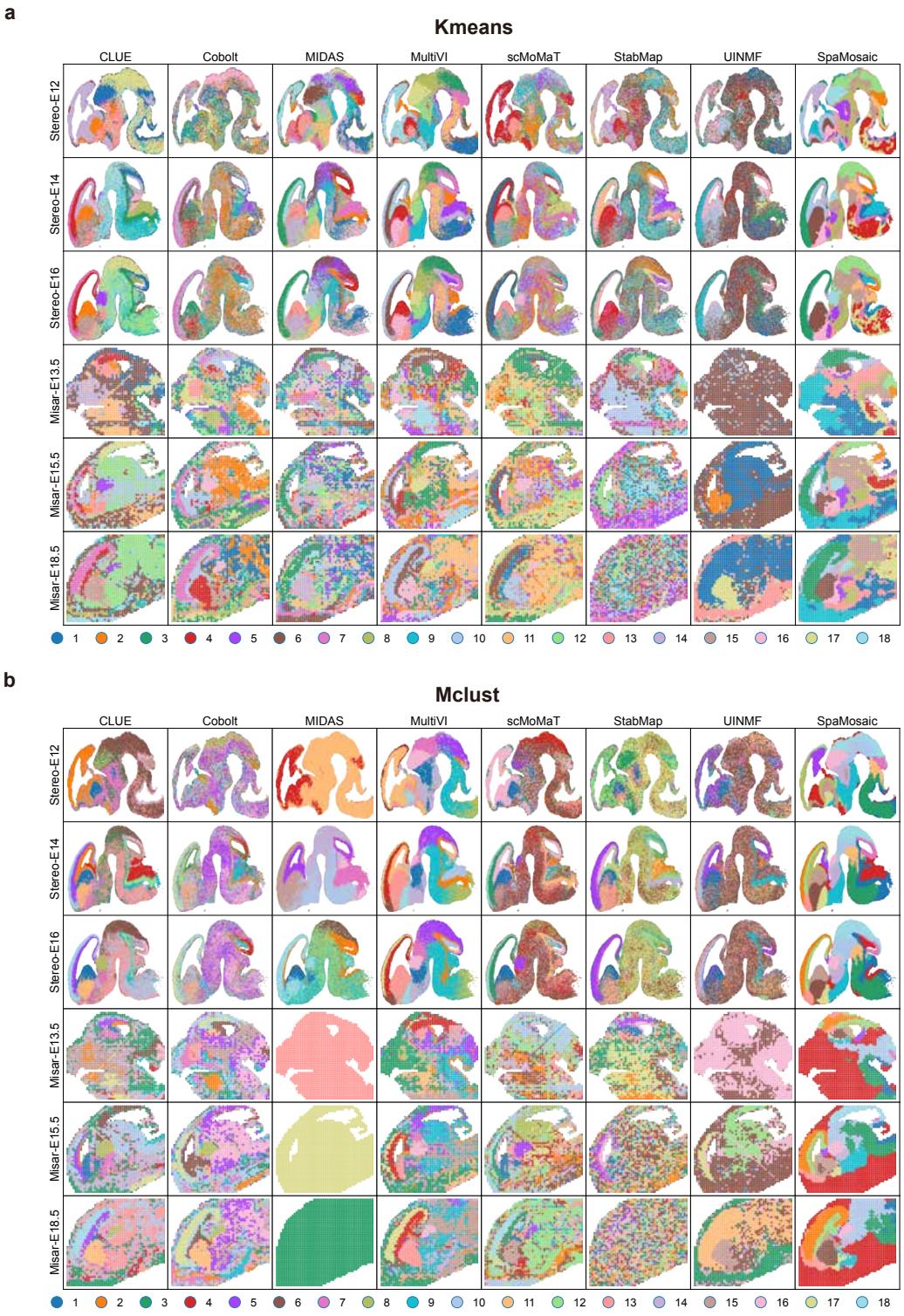
Supplementary Figure S44 Benchmarking of mosaic integration methods using different clustering algorithms on simulated datasets. a-c Spatial plots of clustering results from benchmarked mosaic integration methods using Kmeans (**a**), Mclust (**b**), and Leiden (**c**) for clustering. **d-f** Performance comparison using ARI (**d**), PAS (**e**) and CHAOS (**f**) metrics. **d** Bar plots showing ARI scores across five simulation cases. Bars represent the mean values, with overlaid dots indicating three independent replicates (n=3). **e, f** Box plots displaying PAS (**e**) and CHAOS (**f**) scores derived from three independent replicates, each consisting of three slices (total n=9 data points per box). Center lines indicate the median; box limits represent the interquartile range (IQR); whiskers extend to 1.5× IQR.



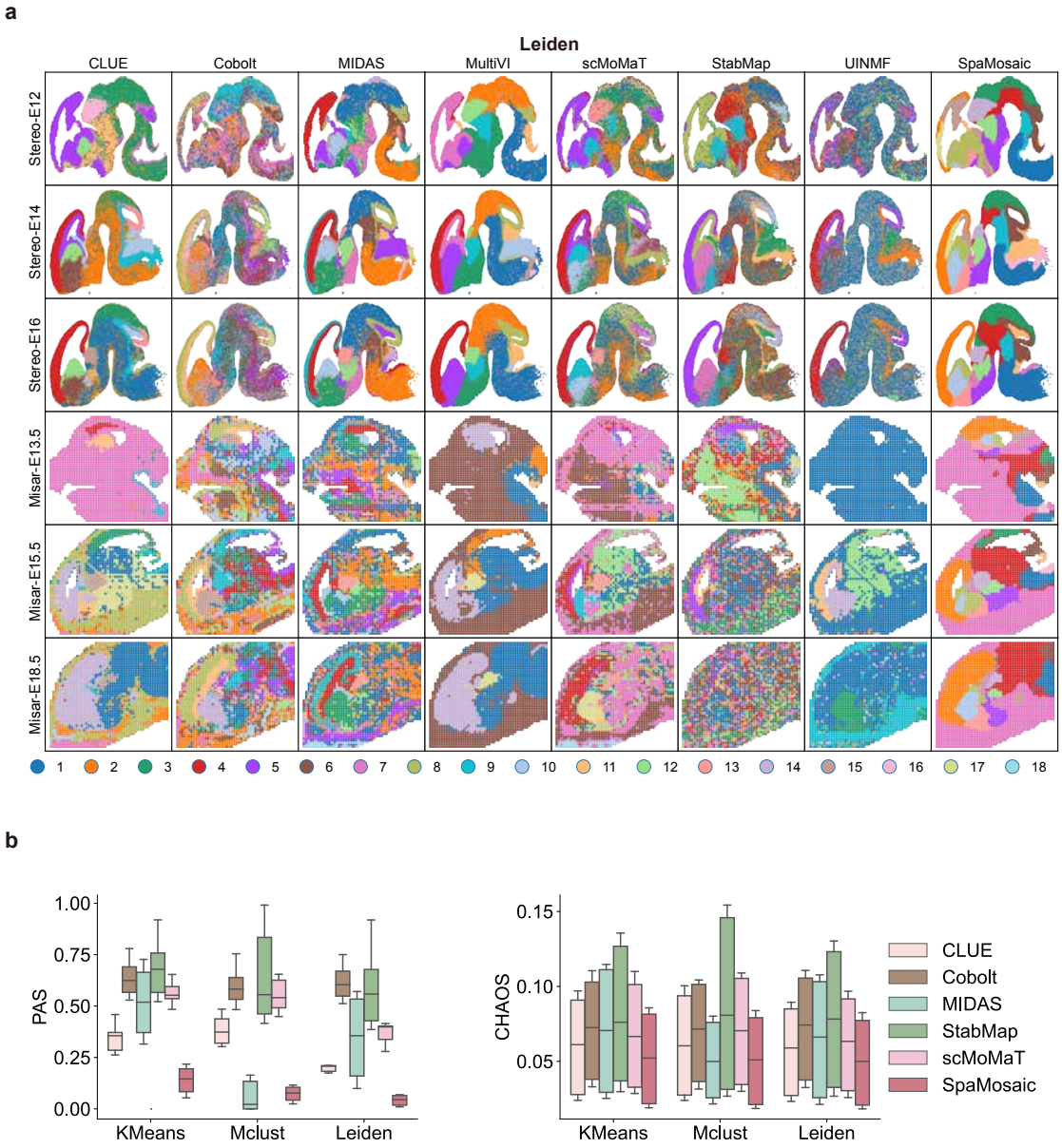
Supplementary Figure S45 Benchmarking of mosaic integration methods using different clustering algorithms on the human lymph node dataset. **a-c** Spatial plots of clustering results from benchmarked mosaic integration methods using Kmeans (**a**), Mclust (**b**), and Leiden (**c**) for clustering. **d** Performance comparison using ARI, PAS and CHAOS metrics. The bar plot for ARI represents scores derived from the entire dataset (n=1). The bars of PAS and CHAOS represent the mean values, with overlaid dots indicating individual sections (n = 3).



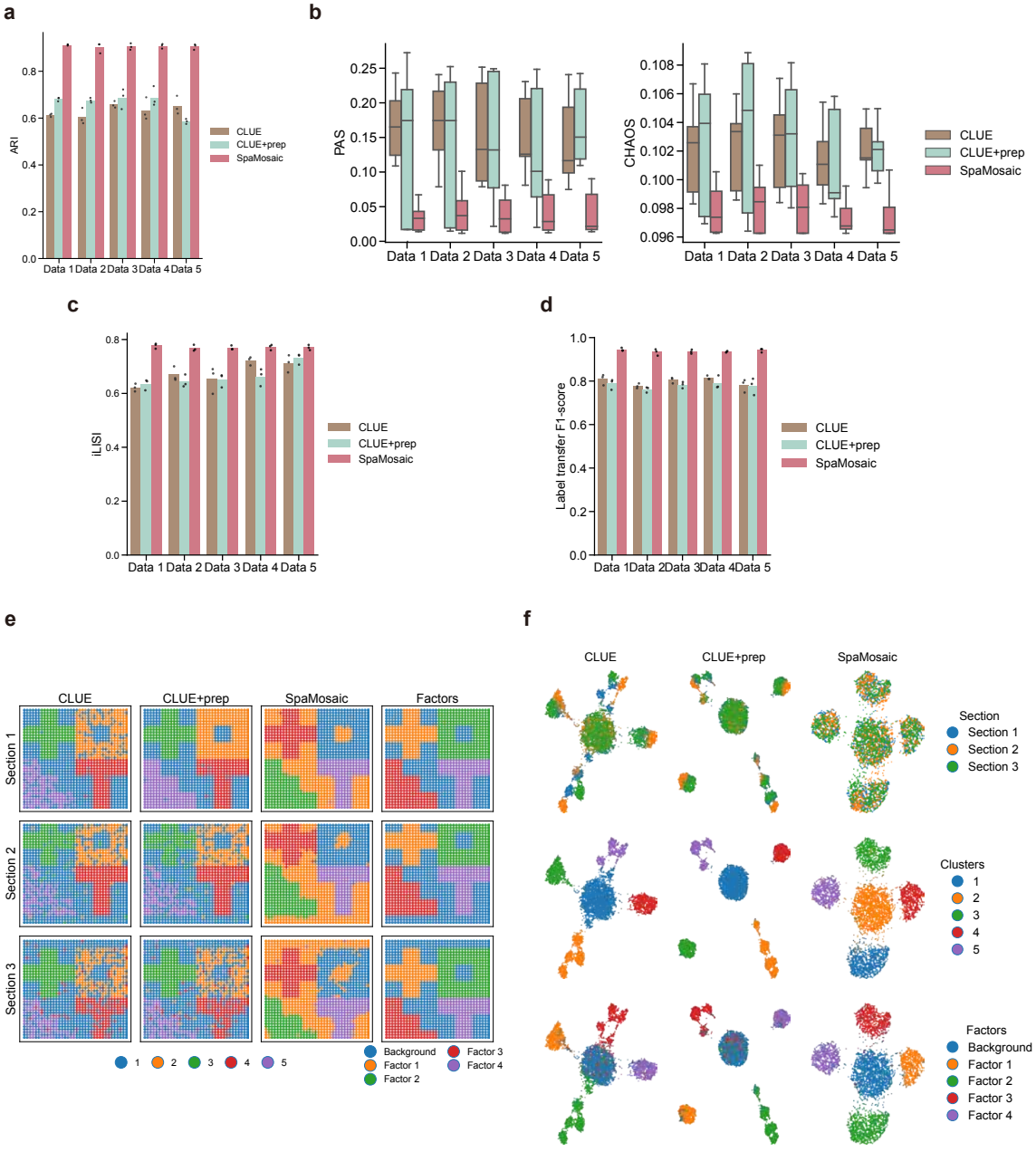
Supplementary Figure S46 Benchmarking of mosaic integration methods using different clustering algorithms on the embryonic mouse brain dataset (Misar). a-c Spatial plots of clustering results from benchmarked mosaic integration methods using Kmeans (**a**), Mclust (**b**), and Leiden (**c**) for clustering. **d** Performance comparison using PAS and CHAOS metrics. Bars represent the mean values, with overlaid dots indicating individual sections (n = 3).



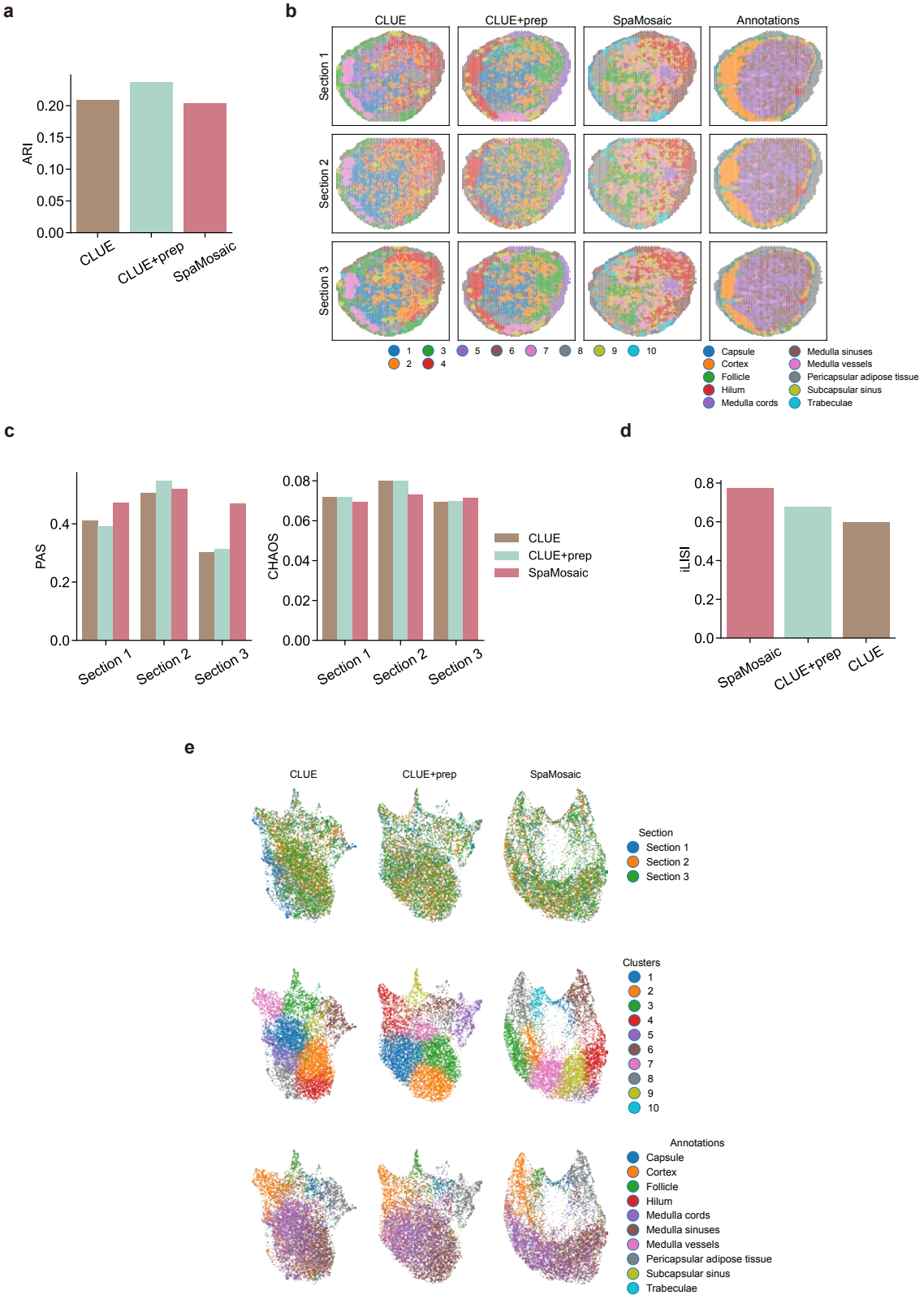
Supplementary Figure S47 Benchmarking of mosaic integration methods using different clustering algorithms on the embryonic mouse brain dataset (Misar+Stereo). a-b Spatial plots of clustering results from all compared mosaic integration methods using Kmeans (**a**) and Mclust (**b**) for clustering.



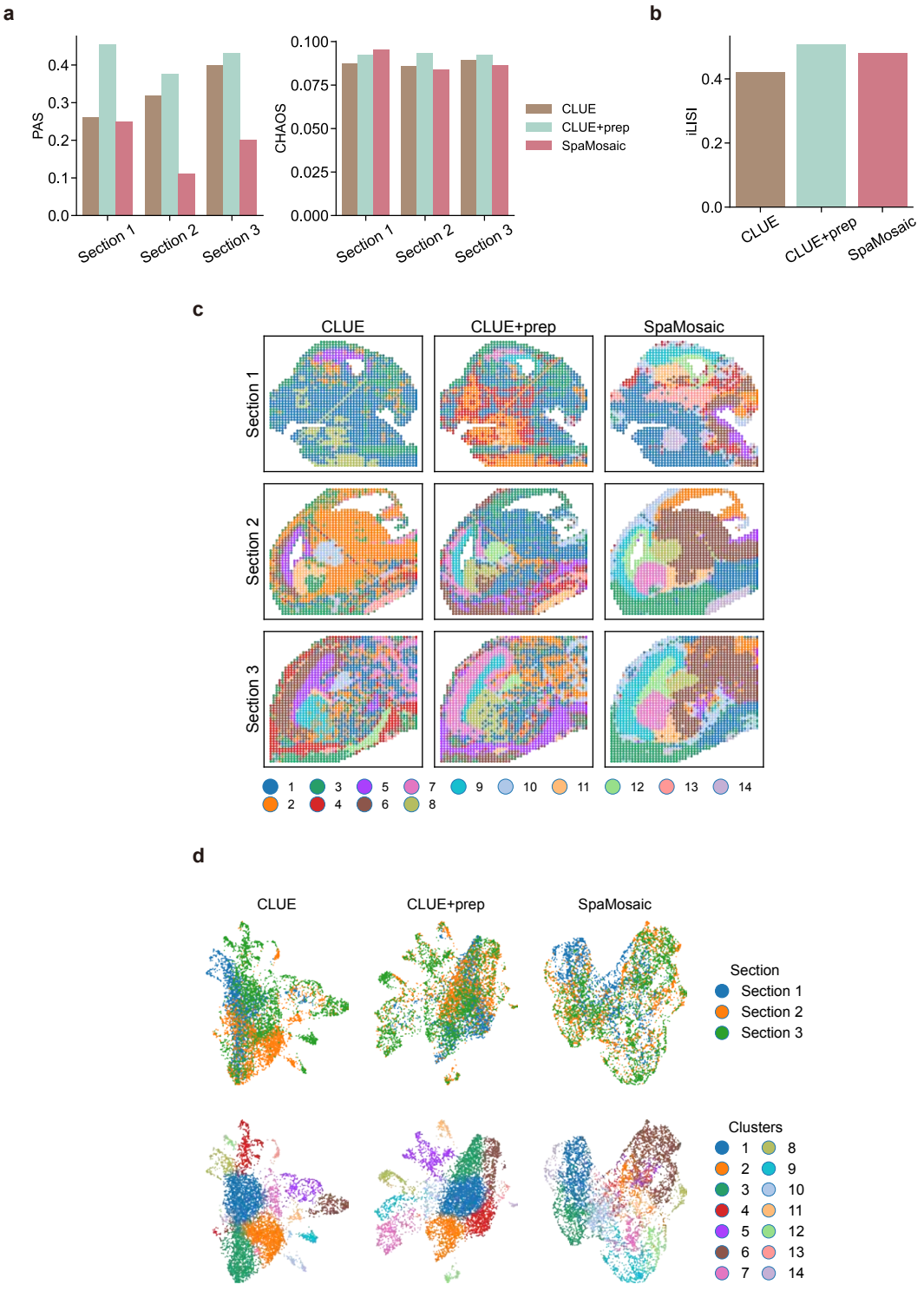
Supplementary Figure S48 Benchmarking of mosaic integration methods using different clustering algorithms on the embryonic mouse brain dataset (Misar+Stereo). **a** Spatial plots of clustering results from benchmarked mosaic integration methods using the Leiden clustering algorithm. **b** Performance comparison using PAS and CHAOS metrics. Box plots for PAS and CHAOS display the distribution of scores calculated across six slices (n=6). Center lines indicate the median; box limits represent the interquartile range (IQR); whiskers extend to 1.5× IQR.



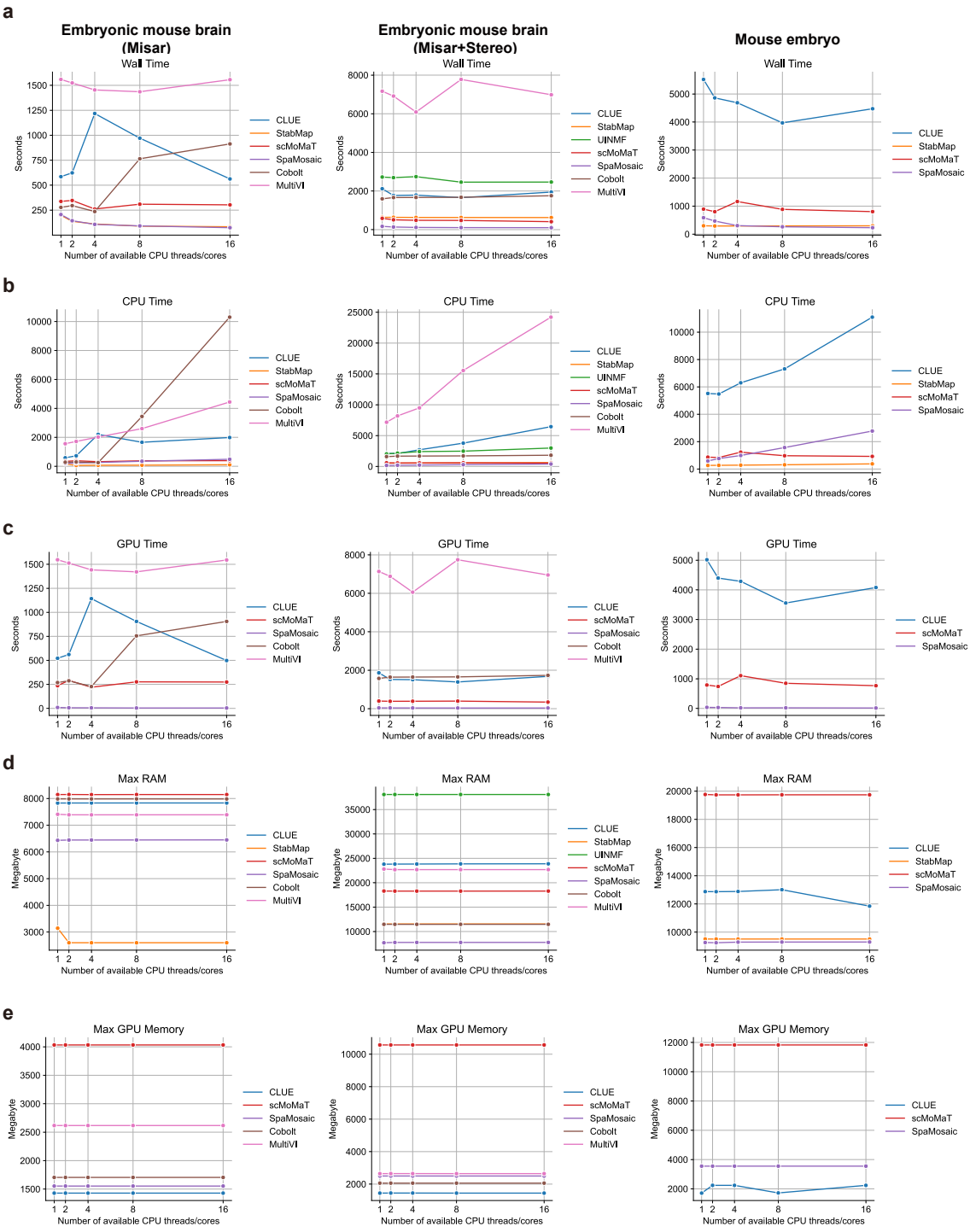
Supplementary Figure S49 Benchmarking results of SpaMosaic, CLUE, and CLUE+prep on the simulated data. **a** ARI scores of the three configurations on five simulation datasets. Bars represent the mean values, with overlaid dots indicating three independent replicates (n=3). **b** PAS and CHAOS scores of the three configurations. Box plots display the distribution of scores derived from three independent replicates, each consisting of three slices (total n=9 data points per box). Center lines indicate the median; box limits represent the interquartile range (IQR); whiskers extend to 1.5× IQR. **c** iLISI scores of the three method configurations. Bars represent the mean values, with overlaid dots indicating three independent replicates (n=3). **d** Label transfer F1-scores of the three configurations. Bars represent the mean values, with overlaid dots indicating independent replicates (n=3). Each replicate is calculated as the average of reciprocal transfers between single-modal slices. **e** Spatial plots of clustering results of the first replicate of the simulation datasets. The rightmost column shows the ground truth of factor patterns. **f** UMAP plots of embeddings of the first replicate of simulation datasets. From top to bottom, the spots are colored by section, cluster and factor labels.



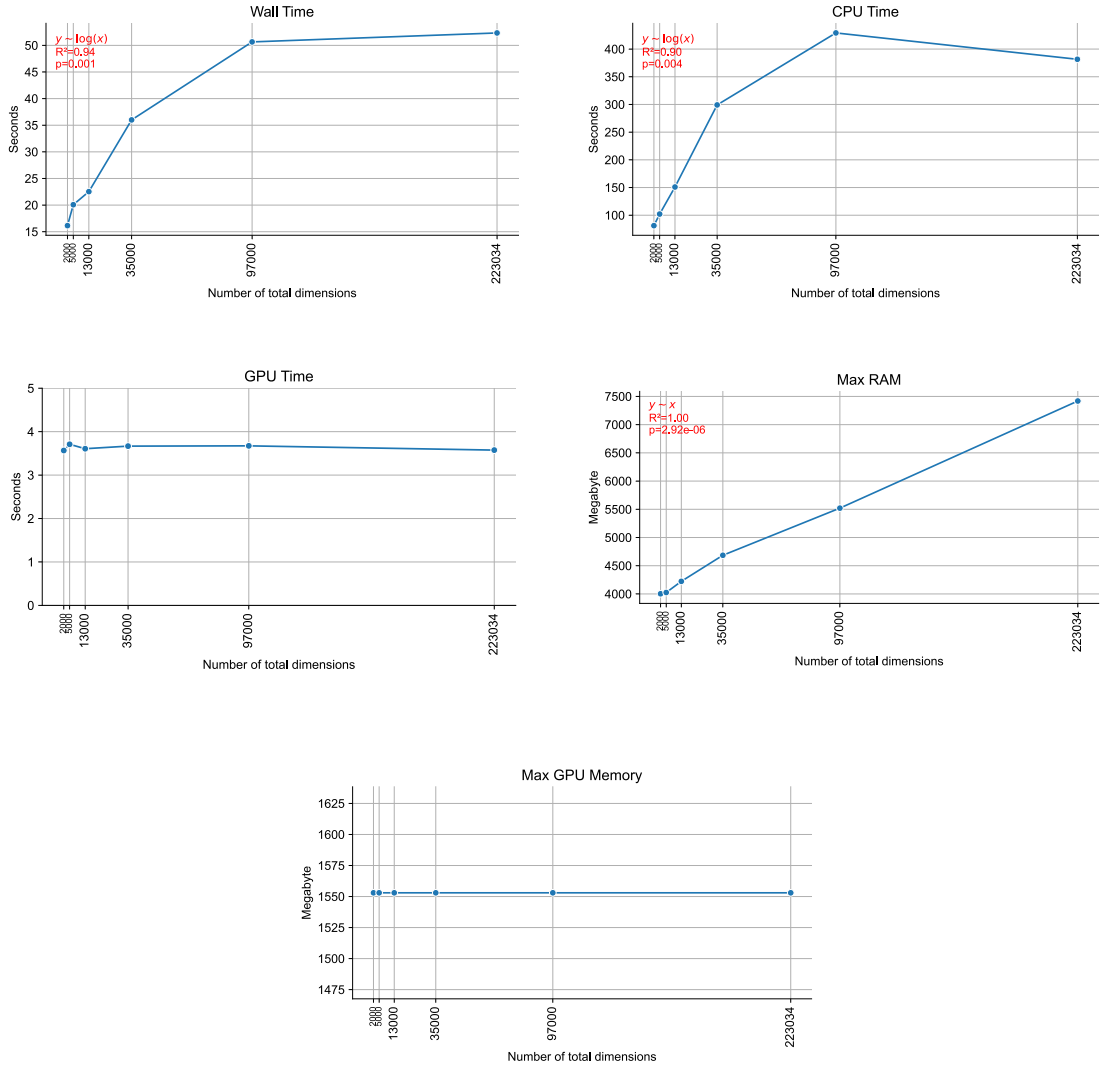
Supplementary Figure S50 Benchmarking results of SpaMosaic, CLUE, and CLUE+prep on the human lymph node dataset. **a** ARI scores of the three method configurations. Bars represent the global scores derived from the entire dataset (n=1). **b** Spatial plots of clustering results. The rightmost column shows the expert manual annotations. **c** PAS and CHAOS scores of the three configurations. Bars represent metric values calculated individually for each section (n=1). **d** iLISI scores of the three configurations. Bars represent the global scores derived from the entire dataset (n=1). **d** Label transfer F1-scores of the three configurations. **e** UMAP plots of embeddings from the three configurations. From top to bottom, the spots are colored by section, cluster and annotation labels.



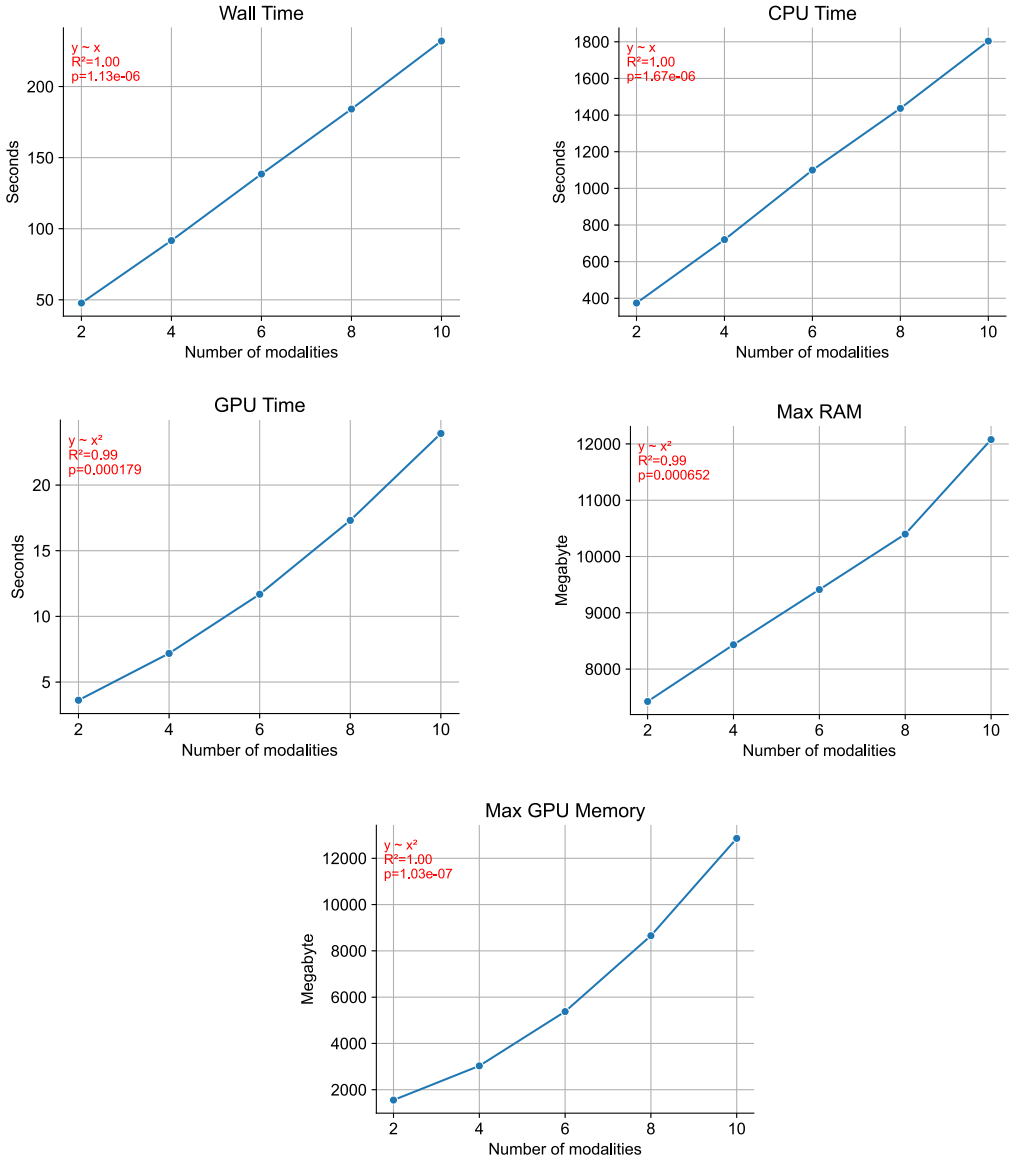
Supplementary Figure S51 Benchmarking results of SpaMosaic, CLUE, and CLUE+prep on the embryonic mouse brain dataset (Misar). **a** PAS and CHAOS scores of the three configurations. Bars represent metric values calculated individually for each section (n=1). **b** iLISI scores of the three configurations. Bars represent the global scores derived from the entire dataset (n=1). **c** Spatial plots of clustering results from the three configurations. **d** UMAP plots of embeddings from three methods. From top to bottom, the spots are colored by section and cluster labels.



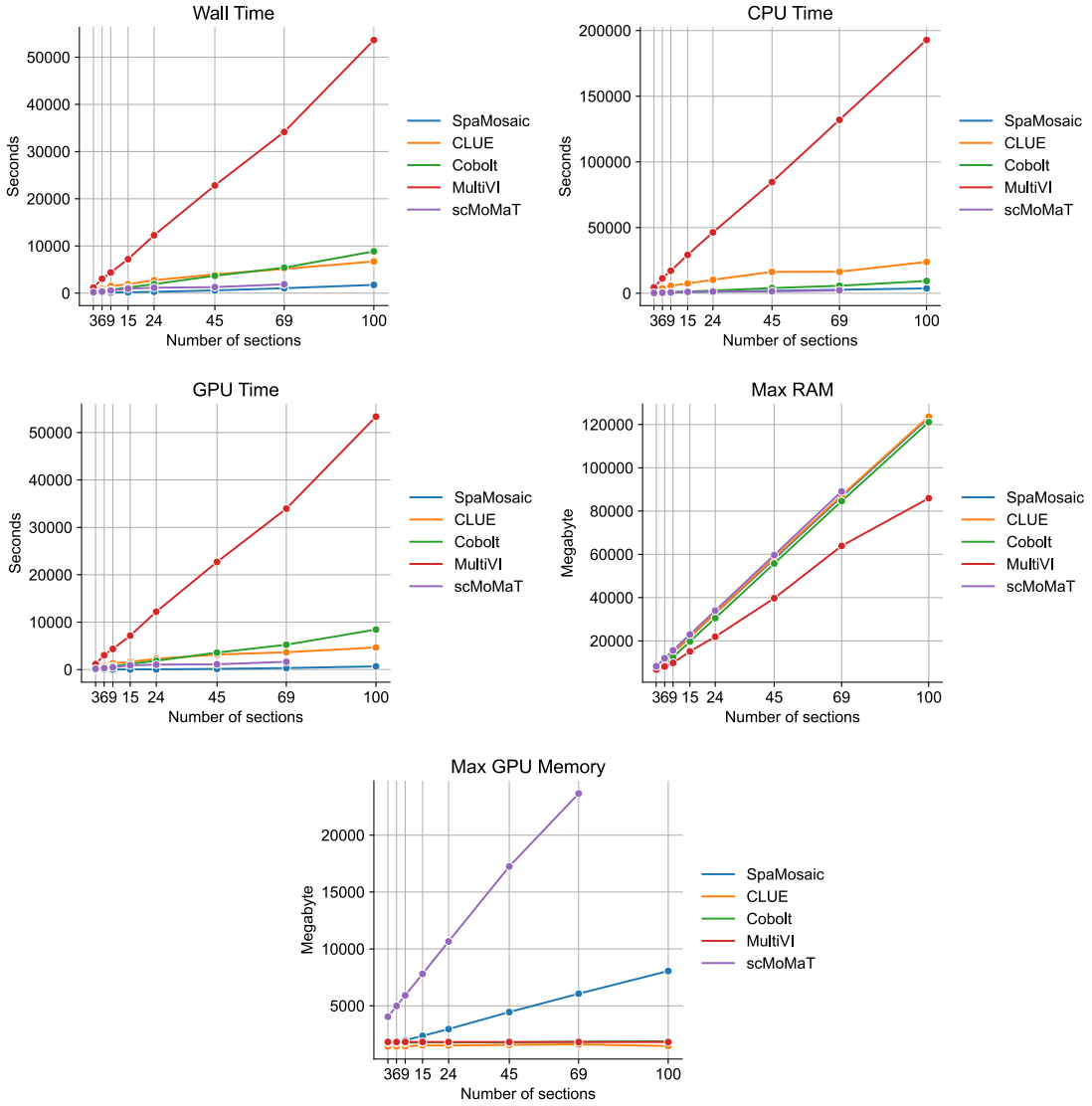
Supplementary Figure S52 Computational cost comparison of mosaic integration methods across different CPU core counts. **a-e** Computational performance comparison of different mosaic integration methods on the embryonic mouse brain (Misar), embryonic mouse brain (Misar+Stereo), and mouse embryo datasets, measuring the wall time (**a**), CPU time (**b**), GPU time (**c**), maximum memory consumption (**d**), and maximum GPU memory consumption (**e**).



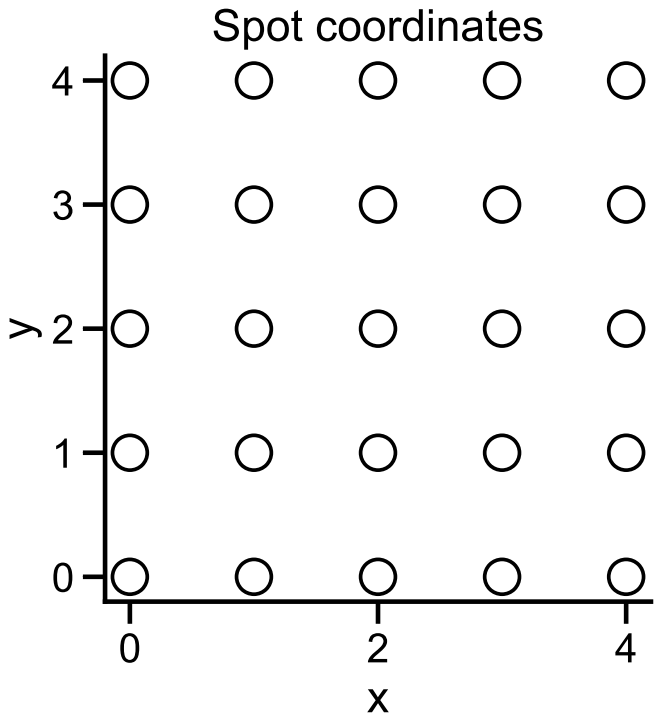
Supplementary Figure S53 Computational cost of SpaMosaic with different input data dimensions. To assess the impact of input feature dimensionality on computational performance, we conducted experiments using the embryonic mouse brain (Misar) dataset, which contains two modalities (RNA and ATAC) across three batches: one batch with both modalities (1949 spots), one with RNA only (1777 spots), and one with ATAC only (2129 spots). We randomly sampled varying numbers of features from the RNA and ATAC modalities. For RNA, 1,000; 2,000; 4,000; 8,000; 16,000; and 32,000 features were sampled; for ATAC, 1,000; 3,000; 9,000; 27,000; 81,000; and 191,034 features were sampled. Each RNA–ATAC feature pair was used to evaluate SpaMosaic's runtime and memory consumption, enabling a systematic analysis of how increasing data dimensionality affects computational cost. The annotations (red) indicate the fitted functional form, coefficient of determination (R^2) and significance level (p). Statistical significance was derived from the Ordinary Least Squares (OLS) regression model, where the p-value corresponds to a two-sided t-test on the slope coefficient (based on the n=6 data points).



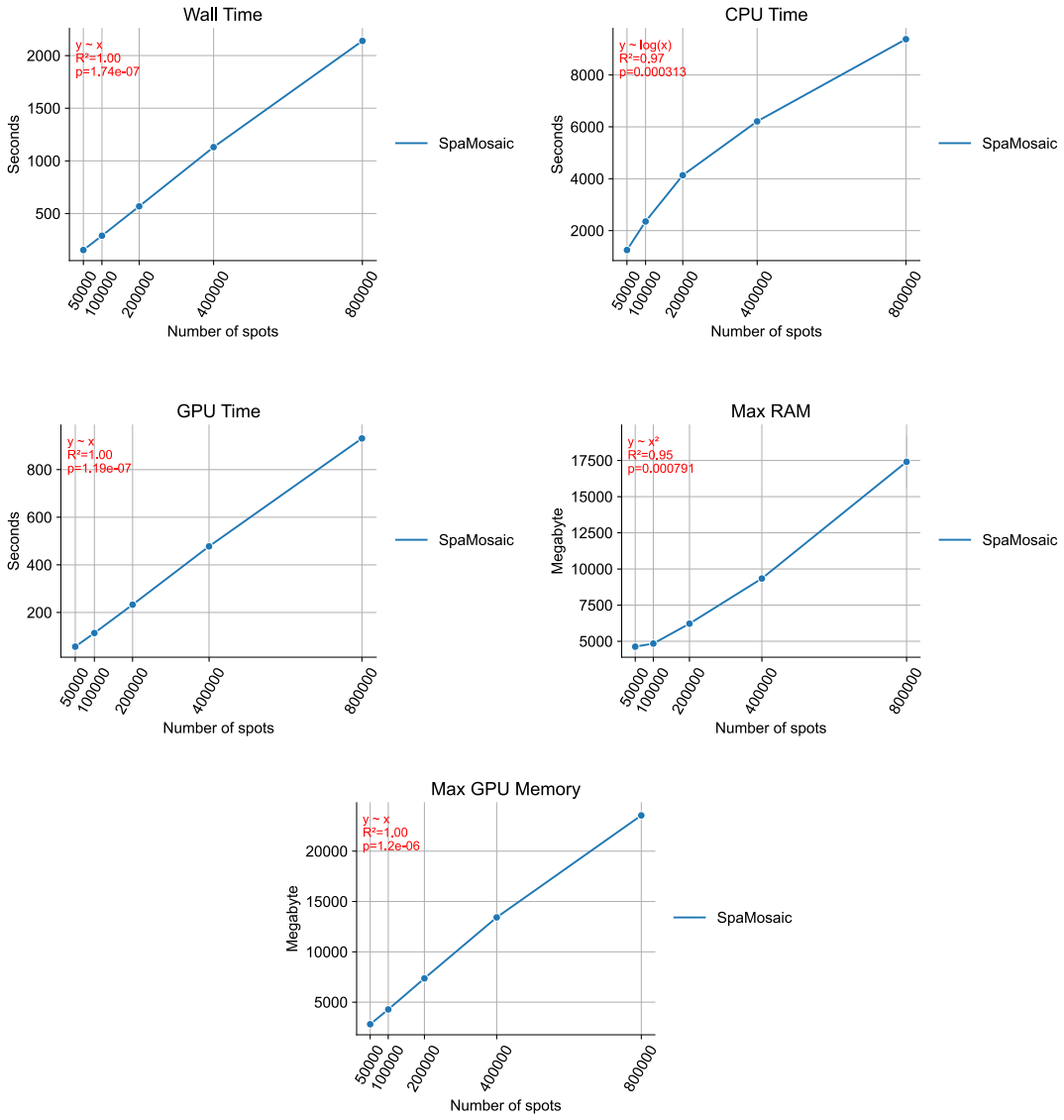
Supplementary Figure S54 Computational cost of SpaMosaic with different numbers of input data modalities. To evaluate how the number of input modalities affects computational performance, we generated simulated datasets from the embryonic mouse brain (Misar) dataset, which originally contains two modalities (RNA and ATAC) across three batches: one batch with both modalities (1949 spots), one with RNA only (1777 spots), and one with ATAC only (2129 spots). Additional modalities were created by replicating each original modality within each batch by a specified multiplication factor and treating each copy as a distinct modality. For example, with a factor of 2, the batch containing both RNA and ATAC becomes {RNA-1, RNA-2, ATAC-1, ATAC-2}, whereas the RNA-only and ATAC-only batches become {RNA-1, RNA-2} and {ATAC-1, ATAC-2}, respectively, yielding 4 modalities in total. Multiplication factors from 1 to 5 were tested, corresponding to 2, 4, 6, 8, and 10 modalities. The annotations (red) indicate the fitted functional form, coefficient of determination (R^2) and significance level (p). Statistical significance was determined using a two-sided t-test on the regression slope coefficient for either the linear or transformed quadratic model, based on n=5 independent data points.



Supplementary Figure S55 Computational cost comparison of mosaic integration methods across different numbers of input sections. To assess how the number of sections influences computational performance, we generated simulated datasets based on the embryonic mouse brain (Misar) dataset, which originally contains three sections (one multimodal section with 1949 spots, one RNA-only section with 1777 spots, and one ATAC-only section with 2129 spots). Additional sections were created by replicating each original section and its corresponding modalities according to predefined multiplication factors. For example, with a factor of 2, each of the three original sections was duplicated once, yielding a dataset with 2 multimodal sections, 2 RNA-only sections, and 2 ATAC-only sections. Replication settings were selected to produce total section counts of 3, 6, 9, 15, 24, 45, 69, and 100. For all configurations except the 100-section setting, the three original sections were replicated equally; the 100-section dataset was generated by replicating the first two sections 33 times and the third section 34 times. Only GPU-based methods were compared.



Supplementary Figure S56 Illustration of spatial coordinates from the largest simulated section for measuring computational cost.



Supplementary Figure S57 Computational scalability of SpaMosaic with increasing number of spots per input section. To evaluate SpaMosaic's scalability with respect to spot density, we generated simulated datasets using the embryonic mouse brain (Misar) dataset as a template. Each dataset configuration consisted of three sections: one multimodal section (RNA+ATAC) and two single-modality sections (RNA-only and ATAC-only), with all three sections containing the same number of spots. RNA profiles were simulated with 10,000 genes using a Poisson distribution with parameter $\lambda = 1$, and ATAC profiles were simulated with 100,000 peaks encoded as binary values with 99.9% sparsity. We created datasets with 50,000, 100,000, 200,000, 400,000, and 800,000 spots per section. The annotations (red) indicate the fitted functional form, coefficient of determination (R^2) and significance level (p). Statistical significance was determined using a two-sided t-test on the regression slope coefficient for the specified functional forms (linear, logarithmic, or quadratic), based on n=5 independent data points.